

Fokker-Planck Equation for Positron Cooling and Annihilation

MSci Project Report

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Abstract

Positron cooling in gases is a fundamental process in antimatter physics, relevant to positron trapping, beam formation and annihilation spectroscopy. At low energies, the positron undergoes many elastic collisions with gas atoms, each producing only a small change in momentum. In this regime, the cumulative dynamics can be described by a Fokker–Planck equation in momentum space. In this dissertation, positron cooling and annihilation are studied in the noble gases helium, neon, argon, krypton and xenon using a Fokker–Planck equation with many-body-theory input for the momentum-transfer cross section and annihilation parameter $Z_{\text{eff}}(p)$.

The Fokker–Planck equation is solved numerically on a uniform momentum grid using a conservative finite-volume discretisation for the cooling operator together with an operator-split exponential update for annihilation. Verification tests show that the transport step conserves total probability to machine precision in the absence of annihilation and that the annihilation sink is implemented consistently.

The results reveal markedly different behaviour across the noble gases. Helium provides a baseline case of weak annihilation and near-thermal cooling, while neon exhibits slow cooling associated with inefficient low-energy momentum transfer. Argon and krypton show intermediate behaviour, with annihilation increasingly affecting the surviving swarm. Xenon is the most striking case: strong low-momentum annihilation prevents the surviving distribution from approaching a true thermal Maxwellian before the population becomes extremely small. Instead, the late-time solution approaches a non-Maxwellian asymptotic form.

To interpret this behaviour, an eigenanalysis of the discrete Fokker–Planck operators is carried out for xenon. The dominant eigenmodes reproduce both the late-time shape and the decay rate obtained from direct time stepping, providing a spectral interpretation of the long-time dynamics.

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Chapter 1

Introduction

Positrons, the antiparticles of electrons, provide a useful probe of low-energy collision physics and are important in applications ranging from positron annihilation spectroscopy to the production of cold antimatter, for example antihydrogen atoms [1, 2]. In gases, a positron injected at non-thermal energies undergoes repeated collisions with surrounding atoms and gradually loses energy until it either thermalises or annihilates. Understanding this cooling process is therefore important both for interpreting experiments and for controlling positron swarms and the creation of positron beams in practical settings [2].

At sufficiently low energies, each elastic collision changes the positron momentum only slightly. The cumulative effect of many such collisions can then be described as drift and diffusion in momentum space, making the Fokker–Planck equation a natural framework for modelling the cooling dynamics. In radial form, this gives a one-dimensional equation for the isotropic momentum distribution, with an additional sink term to account for annihilation.

A central difficulty is that the Fokker–Planck equation is not closed by itself: its coefficients depend on the underlying microscopic collision physics. In the present work, these are supplied by many-body-theory (MBT) calculations for positron–noble-gas interactions. These provide accurate low-energy momentum-transfer cross sections and annihilation rates, including important correlation effects and virtual positronium formation. This makes it possible to combine a relatively simple transport model with realistic atomic collision input.

The noble gases provide a particularly useful set of test systems because the balance between cooling and annihilation changes significantly from helium to xenon. Helium represents a weak-annihilation case in which the positron momentum distribution remains comparatively close

to thermal expectations, whereas in the heavier noble gases, especially xenon, annihilation becomes strong enough to reshape the surviving swarm before thermal equilibrium is reached. This progression makes the noble gases an ideal family for studying how momentum-dependent annihilation modifies low-energy positron cooling.

The aim of this dissertation is to model positron cooling and annihilation in helium, neon, argon, krypton and xenon using a radial momentum-space Fokker–Planck equation informed by MBT input data [3], and to determine how the competition between transport and annihilation changes across the noble-gas sequence. The equation is solved numerically using a conservative finite-volume discretisation for the cooling operator together with an operator-split exponential update for the annihilation sink. From the resulting solutions, quantities such as the survival fraction, averaged annihilation rate, mean energy and cooling time are extracted and compared across the different gases.

A further aim is to understand the late-time behaviour of the surviving positron swarm when annihilation is strong. In a cooling-only problem, the long-time momentum distribution approaches the thermal Maxwellian. With momentum-dependent annihilation present, however, this need not remain true. One of the main results of this dissertation is that xenon provides a clear example of this breakdown: the surviving positron momentum distribution approaches a non-Maxwellian asymptotic shape rather than a true thermal equilibrium. To interpret this, an eigenanalysis of the discrete operators underlying the numerical scheme is carried out.

The dissertation is organised as follows. Chapter 2 develops the theoretical model, including the derivation of the radial Fokker–Planck equation, the inclusion of annihilation, the MBT input data, and the initial and boundary conditions. Chapter 3 describes the numerical method and its verification. Chapter 4 presents the results for the noble gases. Chapter 5 analyses the asymptotic eigenmodes, with particular emphasis on xenon. Finally, Chapter 6 summarises the main conclusions and outlines possible directions for future work.

Chapter 2

Theory

2.1 Physical problem and modelling assumptions

We consider the cooling of a swarm of positrons in a dilute gas of noble-gas atoms at room temperature. After rapid initial energy loss at high energies via ionisation and other inelastic collisions, the positrons enter a low-energy regime in which their evolution is dominated by many elastic collisions with the gas atoms. Since each collision transfers only a small fraction of the positron momentum, the cumulative effect of many such collisions can be described as a drift–diffusion process in momentum space, leading naturally to a Fokker–Planck description. This is the modelling framework adopted in the present work, following the project brief [4] and earlier studies of positron cooling [5].

The gas is assumed to remain in thermal equilibrium at temperature T , with atomic velocities distributed according to the Maxwell–Boltzmann distribution. The positron distribution is taken to be isotropic in momentum space, so that it depends only on the magnitude $p = |\mathbf{p}|$. This allows the problem to be reduced to a one-dimensional radial equation for the momentum distribution. The corresponding radial distribution is written as

$$F(p, t) = 4\pi p^2 f(p, t), \tag{2.1}$$

where $f(p, t)$ is the underlying isotropic distribution function in momentum space, and t is time.

Only two physical processes are included in the model. The first is momentum-space transport due to elastic scattering, which is governed by the momentum-transfer cross section $\sigma_m(p)$.

The second is positron annihilation, which removes particles from the swarm at a momentum-dependent rate determined by the dimensionless effective number of electrons per target atom, $Z_{\text{eff}}(p)$ [6]. Other inelastic channels are neglected in the present formulation. In particular, the model is intended for the low-energy regime below the positronium-formation threshold, where elastic scattering dominates the cooling dynamics and the Fokker–Planck approximation is appropriate.

The required scattering and annihilation input data are not generated within the Fokker–Planck model itself. Instead, they are taken from many-body-theory calculations for positron–noble-gas interactions, which provide accurate low-energy phase shifts, momentum-transfer cross sections, and annihilation [3]. These quantities enter the model through the transport coefficient $B(p)$ and the annihilation rate $\lambda(p)$, and will be described in a later subsection.

With these assumptions in place, the time evolution of the positron swarm can be formulated as a radial Fokker–Planck equation in momentum space. We now derive this equation.

2.2 Derivation of the radial Fokker–Planck equation

2.2.1 Derivation of the radial form

A similar derivation appears in Chapter 2 of Landau and Lifshitz Volume 10 for the isotropic momentum distribution $f(p, t)$ [7]. We assume that, after angular averaging, the positron momentum distribution is isotropic, so that it depends only on the magnitude $p = |\mathbf{p}|$. It is then convenient to work with the radial momentum distribution

$$F(p, t) = 4\pi p^2 f(p, t), \quad (2.2)$$

which is normalized according to

$$\int_0^\infty F(p, t) dp = 1. \quad (2.3)$$

Thus, $F(p, t) dp$ is the probability of finding the positron with momentum magnitude in the interval $[p, p + dp]$.

To derive the evolution equation for F , we consider a one-dimensional jump process in the radial variable p . Let $w(p, q) dq$ be the probability per unit time that a positron with momentum magnitude p undergoes a collision in which its momentum changes from p to $p - q$, where $q > 0$

is small. The corresponding master equation is

$$\frac{\partial F(p, t)}{\partial t} = \int_0^\infty [w(p+q, q) F(p+q, t) - w(p, q) F(p, t)] dq. \quad (2.4)$$

The first term represents gain into the state p from particles previously at $p+q$, while the second term represents loss from the state p .

For sufficiently small momentum changes, the gain term can be expanded in powers of q :

$$w(p+q, q)F(p+q, t) = w(p, q)F(p, t) + q \frac{\partial}{\partial p} [w(p, q)F(p, t)] + \frac{q^2}{2} \frac{\partial^2}{\partial p^2} [w(p, q)F(p, t)] + \mathcal{O}(q^3). \quad (2.5)$$

Substituting Eq. (2.5) into Eq. (2.4), the zeroth-order terms cancel, and retaining terms up to second order gives

$$\frac{\partial F}{\partial t} = \frac{\partial}{\partial p} [\tilde{A}(p) F] + \frac{\partial^2}{\partial p^2} [B(p) F], \quad (2.6)$$

where the first two jump moments are

$$\tilde{A}(p) = \int_0^\infty q w(p, q) dq, \quad B(p) = \frac{1}{2} \int_0^\infty q^2 w(p, q) dq. \quad (2.7)$$

It is useful to rewrite Eq. (2.6) in conservative form. Expanding the second term gives

$$\frac{\partial^2}{\partial p^2} [BF] = \frac{\partial}{\partial p} \left[\frac{dB}{dp} F + B \frac{\partial F}{\partial p} \right]. \quad (2.8)$$

Hence

$$\frac{\partial F}{\partial t} = \frac{\partial}{\partial p} \left[A(p) F + B(p) \frac{\partial F}{\partial p} \right], \quad (2.9)$$

with

$$A(p) = \tilde{A}(p) + \frac{dB}{dp}. \quad (2.10)$$

Introducing the radial flux

$$J(p, t) = - \left[A(p) F(p, t) + B(p) \frac{\partial F(p, t)}{\partial p} \right], \quad (2.11)$$

which casts Eq. (2.9) in the form of the continuity equation

$$\frac{\partial F}{\partial t} + \frac{\partial J}{\partial p} = 0. \quad (2.12)$$

The drift coefficient $A(p)$ is fixed by the requirement that the stationary of Eq. (2.12) be the Maxwell–Boltzmann distribution. In radial form this is

$$F_{\text{th}}(p) = \frac{4\pi p^2}{(2\pi m k_B T)^{3/2}} \exp\left(-\frac{p^2}{2m k_B T}\right). \quad (2.13)$$

where k_B is the Boltzmann constant and m is the mass of the particle (positron).

At equilibrium the flux must vanish, i.e.

$$J_{\text{th}}(p) = 0, \quad (2.14)$$

so that

$$A(p) F_{\text{th}}(p) + B(p) \frac{dF_{\text{th}}}{dp} = 0. \quad (2.15)$$

Therefore

$$A(p) = -B(p) \frac{d}{dp} \ln F_{\text{th}}(p). \quad (2.16)$$

Using

$$\frac{d}{dp} \ln F_{\text{th}}(p) = \frac{2}{p} - \frac{p}{mk_B T}, \quad (2.17)$$

we obtain

$$A(p) = B(p) \left(\frac{p}{mk_B T} - \frac{2}{p} \right). \quad (2.18)$$

Substituting Eq. (2.18) into Eq. (2.9) yields the radial Fokker–Planck equation

$$\frac{\partial F(p, t)}{\partial t} = \frac{\partial}{\partial p} \left\{ B(p) \left[\left(\frac{p}{mk_B T} - \frac{2}{p} \right) F(p, t) + \frac{\partial F(p, t)}{\partial p} \right] \right\}. \quad (2.19)$$

Finally, using $F = 4\pi p^2 f$ shows that Eq. (2.19) is equivalent to

$$\frac{\partial f(p, t)}{\partial t} = \frac{1}{p^2} \frac{\partial}{\partial p} \left[p^2 B(p) \left(\frac{p f(p, t)}{mk_B T} + \frac{\partial f(p, t)}{\partial p} \right) \right]. \quad (2.20)$$

At this stage the coefficient $B(p)$ remains unspecified. We now derive it from the mean-squared momentum change in elastic positron–atom collisions.

2.2.2 Derivation of the diffusion coefficient $B(p)$

We consider elastic collisions between a positron of mass m and velocity $\mathbf{v}$, and a gas atom of mass M and velocity $\mathbf{V}$. Quantities after the collision are denoted by primes.

In the centre-of-mass frame, conservation of kinetic energy for an elastic collision gives

$$|\mathbf{V} - \mathbf{v}|^2 = |\mathbf{V}' - \mathbf{v}'|^2. \quad (2.21)$$

Expanding both sides,

$$V^2 - 2\mathbf{V} \cdot \mathbf{v} + v^2 = V'^2 - 2\mathbf{V}' \cdot \mathbf{v}' + v'^2. \quad (2.22)$$

Since the atom is much heavier than the positron ($M \gg m$), its velocity changes only slightly in a single collision, so to leading order we may set

$$\mathbf{V}' \simeq \mathbf{V}. \quad (2.23)$$

Equation (2.22) then reduces to

$$v'^2 - v^2 = 2\mathbf{V} \cdot (\mathbf{v}' - \mathbf{v}). \quad (2.24)$$

Let

$$\Delta v = v' - v, \quad (2.25)$$

with $|\Delta v| \ll v$. Then

$$v'^2 = (v + \Delta v)^2 = v^2 + 2v \Delta v + \mathcal{O}(\Delta v^2), \quad (2.26)$$

so, neglecting terms of order Δv^2 ,

$$v'^2 - v^2 \simeq 2v \Delta v. \quad (2.27)$$

Substituting Eq. (2.27) into Eq. (2.24) gives

$$\Delta v = \frac{1}{v} \mathbf{V} \cdot (\mathbf{v}' - \mathbf{v}). \quad (2.28)$$

Since $p = mv$, the corresponding change in the positron momentum magnitude is

$$\Delta p = m \Delta v, \quad (2.29)$$

and hence

$$(\Delta p)^2 = \frac{m^2}{v^2} [(\mathbf{V} \cdot \mathbf{v}')^2 + (\mathbf{V} \cdot \mathbf{v})^2 - 2(\mathbf{V} \cdot \mathbf{v}')(\mathbf{V} \cdot \mathbf{v})]. \quad (2.30)$$

We now average over the thermal velocities of the gas atoms. By isotropy,

$$\langle V_i V_j \rangle = \frac{1}{3} \langle V^2 \rangle \delta_{ij}, \quad (2.31)$$

so that for any vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\langle (\mathbf{V} \cdot \mathbf{a})(\mathbf{V} \cdot \mathbf{b}) \rangle = \frac{1}{3} \langle V^2 \rangle \mathbf{a} \cdot \mathbf{b}. \quad (2.32)$$

Using equipartition for the gas atoms,

$$\frac{1}{2} M \langle V^2 \rangle = \frac{3}{2} k_B T, \quad (2.33)$$

we obtain

$$\langle V^2 \rangle = \frac{3k_B T}{M}. \quad (2.34)$$

Therefore Eq. (2.32) becomes

$$\langle (\mathbf{V} \cdot \mathbf{a})(\mathbf{V} \cdot \mathbf{b}) \rangle = \frac{k_B T}{M} \mathbf{a} \cdot \mathbf{b}. \quad (2.35)$$

Applying Eq. (2.35) to Eq. (2.30) gives

$$\langle (\Delta p)^2 \rangle_{\mathbf{v}} = \frac{m^2 k_B T}{v^2 M} (v'^2 + v^2 - 2 \mathbf{v}' \cdot \mathbf{v}). \quad (2.36)$$

For elastic scattering, $v' = v$, and if θ is the scattering angle then

$$\mathbf{v}' \cdot \mathbf{v} = v^2 \cos \theta. \quad (2.37)$$

Hence

$$\langle (\Delta p)^2 \rangle_{\mathbf{v}} = \frac{2m^2 k_B T}{M} (1 - \cos \theta). \quad (2.38)$$

To obtain the mean squared momentum change per unit time, we average over collisions. If n_g is the number density of gas atoms, then the rate of scattering into the solid angle element $d\Omega$ is

$$n_g v \frac{d\sigma}{d\Omega} d\Omega, \quad (2.39)$$

where $\frac{d\sigma}{d\Omega}$ is the positron-atom differential elastic collision cross-section.

Therefore

$$\left\langle \frac{\delta p^2}{\delta t} \right\rangle = n_g v \int \langle (\Delta p)^2 \rangle_{\mathbf{v}} \frac{d\sigma}{d\Omega} d\Omega. \quad (2.40)$$

Using Eq. (2.38),

$$\left\langle \frac{\delta p^2}{\delta t} \right\rangle = \frac{2m^2 k_B T}{M} n_g v \int (1 - \cos \theta) \frac{d\sigma}{d\Omega} d\Omega. \quad (2.41)$$

We now identify the momentum-transfer cross section,

$$\sigma_m(p) = \int (1 - \cos \theta) \frac{d\sigma}{d\Omega} d\Omega. \quad (2.42)$$

Thus

$$\left\langle \frac{\delta p^2}{\delta t} \right\rangle = \frac{2m^2 k_B T}{M} n_g v \sigma_m(p). \quad (2.43)$$

By definition (see Eq. (2.7)), the diffusion coefficient in momentum space is one half of the mean squared momentum change per unit time:

$$B(p) = \frac{1}{2} \left\langle \frac{\delta p^2}{\delta t} \right\rangle. \quad (2.44)$$

Hence

$$B(p) = \frac{m^2 k_B T}{M} n_g v \sigma_m(p). \quad (2.45)$$

Finally, using $p = mv$, we obtain

$$B(p) = n_g \frac{m}{M} k_B T p \sigma_m(p). \quad (2.46)$$

Equation (2.46) is the dimensional form of the diffusion coefficient. In the next section, it will be rewritten in the time-scaled variable used in this work.

2.2.3 Including annihilation and introducing the scaled time variable

The transport equation (2.9) derived above describes positron cooling through many small momentum-transfer collisions. In this form it conserves the total number of positrons,

$$\int_0^\infty F(p, t) dp, \quad (2.47)$$

since it has the structure of a continuity equation in momentum space (2.12). However, for the present problem this is not sufficient, because positrons can also be removed from the swarm through electron–positron annihilation. The Fokker–Planck equation must therefore be modified to include positron depletion due to annihilation.

Starting from the transport-only radial Fokker–Planck equation,

$$\frac{\partial F(p, t)}{\partial t} = \frac{\partial}{\partial p} \left\{ B(p) \left[\left(\frac{p}{m k_B T} - \frac{2}{p} \right) F(p, t) + \frac{\partial F(p, t)}{\partial p} \right] \right\}, \quad (2.48)$$

we include annihilation by adding a local sink term. Physically, annihilation removes positrons from the distribution without redistributing the surviving particles in momentum space, so the natural modification is

$$\frac{\partial F(p, t)}{\partial t} = \frac{\partial}{\partial p} \left\{ B(p) \left[\left(\frac{p}{m k_B T} - \frac{2}{p} \right) F(p, t) + \frac{\partial F(p, t)}{\partial p} \right] \right\} - \lambda_{\text{ann}}(p) F(p, t). \quad (2.49)$$

The annihilation rate for a positron in a gas of number density n_g is conventionally written as

$$\lambda_{\text{ann}}(p) = \pi r_0^2 c n_g Z_{\text{eff}}(p), \quad (2.50)$$

where r_0 is the classical electron radius, c is the speed of light, and $Z_{\text{eff}}(p)$ is the dimensionless effective number of atomic electrons participating in the annihilation process [3].

Using the result derived in the previous subsection,

$$B(p) = n_g \frac{m}{M} k_B T p \sigma_m(p), \quad (2.51)$$

Eq. (2.49) becomes

$$\frac{\partial F}{\partial t} = n_g \frac{m}{M} k_B T \frac{\partial}{\partial p} \left\{ \sigma_m(p) p \left[\left(\frac{p}{m k_B T} - \frac{2}{p} \right) F + \frac{\partial F}{\partial p} \right] \right\} - \lambda_{\text{ann}}(p) F. \quad (2.52)$$

We now rewrite the equation in the scaled time variable used throughout this work. Let the gas number density be written as

$$n_g = n_0 \tilde{n}, \quad (2.53)$$

where n_0 is the number density of the ideal gas at normal conditions, denoted as 1 amagat, and $\tilde{n}$ is the density measured in amagat¹. We also introduce the scaled time

$$\tau = \tilde{n} t_{\text{ns}}, \quad (2.54)$$

where t_{ns} is the physical time measured in nanoseconds. Thus τ has units of ns amagat, which is convenient because it removes the explicit density dependence from the transport term. This choice is also standard in the positron cooling literature [21].

To convert derivatives, note that time in atomic units is related to time in seconds by

$$t_{\text{au}} = \frac{t_s}{t_a}, \quad t_s = 10^{-9} t_{\text{ns}}, \quad (2.55)$$

where t_a is the atomic unit of time. Hence

$$t_{\text{au}} = \frac{10^{-9} t_{\text{ns}}}{t_a}, \quad \frac{\partial}{\partial t_{\text{au}}} = \frac{t_a}{10^{-9}} \frac{\partial}{\partial t_{\text{ns}}} = \frac{t_a}{10^{-9}} \tilde{n} \frac{\partial}{\partial \tau}. \quad (2.56)$$

Substituting $n_g = n_0 \tilde{n}$ and Eq. (2.56) into Eq. (2.52), the factor $\tilde{n}$ cancels from the transport term. For a positron, $m = 1$ in atomic units, so we obtain

$$\frac{\partial F(p, \tau)}{\partial \tau} = \alpha \frac{\partial}{\partial p} \left\{ \sigma_m(p) p \left[\frac{\partial F(p, \tau)}{\partial p} + \left(\frac{p}{k_B T} - \frac{2}{p} \right) F(p, \tau) \right] \right\} - \lambda_{\text{ann}}(p) F(p, \tau), \quad (2.57)$$

where

$$\alpha = \frac{10^{-9} n_0 k_B T}{t_a M}, \quad (2.58)$$

¹Here 1 amagat is defined relative to the ideal-gas number density at $T = 273.15$ K and $p = 1$ atm, so that $n_0 = 2.68678 \times 10^{25} \text{ m}^{-3}$.

and where p and $\sigma_m(p)$ are the positron momentum and momentum transfer cross-section in atomic units. For notational simplicity, the annihilation rate $\lambda_{ann}(p)$ will henceforth be written simply as $\lambda(p)$.

It is convenient to write this as

$$\alpha = D_0 \frac{T}{M}, \quad D_0 = \frac{10^{-9} n_0 k_B}{t_a}, \quad (2.59)$$

with k_B expressed in atomic units per kelvin and M in electron masses.

The scaled annihilation rate $\lambda(p)$ in Eq. (2.57) is defined in units of $(\text{ns amagat})^{-1}$. From Eq. (2.50), this is

$$\lambda(p) = \frac{10^{-9}}{t_a} \pi r_0^2 c n_0 Z_{\text{eff}}(p). \quad (2.60)$$

Equation (2.57) is the form solved numerically in this work. It contains a conservative momentum-space transport operator governed by the momentum-transfer cross section $\sigma_m(p)$, together with a momentum-dependent annihilation sink determined by $Z_{\text{eff}}(p)$. This separation is physically useful: $\sigma_m(p)$ controls cooling, while $Z_{\text{eff}}(p)$ controls depletion of the positron population. The importance of retaining the annihilation term has been demonstrated in Monte-Carlo simulations of noble-gas cooling, which used MBT data and showed that depletion can significantly alter the observed evolution, particularly for xenon [6].

Finally, Eq. (2.57) can be written in the flux form as

$$\frac{\partial F}{\partial \tau} + \frac{\partial J}{\partial p} = -\lambda(p) F, \quad (2.61)$$

with the momentum-space flux given by

$$J(p, \tau) = -\alpha \sigma_m(p) p \left[\frac{\partial F(p, \tau)}{\partial p} + \left(\frac{p}{k_B T} - \frac{2}{p} \right) F(p, \tau) \right]. \quad (2.62)$$

This form is particularly convenient for the finite-volume discretisation used later.

2.3 Many-body-theory input data

The Fokker–Planck equation derived above is not fully defined until the momentum-transfer cross section $\sigma_m(p)$ and annihilation input $Z_{\text{eff}}(p)$ are specified. In the present work these quantities are taken from the MBT calculations of Green, Ludlow and Gribakin for positron scattering and annihilation on noble-gas atoms [3].

The appeal of the MBT data is that they provide an accurate *ab initio* description of low-energy positron–atom interactions, including the important positron–atom and electron–positron correlation effects, as well as virtual positronium formation. These effects have a strong influence on both elastic scattering and annihilation, especially for the heavier noble gases. The MBT calculations provide tabulated phase shifts, elastic cross sections and Z_{eff} values for positron energies below the positronium-formation threshold.

Since the MBT tables are provided only at a discrete set of positron momenta, they must be converted into smooth input data on the numerical momentum grid used by the solver. In practice, this was done separately for the transport cross section $\sigma_m(p)$ and the annihilation quantity $Z_{\text{eff}}(p)$, as described below. A representative xenon implementation of this preprocessing is recorded in Appendix G.

2.3.1 Momentum-transfer cross section $\sigma_m(p)$

The MBT paper tabulates the scattering phase shifts $\delta_\ell(k)$ for the *s*-, *p*- and *d*-waves, i.e. $\ell = 0, 1, 2$, for each noble-gas atom. These are listed in Tables III–VII of Ref. [3]. In atomic units, the wave number k is equal to the positron momentum p , so the tabulated phase shifts can be used directly as functions of p .

For each gas, the tabulated MBT values of $\delta_0(p)$, $\delta_1(p)$ and $\delta_2(p)$ were interpolated onto the solver grid² using cubic splines [8]. The phase shifts for the higher partial waves were then approximated by the asymptotic low-energy polarization formula

$$\delta_\ell(p) \approx \frac{\pi \alpha_d p^2}{(2\ell - 1)(2\ell + 1)(2\ell + 3)}, \quad \ell \geq 3, \quad (2.63)$$

where α_d is the static dipole polarizability of the target atom. This is the same large- ℓ approximation used in the MBT scattering analysis and in the later cooling paper [6].

Once the phase shifts are known, the momentum-transfer cross section is computed from

$$\sigma_m(p) = \frac{4\pi}{p^2} \sum_{\ell=0}^{\ell_{\text{max}}-1} (\ell + 1) \sin^2[\delta_\ell(p) - \delta_{\ell+1}(p)]. \quad (2.64)$$

In the present implementation, the sum was truncated at $\ell_{\text{max}} = 10$, which is sufficient because the higher- ℓ phase shifts are already in their asymptotic regime and become rapidly small. At

²The solver employed an equidistant momentum grid with step size $\Delta p = 0.01$ a.u..

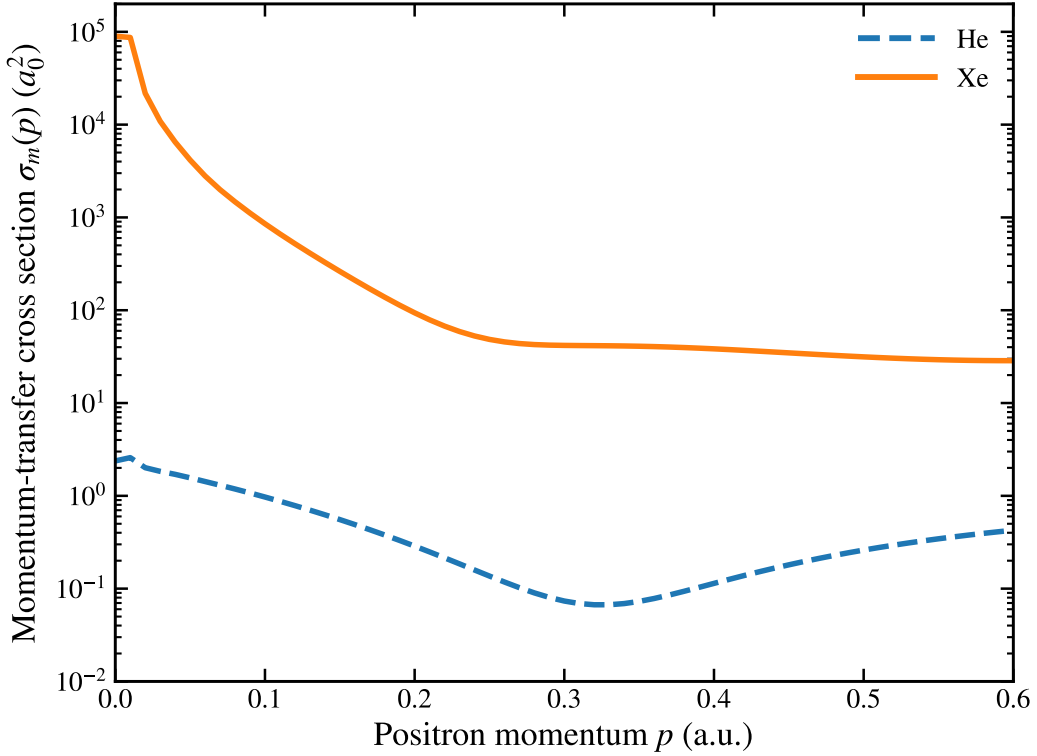


Figure 2.1: MBT momentum-transfer cross sections for helium and xenon. The two gases illustrate the large variation in momentum-space transport strength across the noble-gas sequence: xenon has a much larger $\sigma_m(p)$ than helium over most of the momentum range, while helium shows a pronounced low-momentum minimum.

$p = 0$, the momentum-transfer cross section was set using the scattering-length limit

$$\sigma_m(0) = 4\pi a^2, \quad (2.65)$$

where a is the positron–atom scattering length for the relevant noble gas.

Thus, for each gas, the discrete MBT phase-shift data were converted into a smooth file `gas_sigma_m_01res.txt` containing (p, σ_m) values on a uniform momentum grid. The corresponding xenon preprocessing routine used to generate this input table is given in Appendix G.1. Figure 2.1 shows the resulting momentum-transfer cross sections for helium and xenon, chosen as representative weak- and strong-interaction cases. The contrast is already pronounced at the input level: xenon has a much larger cross section over most of the momentum range, while helium exhibits a pronounced low-energy minimum. The momentum-transfer cross-sections for other atoms are shown in Appendix A.

2.3.2 Annihilation input $Z_{\text{eff}}(p)$

The annihilation part of the model requires the momentum-dependent quantity $Z_{\text{eff}}(p)$. The MBT paper provides partial-wave-resolved values of Z_{eff} for the noble gases, again as tabulated functions of the positron momentum. Depending on the gas and the table, the data are separated into valence and core contributions. In the present work these were first combined to form the total

$$Z_{\text{eff}}(p) = Z_{\text{eff}}^{(s)}(p) + Z_{\text{eff}}^{(p)}(p) + Z_{\text{eff}}^{(d)}(p), \quad (2.66)$$

with the appropriate addition of valence and core terms where both are available.

A practical difficulty is that the MBT tables do not usually include the value at $p = 0$, while the numerical solver is defined on a grid beginning at $p = 0$. To construct a smooth low-momentum anchor, the first three MBT points were fitted to the virtual-level form

$$Z_{\text{eff}}(p) = \frac{K}{x^2 + p^2} + A, \quad (2.67)$$

with A, K and x^2 being the fitting parameters. This fit was motivated by the low-energy behaviour discussed in the MBT paper [3],

$$Z_{\text{eff}}(k) = \frac{K}{\kappa^2 + k^2} + A. \quad (2.68)$$

From the fitted parameters K , x^2 and A , the zero-momentum value is obtained as

$$Z_{\text{eff}}(0) = \frac{K}{x^2} + A. \quad (2.69)$$

In practice, an exact closed-form three-point solve was used, as described in Appendix B.

After inserting this $p = 0$ anchor into the dataset, the total $Z_{\text{eff}}(p)$ was interpolated onto the solver grid using either a cubic spline or, where shape preservation was preferred³, a PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) interpolant [9]. The resulting smooth non-negative total was then saved in files of the form `gas.zeff.total.on.grid.txt`. The corresponding xenon preprocessing routine used to generate the interpolated $Z_{\text{eff}}(p)$ table is given in Appendix G.2. Figure 2.2 compares the resulting $Z_{\text{eff}}(p)$ curves for helium and xenon. The difference between the two gases is dramatic: helium remains nearly flat and small, whereas xenon

³For xenon, a PCHIP interpolant was preferred over a standard cubic spline, since it preserves the local shape of the tabulated data and avoids the small spurious oscillations or overshoots that can arise in spline interpolation when the low-momentum variation is especially steep.

has a very large low-momentum enhancement. This contrast anticipates the much stronger annihilation-induced depletion found later for xenon. The $Z_{\text{eff}}(p)$ data for the other atoms are shown in Appendix C

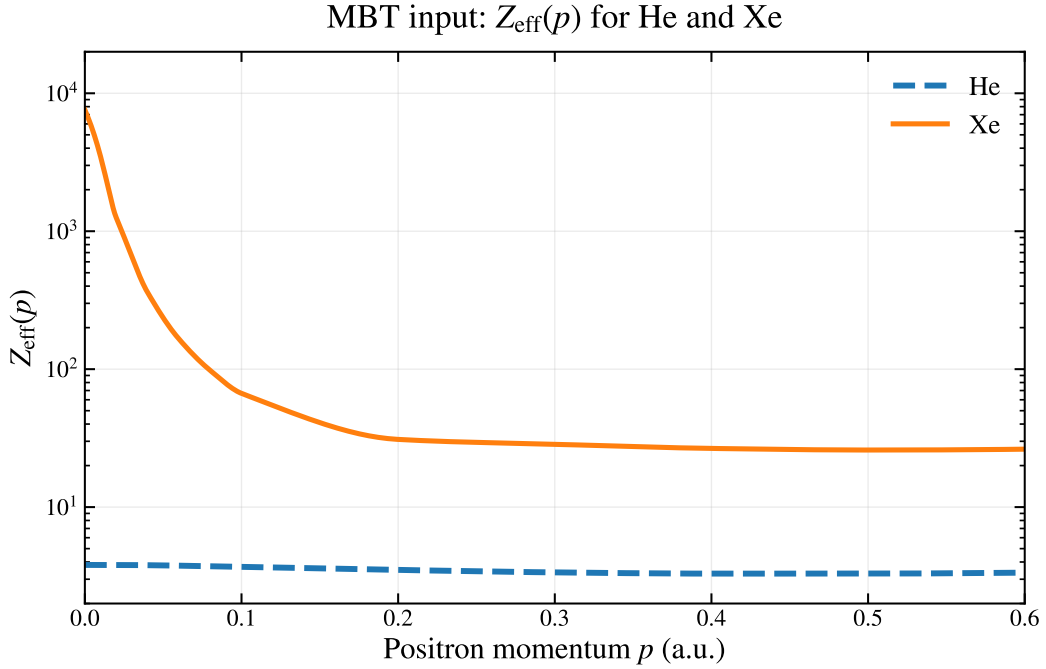


Figure 2.2: MBT annihilation input $Z_{\text{eff}}(p)$ for helium and xenon. Helium shows a weak and slowly varying annihilation parameter, whereas xenon exhibits a very large low-momentum enhancement. This difference is central to the contrasting late-time cooling behaviour of the two gases.

2.3.3 Role of the MBT input in the Fokker–Planck model

The processed MBT input enters the final scaled Fokker–Planck (2.57) equation through the transport and annihilation coefficients. The momentum-transfer cross section determines the cooling operator through the factor $\sigma_m(p)p$, while $Z_{\text{eff}}(p)$ determines the annihilation sink $\lambda(p)$. In this sense, the FP solver evolves the positron swarm (i.e., the momentum-space distribution), while the microscopic collision physics is supplied externally by the MBT calculations.

This separation is physically useful. It allows the numerical scheme to focus on the time evolution of the distribution, while retaining accurate low-energy positron–atom physics from the MBT calculations. The noble-gas cooling study [6] showed that using MBT scattering and annihilation data gives markedly improved agreement with experiment and captures important

effects such as strong annihilation-induced depletion in xenon.

2.4 Initial and boundary conditions

To solve the scaled radial Fokker–Planck (2.57) equation numerically, it is necessary to specify both an initial distribution and appropriate boundary conditions in momentum space. In the present work the momentum domain is truncated to a finite interval

$$0 \leq p \leq p_{\max}, \quad (2.70)$$

with $p_{\max}$ ranging from 0.6 a.u. to 1.0 a.u. for calculations of xenon to helium. The numerical grid is uniform on this interval.

2.4.1 Initial condition

The initial positron distribution is taken to be uniform in kinetic energy up to a cutoff energy $E_{\max}$. This cutoff is chosen as the lowest inelastic scattering threshold, namely the positronium-formation threshold, $E_{\max} \approx I - 6.8\text{eV}$, where I is the ionisation energy of the atom. For a non-relativistic positron,

$$E = \frac{p^2}{2m}, \quad (2.71)$$

and in atomic units $m = 1$, so that

$$E = \frac{p^2}{2}. \quad (2.72)$$

If the distribution is constant in energy, i.e.

$$g(E) = \text{const.}, \quad (2.73)$$

then the corresponding momentum distribution satisfies

$$F(p, 0) dp = g(E) dE. \quad (2.74)$$

Since

$$\frac{dE}{dp} = p, \quad (2.75)$$

it follows that

$$F(p, 0) \propto p. \quad (2.76)$$

Thus a distribution uniform in energy corresponds to a distribution that is linear in momentum.

Writing the cutoff momentum as

$$p_{\max} = \sqrt{2E_{\max}}, \quad (2.77)$$

the initial condition used in the solver is

$$F(p, 0) = \begin{cases} C p, & 0 \leq p < p_{\max}, \\ 0, & p \geq p_{\max}, \end{cases} \quad (2.78)$$

where the constant C is fixed by normalization:

$$\int_0^{p_{\max}} F(p, 0) dp = 1. \quad (2.79)$$

This gives

$$C = \frac{2}{p_{\max}^2}, \quad (2.80)$$

so that

$$F(p, 0) = \frac{2p}{p_{\max}^2}, \quad 0 \leq p < p_{\max}. \quad (2.81)$$

This choice corresponds to a simple broad epithermal initial swarm. It is also consistent with the physical idea that the positrons initially populate a range of energies above the thermal scale before cooling towards equilibrium. In the cooling literature, initial distributions uniform in energy have also been used when modelling positron cooling in noble gases [6], while earlier work likewise treated the initial positron distribution as an explicit modelling input to the slowing-down calculation [21].

For reference, the thermal kinetic-energy scale is

$$E_{\text{th}} = \frac{3}{2} k_B T, \quad (2.82)$$

which corresponds to the momentum scale

$$p_{\text{th}} = \sqrt{3k_B T}. \quad (2.83)$$

For room temperature, $T = 293 \text{ K}$, this gives

$$E_{\text{th}} \approx 6.07 \times 10^{-21} \text{ J} \approx 1.39 \times 10^{-3} \text{ a.u.},$$

and hence

$$p_{\text{th}} \approx 5.28 \times 10^{-2} \text{ a.u.}$$

Thus, in the calculations considered here, the initial distribution is far broader than the final thermalized distribution.

2.4.2 Boundary conditions

The physically appropriate boundary condition for the transport part of the equation is that there is no net probability flux across the endpoints of the momentum interval $[0, p_{\max}]$. Writing the equation in flux form,

$$\frac{\partial F}{\partial \tau} + \frac{\partial J}{\partial p} = -\lambda(p)F, \quad (2.84)$$

with

$$J(p, \tau) = -\alpha \sigma_m(p) p \left[\frac{\partial F}{\partial p} + \left(\frac{p}{k_B T} - \frac{2}{p} \right) F \right], \quad (2.85)$$

the boundary conditions imposed are

$$J(0, \tau) = 0, \quad J(p_{\max}, \tau) = 0. \quad (2.86)$$

These are no-flux boundary conditions, ensuring that the transport operator conserves total positron number on the truncated domain in the absence of annihilation.

In the discrete implementation, the endpoint values of the distribution are also set to zero,

$$F(0, \tau) = 0, \quad F(p_{\max}, \tau) = 0. \quad (2.87)$$

At $p = 0$, this is consistent with the behaviour of the radial distribution, since $F(p, \tau) = 4\pi p^2 f(p, \tau)$ must vanish as $p \rightarrow 0$ provided the underlying isotropic distribution f remains finite. At the upper boundary, the condition $F(p_{\max}, \tau) = 0$ is compatible with the assumption that the chosen computational cutoff is sufficiently large that the distribution remains negligible there throughout the evolution.

Together, the initial condition (2.81) and the no-flux boundary conditions (2.86) define the cooling problem solved in the subsequent numerical calculations.

The positron-cooling problem is now fully specified: the governing equation is given by Eq. (2.57), the transport and annihilation input are supplied by the MBT data, and the initial and boundary conditions have been stated above. The remaining task is therefore numerical: to construct a stable and conservative discretisation capable of evolving the distribution in time. This is described in the next chapter.

Chapter 3

Numerical scheme

This chapter describes the numerical method used to solve the scaled radial Fokker–Planck equation derived in Chapter 2. The governing equation is

$$\frac{\partial F}{\partial \tau} = \alpha \frac{\partial}{\partial p} \left\{ \sigma_m(p) p \left[\frac{\partial F}{\partial p} + \left(\frac{p}{k_B T} - \frac{2}{p} \right) F \right] \right\} - \lambda(p) F, \quad (3.1)$$

or, equivalently, in flux form,

$$\frac{\partial F}{\partial \tau} + \frac{\partial J}{\partial p} = -\lambda(p) F, \quad (3.2)$$

with

$$J(p, \tau) = -\alpha \sigma_m(p) p \left[\frac{\partial F}{\partial p} + \left(\frac{p}{k_B T} - \frac{2}{p} \right) F \right]. \quad (3.3)$$

The transport discretisation follows the standard finite-volume construction for divergence-form equations, in which the cell average evolves through the difference of numerical fluxes at neighbouring cell faces [10, 11]. In the present work this framework is adapted to the radial momentum-space Fokker–Planck equation and combined with a separate exponential update for the local annihilation sink. This choice is convenient because the transport part has divergence form and should conserve total probability in the absence of annihilation, whereas the sink term is purely local in momentum space.

An earlier version of the solver was implemented using a finite-difference discretisation of the momentum derivatives. While this was useful for initial experimentation, it does not make the conservative structure of the transport equation as transparent as a flux-based formulation. Since the cooling operator naturally has divergence form, a finite-volume discretisation was ultimately adopted. This allows the no-flux boundary conditions to be imposed directly at cell faces and ensures exact discrete conservation of total probability for the transport step.

3.1 Momentum grid and discrete variables

As explained in Section 2.4, the momentum domain is truncated to a finite interval

$$0 \leq p \leq p_{\max}, \quad (3.4)$$

and discretised using a uniform grid with $N_p + 1$ nodal points,

$$p_i = i \Delta p, \quad i = 0, 1, \dots, N_p, \quad (3.5)$$

where

$$\Delta p = \frac{p_{\max}}{N_p}, \quad (3.6)$$

is the step size.

In the calculations presented later, typical values are $N_p = 400$, with $p_{\max} = 1.0$ a.u. for helium and $p_{\max} = 0.6$ a.u. for xenon.

The finite-volume method is based on control volumes

$$\left[p_{i-\frac{1}{2}}, p_{i+\frac{1}{2}} \right], \quad (3.7)$$

centred on p_i , where the cell faces are defined by

$$p_{i+\frac{1}{2}} = \frac{p_i + p_{i+1}}{2}. \quad (3.8)$$

The numerical solution $F_i(\tau)$ is regarded as the cell-centred approximation to the radial distribution $F(p, \tau)$.

For convenience, define

$$G(p) = \sigma_m(p) p, \quad A(p) = \frac{p}{k_B T} - \frac{2}{p}, \quad (3.9)$$

so that the flux can be written as

$$J(p, \tau) = -\alpha G(p) \left(\frac{\partial F}{\partial p} + A(p) F \right). \quad (3.10)$$

3.2 Finite-volume discretisation of the transport term

3.2.1 Cell-integrated conservation law

The transport part of the equation is obtained by omitting annihilation from Eq. (3.2):

$$\frac{\partial F}{\partial \tau} + \frac{\partial J}{\partial p} = 0. \quad (3.11)$$

Integrating Eq. (3.11) over the i th control volume gives

$$\frac{d}{d\tau} \int_{p_{i-\frac{1}{2}}}^{p_{i+\frac{1}{2}}} F(p, \tau) dp + \int_{p_{i-\frac{1}{2}}}^{p_{i+\frac{1}{2}}} \frac{\partial J}{\partial p} dp = 0. \quad (3.12)$$

Applying the fundamental theorem of calculus to the flux term yields

$$\frac{d}{d\tau} \int_{p_{i-\frac{1}{2}}}^{p_{i+\frac{1}{2}}} F(p, \tau) dp = J_{i-\frac{1}{2}} - J_{i+\frac{1}{2}}, \quad (3.13)$$

where

$$J_{i\pm\frac{1}{2}} = J(p_{i\pm\frac{1}{2}}, \tau). \quad (3.14)$$

This is the discrete conservation statement: the change in the mass of a cell is equal to inflow minus outflow.

Approximating the cell average by the cell-centred value F_i , one obtains the semidiscrete finite-volume form

$$\frac{dF_i}{d\tau} = -\frac{J_{i+\frac{1}{2}} - J_{i-\frac{1}{2}}}{\Delta p}. \quad (3.15)$$

3.2.2 Face flux approximation

To close Eq. (3.15), the face fluxes must be approximated in terms of the neighbouring cell values. At the face $p_{i+\frac{1}{2}}$, a centred approximation is used for the derivative,

$$\left. \frac{\partial F}{\partial p} \right|_{i+\frac{1}{2}} \approx \frac{F_{i+1} - F_i}{\Delta p}, \quad (3.16)$$

and the distribution itself is approximated by the arithmetic mean

$$F_{i+\frac{1}{2}} \approx \frac{F_i + F_{i+1}}{2}. \quad (3.17)$$

The transport coefficient is evaluated directly at the face,

$$G_{i+\frac{1}{2}} = G(p_{i+\frac{1}{2}}) = \sigma_m(p_{i+\frac{1}{2}}) p_{i+\frac{1}{2}}, \quad (3.18)$$

and similarly

$$A_{i+\frac{1}{2}} = A(p_{i+\frac{1}{2}}) = \frac{p_{i+\frac{1}{2}}}{k_B T} - \frac{2}{p_{i+\frac{1}{2}}}. \quad (3.19)$$

The numerical face flux is therefore

$$J_{i+\frac{1}{2}} = -\alpha G_{i+\frac{1}{2}} \left[\frac{F_{i+1} - F_i}{\Delta p} + A_{i+\frac{1}{2}} \frac{F_i + F_{i+1}}{2} \right]. \quad (3.20)$$

This is the flux used in the code.

Substituting Eq. (3.20) into Eq. (3.15) gives the conservative semidiscrete transport operator. The explicit time update for the cooling step is then

$$F_i^* = F_i^n - \Delta\tau \frac{J_{i+\frac{1}{2}}^n - J_{i-\frac{1}{2}}^n}{\Delta p}. \quad (3.21)$$

Here F_i^n denotes the numerical solution at time level n , and F_i^* is the intermediate value after the transport step and before annihilation is applied.

3.2.3 No-flux boundary conditions and conservation

The theory chapter imposed no-flux boundary conditions,

$$J(0, \tau) = 0, \quad J(p_{\max}, \tau) = 0. \quad (3.22)$$

In the discrete scheme these become

$$J_{\frac{1}{2}} = 0, \quad J_{N_p - \frac{1}{2}} = 0. \quad (3.23)$$

Thus no probability is allowed to leave the domain through either boundary.

A key advantage of the finite-volume form is that the transport step is exactly conservative on the discrete grid. Summing Eq. (3.15) over all cells gives

$$\sum_i \Delta p \frac{dF_i}{d\tau} = - \sum_i \left(J_{i+\frac{1}{2}} - J_{i-\frac{1}{2}} \right). \quad (3.24)$$

The flux differences telescope, leaving only the boundary contributions,

$$\frac{d}{d\tau} \sum_i F_i \Delta p = - \left(J_{N_p - \frac{1}{2}} - J_{\frac{1}{2}} \right). \quad (3.25)$$

Using Eq. (3.23), this becomes

$$\frac{d}{d\tau} \sum_i F_i \Delta p = 0. \quad (3.26)$$

Hence the transport step conserves total probability exactly in the absence of annihilation. This discrete conservation property is one of the main reasons for writing the scheme in flux form.

It is also natural to impose

$$F(0, \tau) = F(p_{\max}, \tau) = 0 \quad (3.27)$$

after each update. At $p = 0$, this is consistent with the radial form $F = 4\pi p^2 f$, while at the upper boundary it reflects the assumption that the computational cutoff is chosen large enough that the distribution is negligible there.

3.3 Operator splitting and annihilation update

After the conservative transport step, the annihilation term is applied separately. The sink equation is

$$\frac{\partial F}{\partial \tau} = -\lambda(p)F. \quad (3.28)$$

Since this is a local decay equation at each momentum node, it can be integrated exactly over one timestep $\Delta\tau$, giving

$$F_i^{n+1} = F_i^* \exp[-\lambda_i \Delta\tau]. \quad (3.29)$$

This update is exact for the isolated sink term and preserves non-negativity whenever $F_i^* \geq 0$.

The full scheme is therefore

$$F_i^* = F_i^n - \Delta\tau \frac{J_{i+\frac{1}{2}}^n - J_{i-\frac{1}{2}}^n}{\Delta p}, \quad (3.30)$$

followed by

$$F_i^{n+1} = F_i^* \exp[-\lambda_i \Delta\tau]. \quad (3.31)$$

This is a first-order operator-splitting method of Lie type [12].

The splitting is natural because the two parts of the equation are mathematically different. The transport term redistributes probability between neighbouring momentum cells through fluxes, whereas the annihilation term is purely local and does not couple neighbouring cells. For sufficiently small $\Delta\tau$, the split scheme is consistent with the full semidiscrete equation

$$\frac{dF_i}{d\tau} = -\frac{J_{i+\frac{1}{2}} - J_{i-\frac{1}{2}}}{\Delta p} - \lambda_i F_i. \quad (3.32)$$

Indeed, expanding the exponential in Eq. (3.31) for small $\Delta\tau$,

$$e^{-\lambda_i \Delta\tau} = 1 - \lambda_i \Delta\tau + \mathcal{O}(\Delta\tau^2), \quad (3.33)$$

shows that the split update reproduces the standard explicit Euler discretisation of Eq. (3.32) to first order in time.

3.4 Diagnostic quantities

Several diagnostic quantities are monitored during the evolution. The total surviving fraction is

$$N(\tau) = \int_0^{p_{\max}} F(p, \tau) dp, \quad (3.34)$$

or, in discrete form,

$$N(\tau) \approx \sum_{i=0}^{N_p} F_i(\tau) \Delta p. \quad (3.35)$$

Because the transport step is conservative, changes in $N(\tau)$ arise only from the annihilation sink.

The average annihilation rate is defined by

$$\bar{\lambda}(\tau) = \frac{\int \lambda(p) F(p, \tau) dp}{\int F(p, \tau) dp} \approx \frac{\sum_{i=0}^{N_p} \lambda_i F_i(\tau) \Delta p}{N(\tau)}, \quad (3.36)$$

and the averaged annihilation parameter by

$$\bar{Z}_{\text{eff}}(\tau) = \frac{\int Z_{\text{eff}}(p) F(p, \tau) dp}{\int F(p, \tau) dp} \approx \frac{\sum_{i=0}^{N_p} Z_{\text{eff},i} F_i(\tau) \Delta p}{N(\tau)}, \quad (3.37)$$

The mean energy of the surviving positrons is

$$\bar{E}(\tau) = \frac{\int \frac{p^2}{2} F(p, \tau) dp}{\int F(p, \tau) dp} \approx \frac{\sum_{i=0}^{N_p} \frac{p_i^2}{2} F_i(\tau) \Delta p}{N(\tau)}. \quad (3.38)$$

In the present work the cooling time is defined from the mean energy. The thermal mean energy is

$$E_{\text{th}} = \frac{3}{2} k_B T, \quad (3.39)$$

and a threshold

$$E_{\text{thr}} = 1.01 E_{\text{th}} \quad (3.40)$$

is used to identify practical thermalization. The cooling time τ_{cool} is taken to be the first diagnostic time at which

$$\bar{E}(\tau_{\text{cool}}) = E_{\text{thr}}. \quad (3.41)$$

If the crossing occurs between two discretised diagnostic times $\tau_n = n\Delta\tau$ and τ_{n+1} , then τ_{cool} is estimated by linear interpolation.

3.5 Accuracy and stability

The transport discretisation uses centred differences and centred face averages, so for sufficiently smooth solutions it is formally second order in momentum space. The time integration, however, is first order, since the transport step uses forward Euler and the full transport-plus-sink update

is performed with first-order Lie splitting. Thus the overall scheme is approximately second order in space and first order in time.

The transport step is conservative by construction, while the exponential sink update is positivity preserving. However, the explicit centred transport update is not unconditionally positivity preserving for arbitrary $\Delta\tau$. If the timestep is chosen too large, small negative undershoots may in principle appear. In practice, this is controlled by taking $\Delta\tau$ sufficiently small. The values used in the simulations reported later were chosen so that the numerical solution remained stable and physically well behaved throughout the evolution.

3.6 Verification checks

To verify that the numerical scheme reproduces the key structural properties of the governing equation, two simple checks were performed. The first tests conservation of total probability when annihilation is switched off. The second tests the internal consistency of the annihilation sink when it is included. These checks were carried out for both helium and xenon, representing a weak-annihilation case and a much more demanding strong-annihilation case, respectively.

3.6.1 Conservation check without annihilation

If the annihilation term is set to zero, the Fokker–Planck equation (3.1) reduces to the conservative transport problem

$$\frac{\partial F}{\partial \tau} + \frac{\partial J}{\partial p} = 0, \quad (3.42)$$

subject to no-flux boundary conditions at $p = 0$ and $p = p_{\max}$. Integrating over the whole momentum domain gives

$$\frac{d}{d\tau} \int_0^{p_{\max}} F(p, \tau) dp = 0, \quad (3.43)$$

so the total probability should remain constant in time.

In the numerical scheme, the discrete total probability is given by Eq. (3.35). To test conservation, the solver was run with

$$\lambda(p) = 0 \quad (3.44)$$

for all p , and the relative positron number error

$$\varepsilon_N(\tau) = \frac{N(\tau) - N(0)}{N(0)} \quad (3.45)$$

was monitored.

For the helium test, the calculation used $\Delta p = 1.0/400 = 2.5 \times 10^{-3}$ a.u., $\Delta\tau = 3.0 \times 10^{-2}$ ns amagat, and $N_\tau = 133334$ timesteps, corresponding to a final time $\tau_{\max} = 4000$ ns amagat. The maximum observed absolute relative error was

$$\max_{\tau} |\varepsilon_N(\tau)| = 3.775 \times 10^{-15}, \quad (3.46)$$

while for xenon, using $\Delta p = 0.6/400 = 1.5 \times 10^{-3}$ a.u., $\Delta\tau = 1.0 \times 10^{-3}$ ns amagat, and $N_\tau = 300001$ timesteps, corresponding to $\tau_{\max} = 300$ ns amagat, the corresponding value was

$$\max_{\tau} |\varepsilon_N(\tau)| = 1.288 \times 10^{-14}. \quad (3.47)$$

Both values are at the level of floating-point roundoff, confirming that the finite-volume transport update is conservative to machine precision.

3.6.2 Self-consistency of the annihilation sink

When annihilation is included, the total surviving fraction (3.34) is no longer constant. Integrating the full equation

$$\frac{\partial F}{\partial \tau} + \frac{\partial J}{\partial p} = -\lambda(p)F \quad (3.48)$$

over the momentum domain, and again using the no-flux boundary conditions, gives

$$\frac{dN}{d\tau} = - \int_0^{p_{\max}} \lambda(p)F(p, \tau) dp. \quad (3.49)$$

Dividing by $N(\tau)$ yields

$$-\frac{d}{d\tau} \ln N(\tau) = \bar{\lambda}(\tau), \quad (3.50)$$

where the distribution-averaged annihilation rate is given in Eq. (3.36).

This identity provides a useful check on the annihilation implementation. In the numerical calculation, $\bar{\lambda}(\tau)$ was computed directly from the evolving distribution, while

$$-\frac{d}{d\tau} \ln N(\tau) \quad (3.51)$$

was estimated from the discrete survival fraction using a numerical derivative. The two quantities were then compared over the full run, as shown in Figure 3.1.

For helium, the comparison showed excellent agreement. The mean relative mismatch was

$$1.293 \times 10^{-6}, \quad (3.52)$$

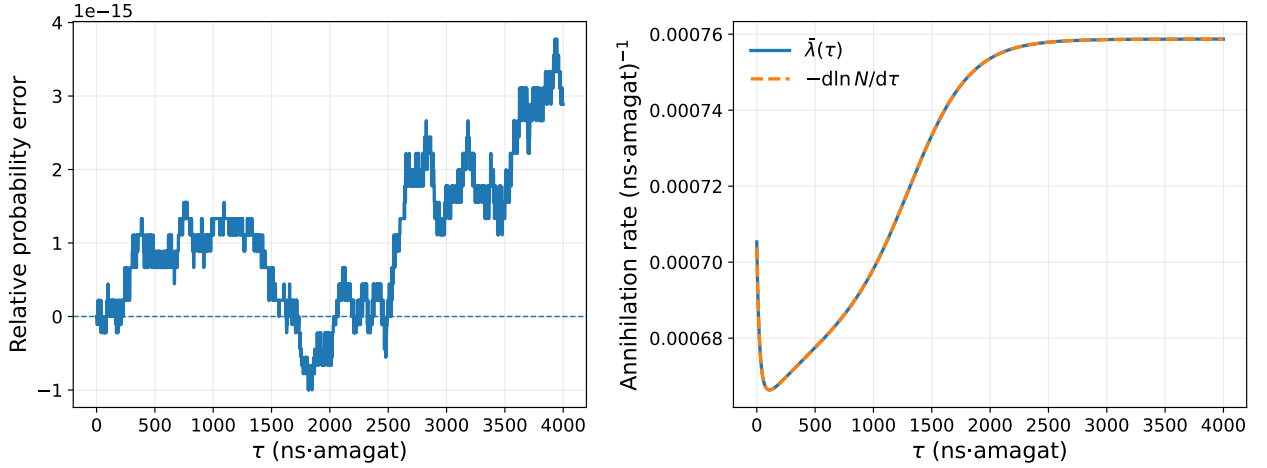


Figure 3.1: Verification checks for the helium implementation. Left: relative error in the total probability for the transport-only case $\lambda(p) = 0$, showing conservation to machine precision. Right: comparison of $\bar{\lambda}(\tau)$ with $-d \ln N / d\tau$, showing that the annihilation sink is implemented consistently. These results were obtained using $\Delta p = 2.5 \times 10^{-3}$ a.u. and $\Delta \tau = 3.0 \times 10^{-2}$ ns amagat.

while the maximum relative mismatch was

$$2.016 \times 10^{-3}. \quad (3.53)$$

The small discrepancy is attributable mainly to the numerical differentiation used to estimate $d \ln N / d\tau$, rather than to any inconsistency in the solver itself.

Xenon provides a more demanding verification case than helium because its annihilation rate is much larger, leading to more rapid variation of both $N(\tau)$ and $\bar{\lambda}(\tau)$. For the xenon calculation shown here, $\lambda(p)$ lies in the interval 5.2×10^{-3} to 1.52 (ns · amagat) $^{-1}$, so that

$$\max(\lambda \Delta \tau) = 1.52 \times 10^{-3} \quad (3.54)$$

for the timestep $\Delta \tau = 10^{-3}$. This remains comfortably small, so the exponential sink update is well resolved.

For xenon, the comparison shown in Figure 3.2 again gave very good agreement. The mean relative mismatch between $\bar{\lambda}(\tau)$ and $-d \ln N / d\tau$ was

$$6.760 \times 10^{-5}, \quad (3.55)$$

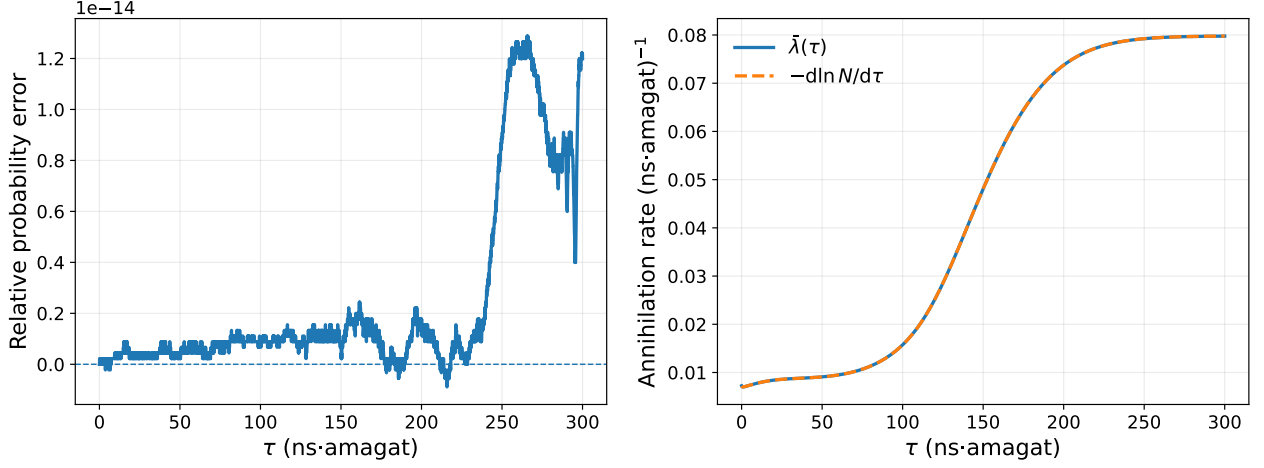


Figure 3.2: Verification checks for the xenon implementation. Left: relative error in the total probability for the transport-only case $\lambda(p) = 0$, again showing conservation to machine precision. Right: comparison of $\bar{\lambda}(\tau)$ with $-d \ln N / d\tau$ for the strongly annihilating xenon case. The close agreement shows that the sink remains internally consistent even when $\lambda(p)$ is much larger and more strongly momentum dependent than in helium. These results were obtained using $\Delta p = 1.5 \times 10^{-3}$ a.u. and $\Delta \tau = 1.0 \times 10^{-3}$ ns amagat.

while the maximum relative mismatch was

$$5.807 \times 10^{-3}. \quad (3.56)$$

Although these errors are larger than in helium, they remain sufficiently small and are fully consistent with the greater sensitivity of the numerical differentiation in the strong-annihilation regime.

Together, the helium and xenon tests confirm that the numerical method preserves the conservative structure of the transport operator and implements the annihilation sink consistently across both weak- and strong-annihilation regimes.

With the numerical method verified, we now apply the scheme to positron cooling and annihilation in the noble gases and examine how the balance between transport and annihilation changes from helium to xenon. A representative implementation of the production xenon solver is included in Appendix F.

Chapter 4

Results

4.1 Overview of computed observables

With the numerical method verified in the previous chapter [3](#), we now turn to the physical results for positron cooling and annihilation in the noble gases helium, neon, argon, krypton and xenon. The purpose of this chapter is twofold. First, it is to describe how the positron swarm evolves in each gas under the combined action of momentum-space transport and annihilation. Second, it is to identify the systematic trends that emerge across the noble-gas sequence as the momentum-transfer cross section and annihilation parameter change from the weakly annihilating helium case to the strongly annihilating xenon case.

Throughout this chapter, all calculations use the Fokker–Planck equation, MBT input data and diagnostic definitions introduced earlier. In particular, the results are discussed in terms of the scaled time variable τ (in ns amagat), so that the quoted evolution is independent of the gas density, and all gases are considered at room temperature. The initial positron distribution is the broad epithermal distribution specified in Section [2.4](#), and the same diagnostic framework is used throughout in order to allow direct comparison between gases.

The most important quantity is the evolving radial momentum distribution $F(p, \tau)$ itself. This shows how the surviving positron population is redistributed in momentum space as cooling proceeds, allowing to distinguish genuine approach towards thermal equilibrium from cases in which annihilation removes particles before such equilibration is reached. In practice, however, it is more convenient to analyse the time evolution using a small number of scalar observables derived from $F(p, \tau)$, as specified in Section [3.4](#).

The first of these is the surviving fraction $N(\tau)$, (Eq. (3.34)), i.e. the total probability remaining in the distribution at time τ . Since the transport operator is conservative, any decrease in $N(\tau)$ is entirely due to annihilation. This quantity therefore gives a direct measure of the strength of positron depletion in each gas. In weak-annihilation cases, $N(\tau)$ remains close to unity over the cooling timescale, whereas in strongly annihilating gases it can fall rapidly as the swarm approaches a thermal distribution.

The second key observable is the averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$, (Eq. (3.37)). Because $Z_{\text{eff}}(p)$ is strongly momentum dependent, especially for the heavier noble gases, $\bar{Z}_{\text{eff}}(\tau)$ provides a sensitive probe of where the positron population is concentrated in momentum space. It is also the quantity most directly connected to positron lifetime measurements and earlier cooling studies reported in the literature. A rise in $\bar{Z}_{\text{eff}}(\tau)$ generally signals that the swarm is moving towards lower momenta, where annihilation is more probable, but the detailed behaviour depends on the competition between continued cooling and preferential removal of the lowest-momentum positrons.

Closely related to this is the averaged annihilation rate $\bar{\lambda}(\tau)$, (Eq. (3.36)). Since $\lambda(p) \propto Z_{\text{eff}}(p)$, the two quantities carry essentially the same physical information, but $\bar{\lambda}(\tau)$ is useful when discussing depletion directly through the evolution of $N(\tau)$. In particular, it links the shape of the distribution to the observed decay of the surviving population.

To follow thermalization more directly, the mean kinetic energy $\bar{E}(\tau)$, (Eq. (3.38)), of the surviving positrons is also monitored. This gives a compact description of the overall cooling rate and provides the operational definition of the cooling time used in this dissertation. The cooling time τ_{cool} , (Eq. (3.41)), is defined as the earliest time at which $\bar{E}(\tau)$ falls to within 1% of the thermal mean energy. This criterion is convenient because it gives a single quantitative measure of the speed of cooling, while still remaining closely connected to the full momentum-space evolution.

In the gas-by-gas sections below, these observables are used together rather than in isolation. The distribution $F(p, \tau)$ shows the detailed evolution in momentum space; $N(\tau)$ measures the extent of annihilation losses; $\bar{Z}_{\text{eff}}(\tau)$ and $\bar{\lambda}(\tau)$ characterise how the annihilation response changes during cooling; and $\bar{E}(\tau)$ provides a simple measure of approach to thermal energies. Taken together, they give a comprehensive picture of whether a positron swarm cools efficiently, stalls in an intermediate momentum range, or is substantially depleted before thermal equilibrium

can be reached.

The discussion begins with helium, which serves as a baseline weak-annihilation case. It then moves through neon, argon and krypton to xenon, where the competition between cooling and annihilation is strongest and the late-time behaviour departs most clearly from the thermal Maxwellian picture suggested by the cooling-only problem.

4.2 Helium: baseline cooling behaviour

Helium provides the clearest baseline case for the present study. Among the noble gases considered here, it combines relatively weak annihilation with comparatively simple cooling behaviour, so that the positron swarm is able to approach a near-thermal distribution without strong distortion by depletion. This makes helium a useful reference system against which the more strongly annihilating and more weakly cooling gases can later be compared.

For the helium calculations shown in this section, the momentum domain was taken as $0 \leq p \leq 1.0$ a.u. and discretised with $N_p = 400$, giving $\Delta p = 2.5 \times 10^{-3}$ a.u. The timestep was $\Delta\tau = 3.0 \times 10^{-2}$ ns amagat, and the evolution was followed up to $\tau_{\max} = 4000$ ns amagat. These are the same numerical parameters used in the helium verification tests of Chapter 3.

Figure 4.1 shows the radial momentum distribution $F(p, \tau)$ for helium at the initial time, at an intermediate stage of the evolution, and at a later stage, together with the corresponding Maxwellian equilibrium shape for reference. The initial distribution is broad and epithermal, with support extending across much of the computational interval. By $\tau = 500$ ns amagat, the distribution has already shifted strongly towards lower momenta, but it still exhibits a pronounced peak at intermediate momentum. By $\tau = 2000$ ns amagat, this peak has moved further towards the thermal region and the distribution is much more concentrated at small p . The final profile is not identical to the Maxwellian reference, but it is clearly much closer to it than to the initial state, indicating that the distribution has cooled substantially and that annihilation in helium is not strong enough to severely distort the surviving swarm.

This behaviour is seen more fully in the time-dependent heat map in Figure 4.2. The broad initial population at high momenta is rapidly depleted as probability is transported towards lower p , while the bulk of the swarm progressively contracts towards the thermal momentum scale $p_{\text{th}} = \sqrt{3k_{\text{B}}T} \approx 0.053$ a.u.. The evolution is smooth and overall monotonic, but the

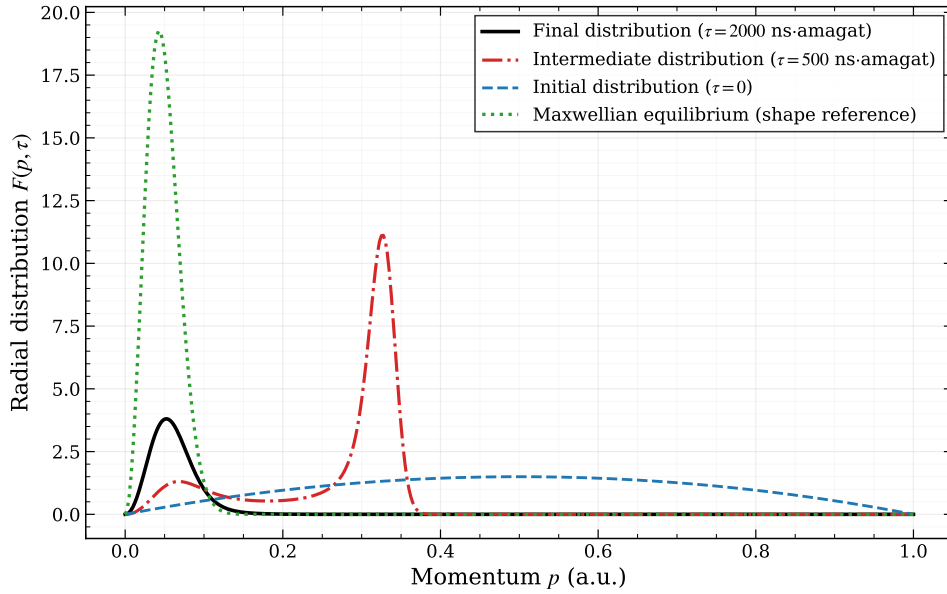


Figure 4.1: Comparison of the helium radial momentum distribution $F(p, \tau)$ at the initial time, at an intermediate stage of the evolution, and at a later stage, together with the Maxwellian equilibrium shape for reference. The initial distribution is broad and epithermal, the intermediate profile shows a transient accumulation at intermediate momentum, and the late-time distribution is concentrated at much lower momenta and lies much closer to the thermal form.

intermediate distribution in Figure 4.1 shows that the cooling is not completely featureless: there is a mild transient accumulation at intermediate momentum. This is consistent with the shallow minimum in the helium momentum-transfer cross section shown in Figure 2.1, which slightly reduces the efficiency of momentum relaxation in that region. However, unlike neon, this effect is weak and does not produce a severe or long-lived bottleneck. Helium therefore still behaves close to the intuitive cooling-only picture, with annihilation acting mainly as a weak perturbation rather than qualitatively changing the dynamics.

The same conclusion is supported by the mean energy shown in Figure 4.3. The average kinetic energy decreases rapidly at early times and then more gradually as the distribution approaches thermal energies. Using the criterion defined in Section 3.4, the cooling time is found to be

$$\tau_{\text{cool}} \approx 3197.5 \text{ ns amagat.}$$

Thus, while helium cools comparatively slowly in absolute terms, the evolution remains regular and the swarm does reach the practical thermalization threshold within the simulated time

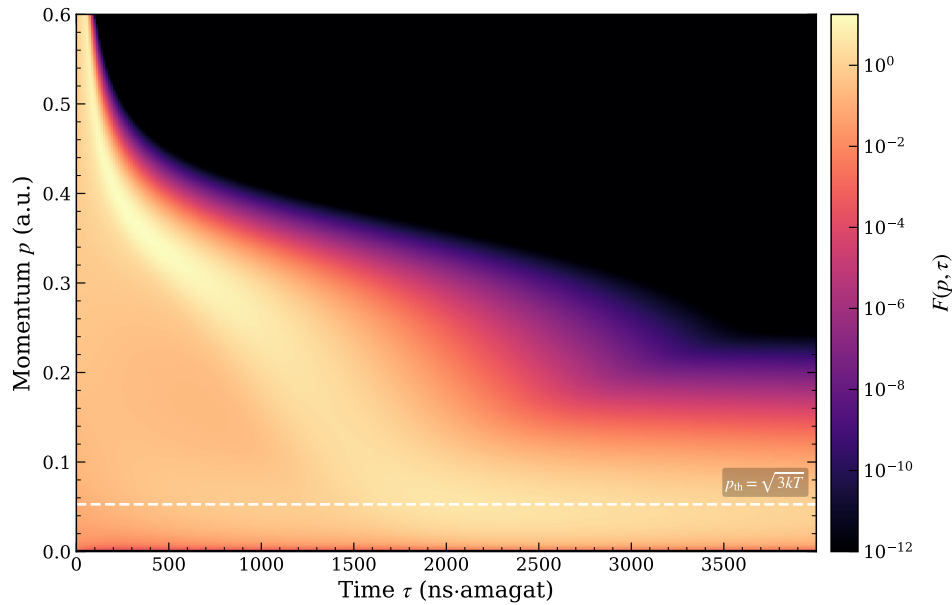


Figure 4.2: Time evolution of the helium radial momentum distribution $F(p, \tau)$. Probability is steadily transported from the broad epithermal initial state towards the low-momentum thermal region. The dashed horizontal line marks the thermal momentum scale $p_{\text{th}} = \sqrt{3k_{\text{B}}T}$.

window. This is consistent with helium serving as the weak-annihilation limit of the noble-gas sequence.

A complementary view is provided by the averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$ in Figure 4.4. The unscaled calculation approaches a late-time value of approximately $\bar{Z}_{\text{eff}} \approx 3.78$ and shows an initial decrease followed by a gradual rise towards a steady plateau. Also shown is a curve obtained by multiplying the calculated $\bar{Z}_{\text{eff}}(\tau)$ by a factor of 1.04, included as a simple empirical correction to illustrate the effect of the small underestimation of the MBT-based annihilation input relative to experiment. Physically, the overall shape reflects the redistribution of the swarm in momentum space. At early times the distribution remains broad and contains a substantial epithermal component. As cooling proceeds, the positrons move into the lower-momentum region where annihilation is slightly more favourable, so that $\bar{Z}_{\text{eff}}(\tau)$ rises towards its thermal value. The overall variation is modest, which is itself a characteristic feature of helium: the annihilation parameter is weak and only slowly varying with momentum compared with the heavier noble gases.

The comparison with the measurements of Coleman *et al.* is encouraging. Experimentally, the room-temperature equilibrium value for helium was reported as $Z_{\text{eff}} = 3.94 \pm 0.02$, with

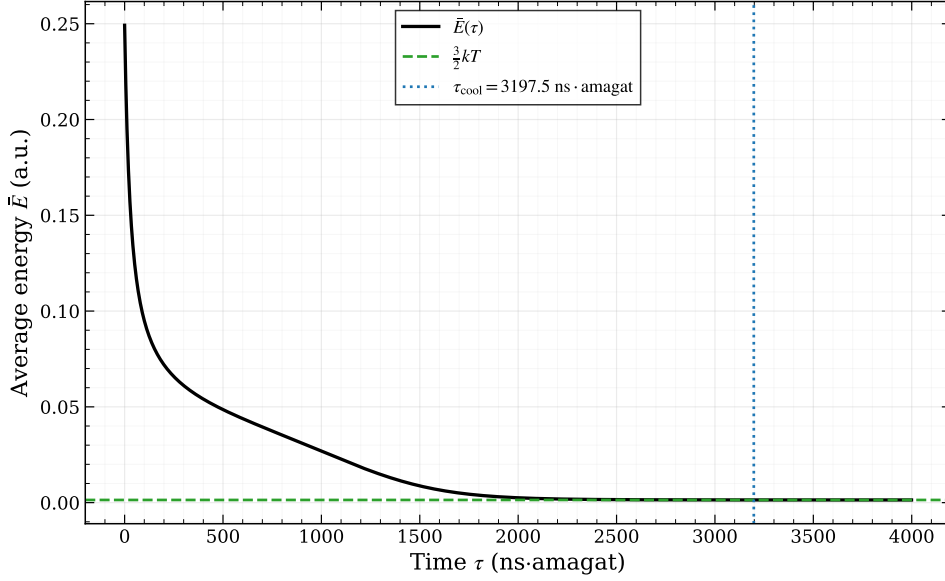


Figure 4.3: Mean positron energy $\bar{E}(\tau)$ for helium. The dashed horizontal line marks the thermal mean energy $3k_{\text{B}}T/2$, and the dotted vertical line marks the cooling time τ_{cool} defined by the 1% thermalization criterion.

a shoulder width of approximately 1700 ± 50 nsamagat in the lifetime spectrum [13]. The unscaled numerical curve in Figure 4.4 approaches a slightly lower asymptotic value, but it reproduces the same overall shoulder-like behaviour and reaches its plateau on a comparable timescale. The empirically rescaled curve, obtained by multiplying the calculated $\bar{Z}_{\text{eff}}(\tau)$ by 1.04, lies in near-coincidence with the experimental trend over most of the plotted range and brings the late-time value into very close agreement with the measured room-temperature result. This behaviour is qualitatively consistent with the more recent MBT-based cooling study of Green [6], which likewise found that helium is only weakly sensitive to annihilation-induced depletion and that the calculated $\bar{Z}_{\text{eff}}(\tau)$ agrees well with experiment in the shoulder region.

Overall, helium behaves as the expected baseline system. The momentum distribution evolves smoothly from the broad epithermal initial state towards a near-thermal low-momentum form; the mean energy falls to the thermal threshold on a finite and well-defined timescale; and the annihilation observable $\bar{Z}_{\text{eff}}(\tau)$ shows good qualitative agreement with the experimental trend, and a simple 4% empirical rescaling brings the calculated curve into very close quantitative agreement with the measured room-temperature value. In this sense helium provides the reference case in which the Fokker–Planck description behaves most nearly as the simple cooling picture would suggest, before the stronger competition between transport and annihilation

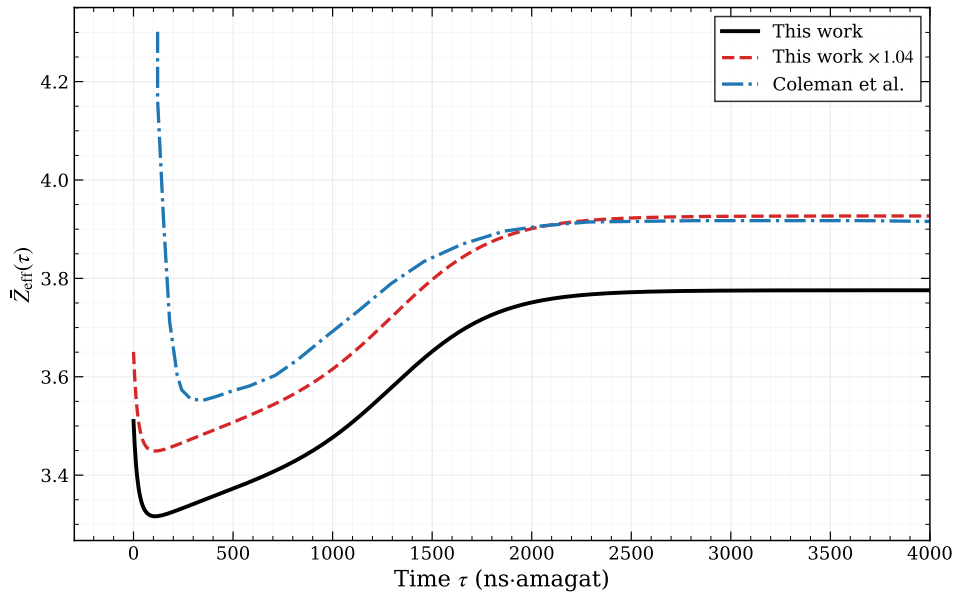


Figure 4.4: Averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$ for helium, compared with the experimental trend reported by Coleman *et al.* [13]. The black curve shows the direct result of the present calculation, while the red dashed curve shows the same result multiplied by 1.04 to illustrate the effect of a small empirical correction for the underestimation of the MBT-based annihilation input. Both theoretical curves reproduce the characteristic shoulder-like evolution, and the rescaled curve gives very close agreement with the experimental late-time value.

emerges in the heavier noble gases.

4.3 Neon: slow cooling and bottleneck behaviour

Neon is the first case in which the cooling dynamics depart markedly from the baseline helium picture. Although annihilation in neon is still relatively weak compared with the heavier noble gases, momentum-space transport is much less effective over an important part of the low-energy range. In particular, the momentum-transfer cross section shown in Appendix Figure A.1 exhibits a pronounced Ramsauer–Townsend minimum. This suppresses the efficiency of momentum relaxation and creates a bottleneck in the cooling process. Neon atoms are also 5 times heavier than helium, which further suppressed the cooling rate.

For the neon calculations shown in this section, the momentum interval was taken as $0 \leq p \leq 1.0$ a.u. and discretised with $N_p = 400$, giving $\Delta p = 2.5 \times 10^{-3}$ a.u. The timestep was $\Delta\tau = 3.0 \times 10^{-2}$ ns amagat, and the evolution was followed up to $\tau_{\text{max}} = 2.0 \times 10^4$ ns amagat.

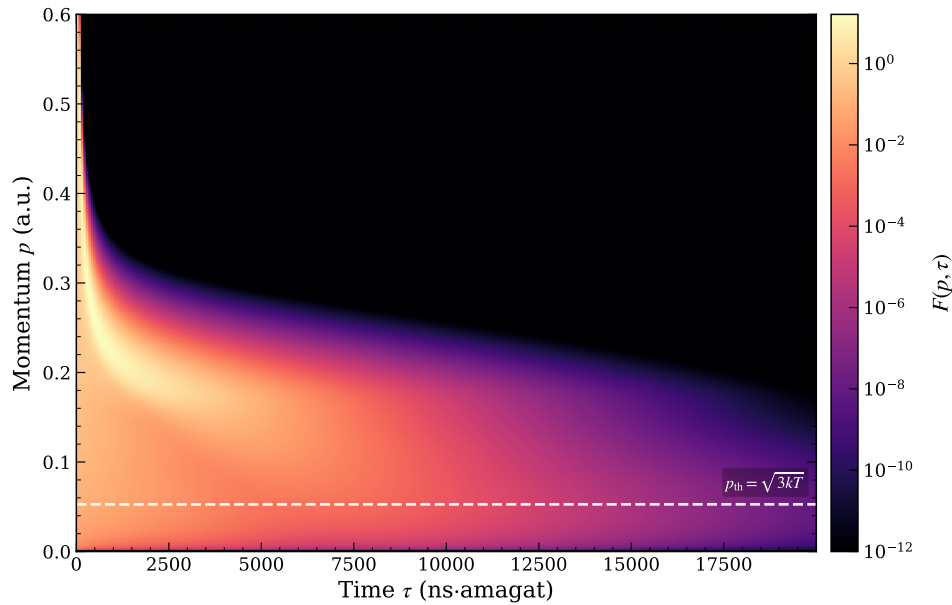


Figure 4.5: Time evolution of the neon radial momentum distribution $F(p, \tau)$. After an initial rapid shift to lower momenta, the bulk of the distribution lingers in an intermediate-momentum band associated with the Ramsauer–Townsend minimum in the momentum-transfer cross section. The dashed horizontal line marks the thermal momentum scale $p_{\text{th}} = \sqrt{3k_{\text{B}}T}$.

The effect of this bottleneck is seen clearly in the time-dependent momentum distribution, shown in Figure 4.5. As in helium, the initially broad epithermal distribution first moves rapidly towards lower momenta. However, instead of continuing smoothly towards the thermal region, the bulk of the distribution becomes concentrated in an intermediate momentum band, $p \approx 0.2$ a.u., centred roughly around the minimum in the transport cross section. The subsequent evolution towards thermal momenta is then very slow. In this sense, neon does not simply cool more slowly than helium; rather, the distribution becomes transiently trapped in a momentum range where elastic collisions are inefficient at removing further energy.

This observation is reinforced by the mean energy behaviour in Figure 4.6. After a rapid initial drop, $\bar{E}(\tau)$ enters a long, slowly decaying tail. The evolution towards the thermal mean energy $3k_{\text{B}}T/2$ is therefore much more gradual than in helium. In the present run, the distribution approaches thermal energies only very late in the simulation window, reflecting the fact that the transport bottleneck dominates the overall timescale. To test whether this was simply a consequence of the finite plotting interval, the calculation was extended to $\tau =$

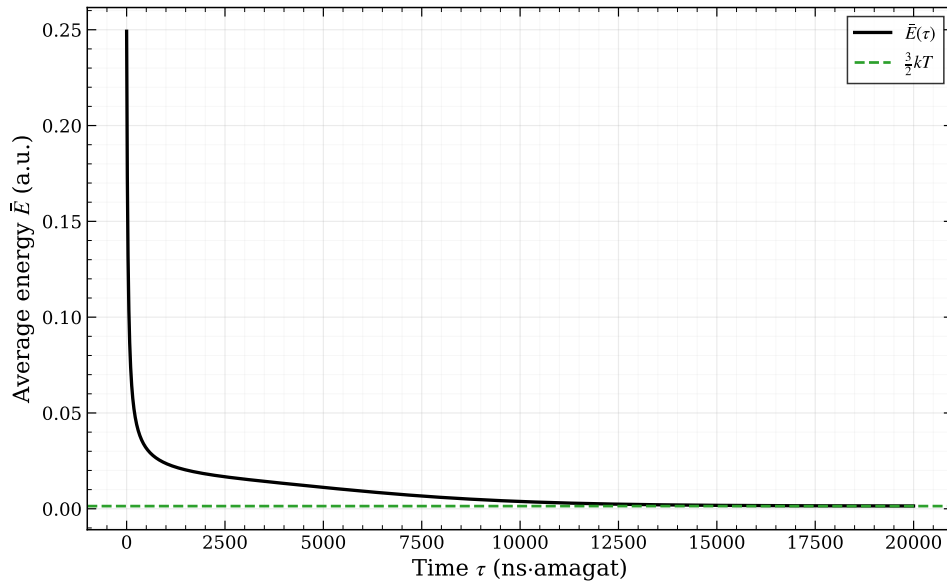


Figure 4.6: Mean positron energy $\bar{E}(\tau)$ for neon. After a rapid initial decrease, the energy relaxes only very slowly towards the thermal value $3k_{\text{B}}T/2$, reflecting the cooling bottleneck created by the Ramsauer–Townsend minimum in the transport cross section.

5.0×10^4 ns amagat. In this longer run, the mean energy approached

$$\bar{E} \approx 1.4197 \times 10^{-3} \text{ a.u.},$$

which corresponds to approximately

$$\frac{\bar{E}}{3k_{\text{B}}T/2} \approx 1.02.$$

Thus, even on timescales much longer than those shown in Figure 4.6, the surviving positron population remains slightly hotter than the true thermal distribution. Since the surviving fraction is already extremely small at such late times, this is best interpreted as evidence of incomplete practical thermalization rather than as a well-populated equilibrium state. This qualitative picture is consistent with the MBT-based Monte Carlo study of Green [6], who likewise found that cooling in neon effectively stalls near the minimum of the momentum diffusion coefficient and that the fraction of positrons surviving to full thermalization is practically zero.

The same conclusion is seen even more starkly in the surviving fraction $N(\tau)$ plotted in Figure 4.7. On the scale of the plot, the population falls very rapidly, and by the time the distribution has evolved significantly through the bottleneck, only a very small fraction of the original swarm remains. Thus, although neon does not have a particularly large annihilation

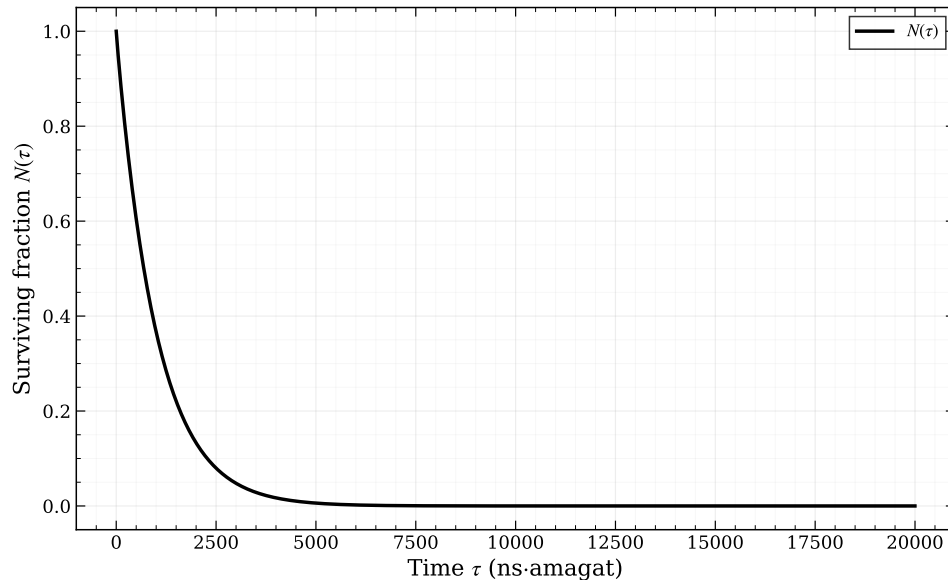


Figure 4.7: Surviving fraction $N(\tau)$ for neon. The population decays rapidly during the long bottleneck cooling stage, showing that most positrons are lost before the swarm thermalizes.

parameter, its very slow cooling means that annihilation has ample time to remove the positrons before genuine thermal equilibrium is reached. Neon is therefore an important example showing that substantial depletion can arise not only from large Z_{eff} , but also from inefficient momentum transfer.

Figure 4.8 compares the calculated $\bar{Z}_{\text{eff}}(\tau)$ with digitised curves from the diffusion-model study of Campeanu [16]. In the lifetime measurements of Coleman *et al.*, the neon shoulder width was reported as approximately 1700 ± 200 ns amagat, with an equilibrium value $Z_{\text{eff}} = 5.99 \pm 0.06$ [13]. Campeanu’s analysis, using earlier polarized-orbital input, already found that the theoretical shoulder was substantially broader than the experimental one and argued that the momentum-transfer cross sections in the Ramsauer-minimum region were too small to reproduce the observed cooling rate [16]. In particular, Campeanu noted that the theoretical shoulder was much larger than the experimental one, and identified the low transport cross section near the minimum as the source of the slow cooling.

The present calculation shows the same qualitative difficulty from a different and more modern input dataset. The averaged annihilation parameter decreases at early times as the initial epithermal swarm relaxes, and then rises only very slowly as the distribution drifts through the bottleneck region. The rise is noticeably slower than in the digitised curves shown for comparison, and the late-time value remains below the experimental thermal value over the

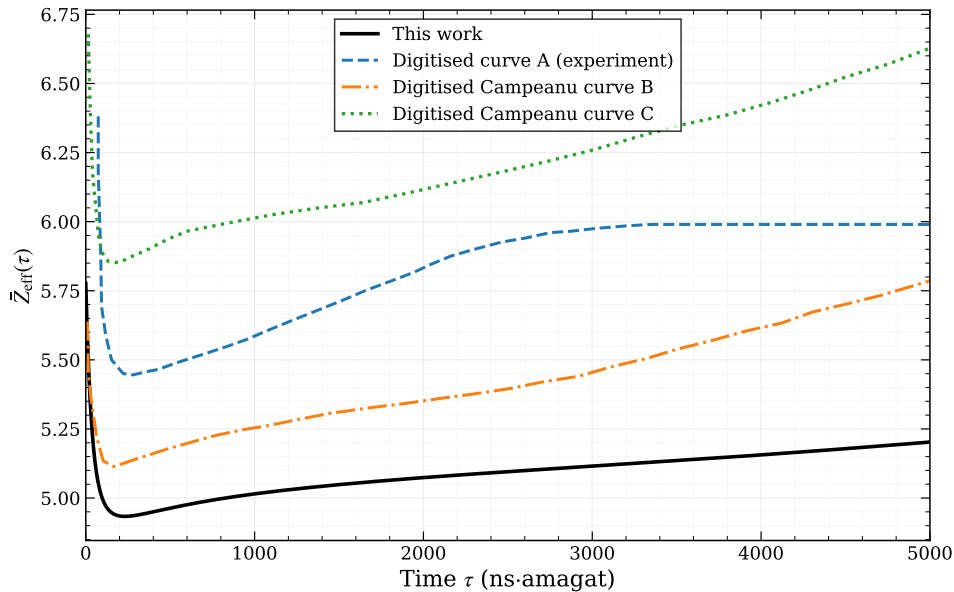


Figure 4.8: Averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$ for neon, compared with digitised curves from Campeanu’s diffusion-model study [16]. Curve A is the experimental trend derived from Coleman *et al.* [13], while curves B and C are Campeanu’s theoretical results based on earlier polarized-orbital input.

plotted interval. This indicates that, within the present Fokker–Planck model, neon remains the most problematic of the noble gases: the calculated cooling is too slow, the surviving fraction becomes extremely small, and the swarm fails to establish a clear thermal plateau on the experimental timescale.

Overall, neon provides the clearest example in this dissertation of a transport-limited cooling regime. The pronounced Ramsauer–Townsend minimum in $\sigma_m(p)$ causes the distribution to linger at intermediate momenta, strongly delays approach to thermal equilibrium, and allows annihilation to deplete most of the swarm before thermalization is effectively complete. This makes neon qualitatively different from helium, and marks the first point in the noble-gas sequence where the detailed form of the transport cross section becomes the dominant control on the cooling dynamics.

4.4 Argon and krypton: intermediate cooling regimes

Argon and krypton occupy an intermediate position in the noble-gas sequence. In both cases the cooling is appreciably more effective than in neon. Their momentum transfer cross sections

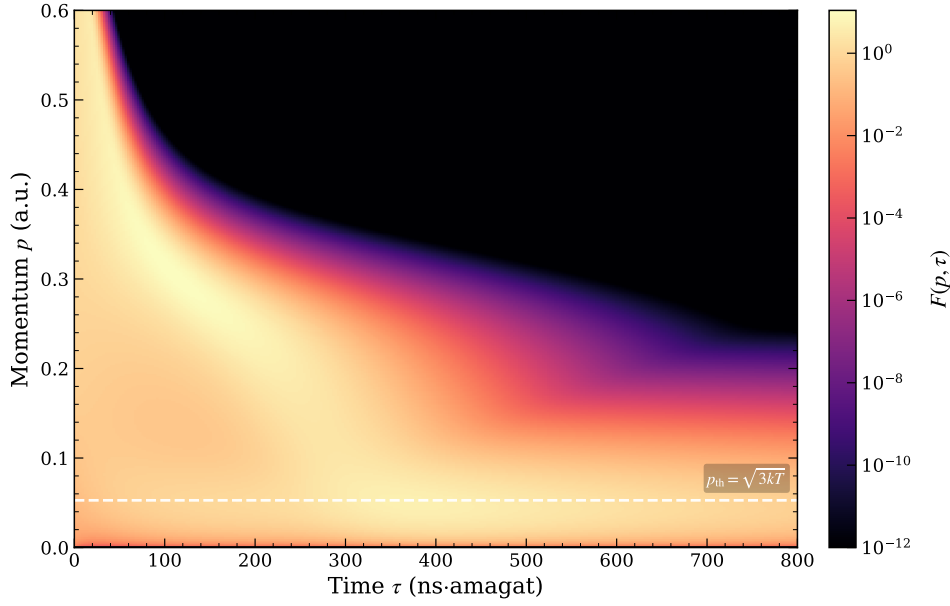


Figure 4.9: Time evolution of the argon radial momentum distribution $F(p, \tau)$. The initially broad epithermal distribution moves steadily towards lower momenta without the pronounced transport bottleneck seen in neon. The dashed horizontal line marks the thermal momentum scale $p_{\text{th}} = \sqrt{3k_{\text{B}}T}$.

are much larger than in neon and the suppressed transport associated with the Ramsauer–Townsend minimum is no longer a bottleneck of the dynamics. At the same time, annihilation is more important than in helium, so the evolution is no longer as close to the weak-depletion limit. These two gases therefore provide a useful bridge between the near-baseline behaviour of helium and the strong annihilation-driven distortion found later in xenon.

For the argon calculations shown here, the momentum range was taken as $0 \leq p \leq 0.8$ a.u. with $N_p = 400$, giving $\Delta p = 2.0 \times 10^{-3}$ a.u. The time step was $\Delta\tau = 5.0 \times 10^{-3}$ ns amagat and the evolution was followed to $\tau_{\text{max}} = 800$ ns amagat. For krypton, the same resolution in grid size was used over the interval $0 \leq p \leq 0.7$ a.u., with the same timestep and a final time $\tau_{\text{max}} = 700$ ns amagat.

For argon, the momentum distribution in Figure 4.9 shows a smooth transfer of probability from the initial epithermal swarm towards lower momenta. In contrast to neon, there is no pronounced long-lived trapping in an intermediate momentum band. The distribution contracts steadily and develops strong weight in the low-momentum region on a timescale of a few hundred ns amagat, indicating that momentum transfer is sufficiently efficient to drive the swarm close

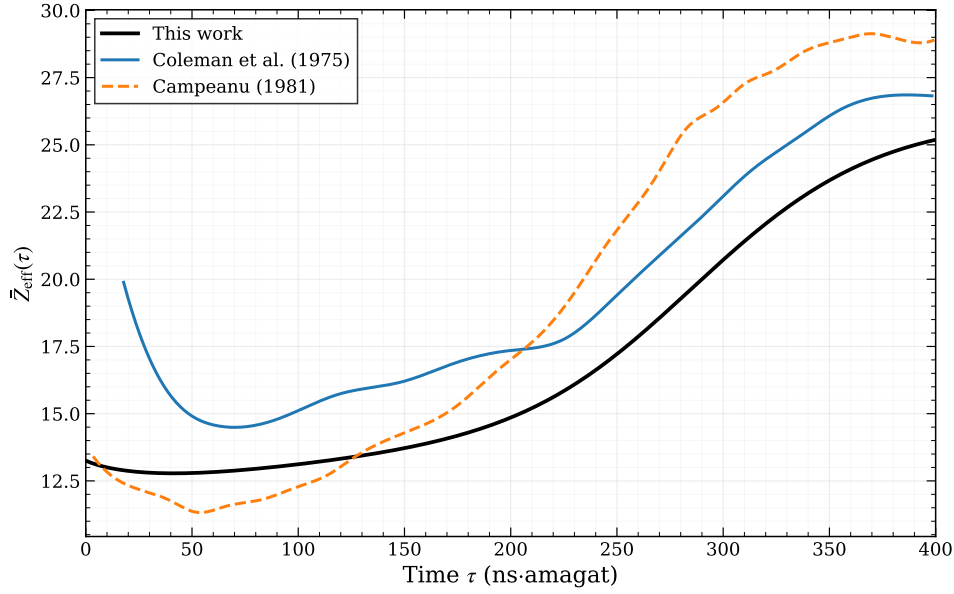


Figure 4.10: Averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$ for argon, compared with digitised curves based on the comparison shown in figure 25(c) of Charlton’s review [14]. The additional curves correspond to the experimental trend of Coleman *et al.* [13] and the diffusion-model result of Campeanu [16].

to thermal energies before annihilation removes most of the population. The corresponding mean-energy diagnostic confirms this impression: by the end of the run the average energy has settled to

$$\bar{E}(\tau_{\text{max}}) = 1.4073 \times 10^{-3} \text{ a.u.},$$

which corresponds to

$$\frac{\bar{E}(\tau_{\text{max}})}{3k_{\text{B}}T/2} \approx 1.011.$$

Thus, argon comes very close to the thermal mean energy within the simulation window, although the surviving distribution remains marginally above the exact Maxwellian value.

This near-thermal behaviour is reflected in the averaged annihilation parameter shown in Figure 4.10. The calculated $\bar{Z}_{\text{eff}}(\tau)$ exhibits the familiar initial decrease as the epithermal swarm relaxes, followed by a gradual rise towards its late-time plateau. At the end of the simulation,

$$\bar{Z}_{\text{eff}}(\tau_{\text{max}}) = 25.92,$$

which is close to the room-temperature experimental value reported for argon. Coleman *et al.* measured $Z_{\text{eff}} \approx 26.77$ and a shoulder width of approximately 362 ± 5 ns amagat [13], while

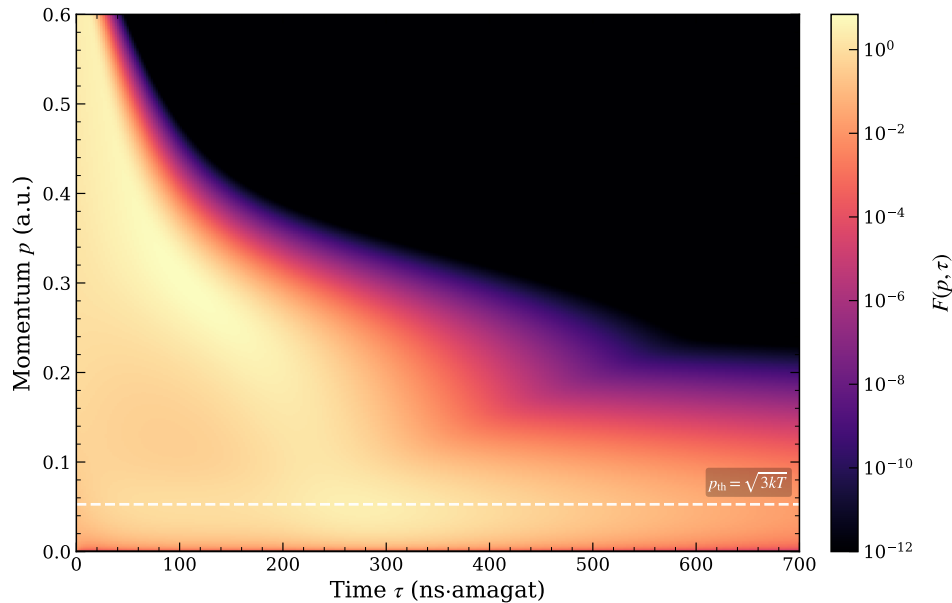


Figure 4.11: Time evolution of the krypton radial momentum distribution $F(p, \tau)$. Cooling remains efficient, but annihilation plays a stronger role than in argon, leading to more pronounced depletion as the swarm approaches low momenta. The dashed horizontal line marks the thermal momentum scale $p_{\text{th}} = \sqrt{3k_{\text{B}}T}$.

the review of Charlton notes that Campeanu’s argon thermalisation calculation gave excellent agreement with both the observed shoulder length and the equilibrium Z_{eff} [14]. The present curve follows the same general trend, supporting the interpretation of argon as a case in which the Fokker–Planck model captures the main balance between cooling and annihilation rather well.

Krypton’s behaviour is broadly similar, but with noticeably stronger depletion and a slightly larger departure from full thermalization. The distribution heat map in Figure 4.11 again shows steady motion of the swarm towards lower momenta without the severe bottleneck seen in neon. However, the low-momentum part of the distribution is built up under conditions of stronger annihilation than in argon, so the evolution is more strongly shaped by particle loss. This is reflected in the supporting diagnostics included in the appendix D, where the surviving fraction falls more rapidly than in argon and the averaged annihilation rate rises more strongly at late times.

The corresponding mean-energy diagnostic shows that krypton also cools close to the ther-

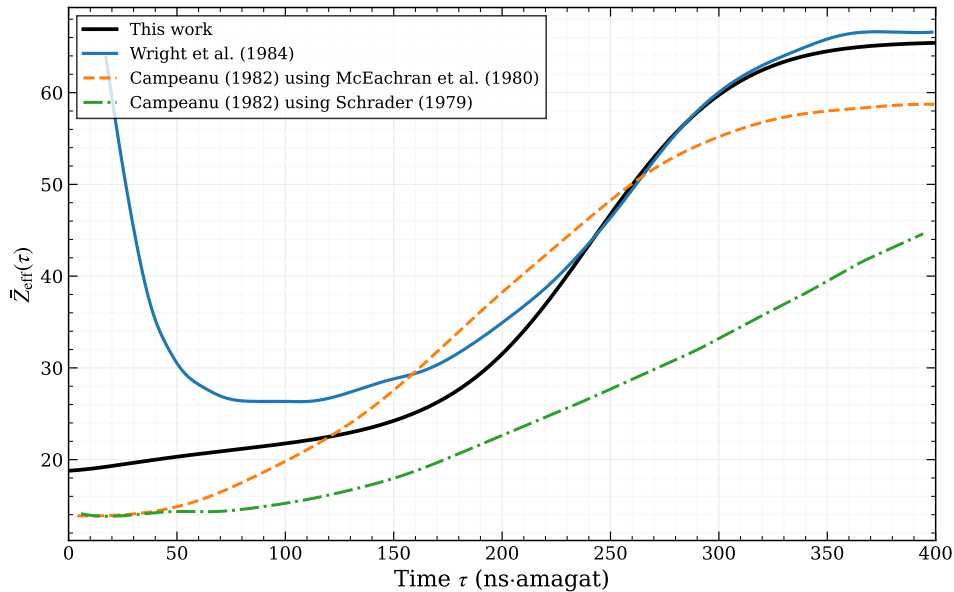


Figure 4.12: Averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$ for krypton, compared with digitised curves based on figure 25(c) of Charlton’s review [14]. The comparison includes the experimental trend of Wright *et al.* [15] and the two theoretical curves of Campeanu [17], based respectively on the input of McEachran *et al.* [18] and Schrader [19].

mal scale, but less completely than argon over the plotted interval. At the end of the run,

$$\bar{E}(\tau_{\text{max}}) = 1.4390 \times 10^{-3} \text{ a.u.},$$

giving

$$\frac{\bar{E}(\tau_{\text{max}})}{3k_{\text{B}}T/2} \approx 1.034.$$

Thus krypton remains a few percent above the thermal mean energy, indicating that annihilation has begun to compete more seriously with the final stages of cooling.

This is consistent with the behaviour of $\bar{Z}_{\text{eff}}(\tau)$ in Figure 4.12. The present calculation rises towards a late-time value

$$\bar{Z}_{\text{eff}}(\tau_{\text{max}}) = 65.55,$$

which is in very good agreement with the experimental room-temperature value for krypton. In the summary given by Charlton, the krypton data of Wright *et al.* are compared with two curves from Campeanu (1982): one based on the cross sections of McEachran *et al.* [18] and the other on those of Schrader [19]. The review notes that the McEachran-based curve reproduces the krypton shoulder much better, whereas the Schrader-based calculation predicts

a shoulder that is too long [14]. The present result follows the experimental trend closely and reaches essentially the correct late-time magnitude, which suggests that krypton remains a case where the cooling and annihilation dynamics are still described quite successfully by accurate transport input.

Taken together, argon and krypton illustrate the transition from the weak-annihilation regime of helium to the strongly depletion-dominated regime of xenon. Argon cools efficiently and approaches a near-thermal state with only modest depletion, while krypton shows a stronger influence of annihilation but still retains much of the same qualitative structure. In both gases the competition between transport and annihilation is balanced well enough that the time-dependent $\bar{Z}_{\text{eff}}(\tau)$ remains in good overall accord with the experimental trends, unlike the more problematic neon case and the more strongly distorted xenon case considered next.

4.5 Xenon: strong annihilation and non-thermal late-time behaviour

Xenon represents the opposite extreme from helium. While momentum transfer remains efficient, the annihilation parameter is very strongly enhanced at low momentum, so that annihilation competes directly with cooling already at epithermal positron energies. In this regime the positron swarm is no longer expected to approach a simple thermal Maxwellian before depletion becomes important. Instead, the late-time behaviour is set by a competition between the cooling operator, which drives the distribution towards thermal equilibrium, and the annihilation sink, which selectively removes particles from the low-momentum region where $Z_{\text{eff}}(p)$ is largest. This is precisely the special feature of xenon identified in the MBT-based cooling study of Green, who showed that xenon is uniquely sensitive to positron depletion and that this effect explains the long-standing discrepancy between gas-cell and trap-based measurements of $\bar{Z}_{\text{eff}}$ [6].

For the xenon calculations shown here, the momentum interval was taken as $0 \leq p \leq 0.6$ a.u. and discretised with $N_p = 400$, giving $\Delta p = 1.5 \times 10^{-3}$ a.u. The timestep was $\Delta\tau = 1.0 \times 10^{-3}$ ns amagat, and the evolution was followed up to $\tau_{\text{max}} = 500$ ns amagat.

The distribution heat map in Figure 4.13 already shows that xenon behaves qualitatively differently from the lighter noble gases. The initial epithermal population is driven quickly

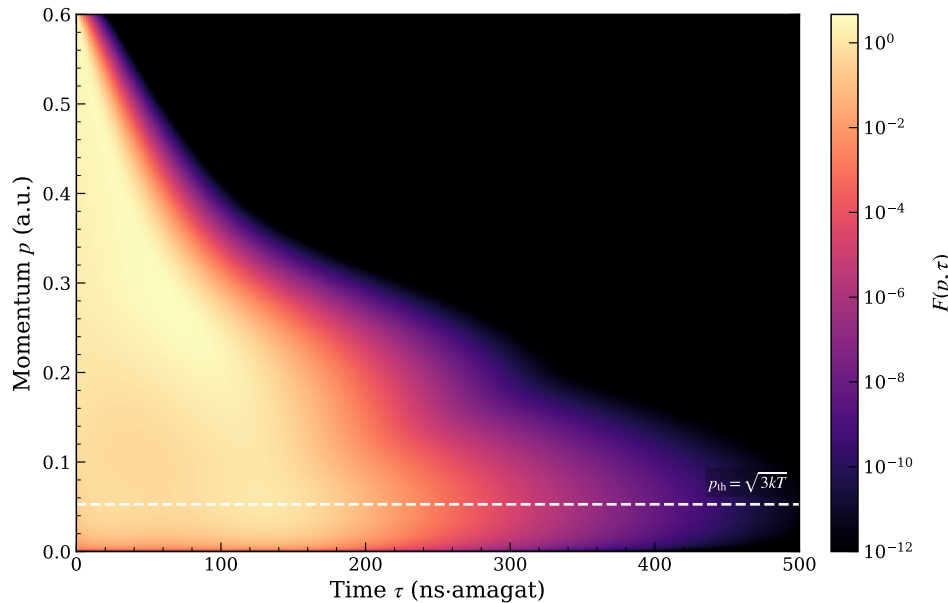


Figure 4.13: Time evolution of the xenon radial momentum distribution $F(p, \tau)$. The swarm is driven rapidly towards low momentum, but the low- p region is strongly depleted by annihilation, preventing approach to a simple Maxwellian-like thermal shape. The dashed horizontal line marks the thermal momentum scale $p_{\text{th}} = \sqrt{3k_{\text{B}}T}$.

towards lower momenta, but the low-momentum part of the distribution is simultaneously depleted so strongly that a true Maxwellian-like accumulation near thermal momentum never develops. Instead, the surviving swarm is biased away from the most strongly annihilating momenta. The evolution therefore does not resemble simple cooling towards equilibrium, but rather relaxation towards a deformed late-time shape controlled by the competition between cooling and annihilation.

This distortion is reflected directly in the mean energy shown in Figure 4.14. The average energy falls rapidly at early times, but then levels off well above the thermal value $3k_{\text{B}}T/2$. In the present run, the late-time mean energy settles to

$$\bar{E}(\tau_{\text{max}}) = 1.6979 \times 10^{-3} \text{ a.u.},$$

which gives

$$\frac{\bar{E}(\tau_{\text{max}})}{3k_{\text{B}}T/2} \approx 1.22.$$

Thus the surviving distribution remains substantially hotter than the thermal Maxwellian. This is not merely a matter of slow convergence: it is evidence that the xenon late-time state

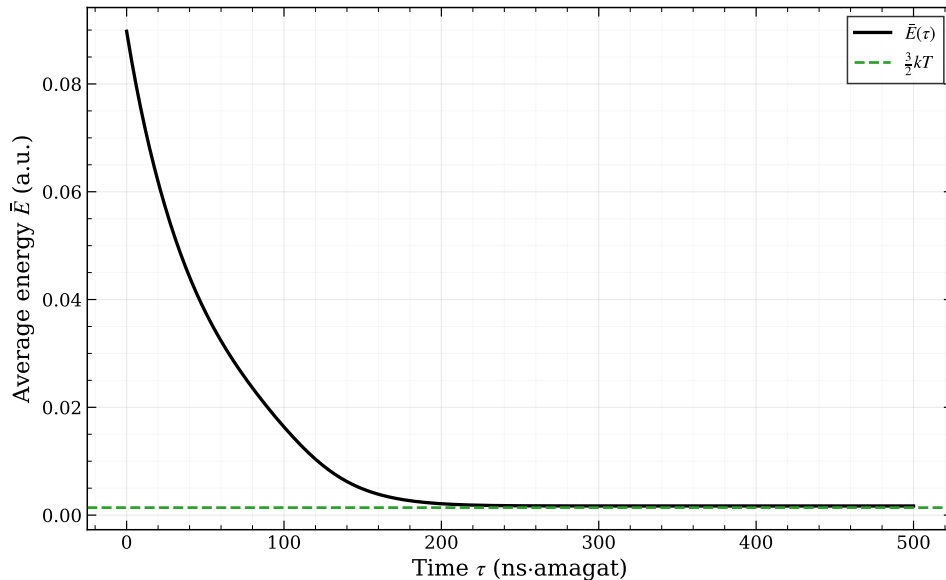


Figure 4.14: Mean positron energy $\bar{E}(\tau)$ for xenon. The late-time mean energy remains well above the thermal value $3k_{\text{B}}T/2$, showing that the surviving distribution is hotter than a thermal Maxwellian.

is genuinely non-thermal. This is also consistent with the asymptotic behaviour discussed in Chapter 5, where the dominant late-time mode is likewise found to be hotter than thermal.

The strength of depletion is made explicit by the surviving fraction $N(\tau)$ and the averaged annihilation rate $\bar{\lambda}(\tau)$, shown in Appendix, Figures D.10 and D.11. The population falls extremely rapidly, with only a small amount surviving once the distribution has reached the low-momentum region, while $\bar{\lambda}(\tau)$ rises steeply towards a high plateau. This is consistent with the Monte Carlo cooling study of Green, which found that the fraction of positrons surviving to thermalization in xenon is only of order 10^{-2} [6]. In xenon, therefore, the annihilation sink is not a small correction to the cooling dynamics: it is one of the dominant processes shaping the evolution.

Figure 4.15 compares the calculated $\bar{Z}_{\text{eff}}(\tau)$ with digitised experimental and earlier theoretical curves. The experimental comparison curve is taken from Wright *et al.* [15], their figure 4(b), while the two older theoretical curves correspond to the diffusion-model calculations of Campeanu [17] using the scattering input of McEachran *et al.* [18] and Schrader [19], respectively. This is the same attribution given in Charlton’s review of the lifetime shoulder data [14].

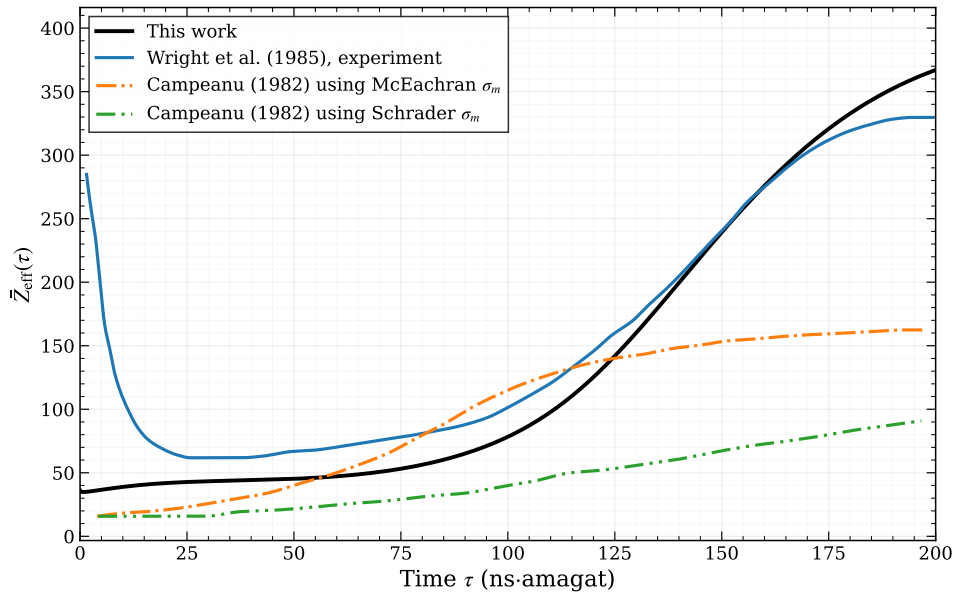


Figure 4.15: Averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$ for xenon, compared with the experimental curve of Wright *et al.* [15] and the earlier diffusion-model calculations of Campeanu [17] using the cross-section input of McEachran *et al.* [18] and Schrader [19].

Quantitatively, the present result rises to a late-time value

$$\bar{Z}_{\text{eff}}(\tau_{\text{max}}) = 397.12,$$

which is substantially larger than the older low-density gas-cell value $Z_{\text{eff}} \approx 320$ reported for xenon [15], but much closer to the higher 400–450 range found in later measurements and many-body calculations [3, 6]. Wright *et al.* also reported a xenon shoulder width of

$$\tau_s = 178 \pm 3 \text{ ns amagat},$$

while the older diffusion-model calculations of Campeanu were in serious disagreement with both the shoulder shape and magnitude [15, 14]. More recent MBT work showed that once positron depletion is included, the xenon shoulder region and shoulder length come into excellent agreement with experiment [6].

The physical reason is clear. Because $Z_{\text{eff}}(p)$ in xenon is so strongly peaked at low momentum, the very act of cooling the swarm towards thermal energies makes it more vulnerable to annihilation. The low-momentum component is therefore systematically suppressed relative to a Maxwellian. Green identified this as a quasi-steady-state long-time distribution with reduced low-momentum weight and a steady annihilation rate significantly below that obtained

by simply averaging $\lambda(p)$ over a thermal distribution [6]. The present calculations show the same effect in a particularly transparent way through the simultaneous behaviour of $F(p, \tau)$, $\bar{E}(\tau)$, $N(\tau)$, $\bar{\lambda}(\tau)$, and $\bar{Z}_{\text{eff}}(\tau)$.

Overall, xenon is the clearest example in this dissertation of annihilation-controlled cooling. Unlike helium, argon and even krypton, the positron distribution in xenon does not evolve towards a nearly thermal swarm before most particles are lost. Instead, the surviving positron population approaches a hotter-than-thermal late-time state whose shape is determined by the competition between transport and strongly momentum-dependent annihilation. This makes xenon the natural case for the asymptotic analysis of the next chapter.

4.6 Comparison across the noble gases

Taken together, the results for helium, neon, argon, krypton and xenon show a clear progression in the balance between momentum-space transport and annihilation across the noble-gas sequence. Helium lies closest to the weak-annihilation limit: the positron swarm cools relatively smoothly, the late-time distribution remains close to thermal, and the averaged annihilation parameter evolves in good qualitative agreement with experiment. Neon departs from this simple picture in a different way. The dominant feature here is not particularly strong annihilation, but rather the severe transport bottleneck associated with the deep Ramsauer–Townsend minimum in the momentum-transfer cross section. As a result, cooling becomes extremely slow, the positron population lingers at intermediate momenta, and practical thermalization is not achieved before the swarm is almost completely depleted.

Argon and krypton occupy an intermediate regime between these two extremes. In both cases, cooling is much more effective than in neon, so the positron distribution can move towards the thermal region on experimentally relevant timescales. At the same time, annihilation is no longer negligible, especially in krypton, and the resulting evolution is more strongly shaped by depletion than in helium. Nevertheless, both gases still show broadly good agreement with the observed shoulder behaviour and with the corresponding room-temperature values of $\bar{Z}_{\text{eff}}$, making them the cases in which the combined transport–annihilation description appears to be most successful overall.

Xenon is qualitatively different from all of the other gases. Here the strong low-momentum

enhancement of $Z_{\text{eff}}(p)$ means that annihilation competes directly with cooling already at epithermal momenta. The consequence is that the low-momentum part of the distribution is selectively removed before a Maxwellian swarm can be established. The surviving positrons therefore approach a late-time state that is distinctly hotter than thermal, with a reduced low-momentum weight and a correspondingly modified annihilation response. In this sense xenon is not simply the heaviest member of the sequence, but the first case in which the long-time state is genuinely non-Maxwellian.

These trends are fully consistent with the broader picture established in the recent MBT-based Monte Carlo cooling study of Green [6]. That work found the fraction of positrons surviving to thermalization to be strikingly small across the noble gases, with approximate values of 0.1 in He, ~ 0 in Ne, 0.15 in Ar, 0.05 in Kr, and 0.01 in Xe, and showed that xenon is uniquely sensitive to the distortion of the momentum distribution by annihilation. The same overall hierarchy is seen in the present calculations. Helium remains the case closest to the weak-annihilation, near-thermal limit; neon is the most transport-limited; argon and krypton are intermediate cases in which both cooling and annihilation matter; and xenon is the most strongly annihilation-dominated case.

The comparison with earlier lifetime experiments follows the same pattern. The classic shoulder widths compiled in the review of Charlton, namely approximately 1700 ± 50 ns amagat for He, 2300 ± 200 ns amagat for Ne, 362 ± 5 ns amagat for Ar, 325 ± 6 ns amagat for Kr, and 178 ± 3 ns amagat for Xe, already suggest that the noble gases do not form a simple monotonic sequence in cooling time alone [14, 15]. Instead, the observed behaviour reflects the interplay of two distinct effects: the efficiency of momentum transfer, which is especially poor in neon, and the growth of annihilation at low momentum, which becomes increasingly important from helium to xenon.

Overall, the noble-gas sequence provides a clear demonstration of how momentum-dependent annihilation modifies cooling. Helium shows that when annihilation is weak, the swarm remains close to the thermal Maxwellian. Neon shows that even moderate annihilation can become decisive when transport is inefficient. Argon and krypton illustrate the intermediate regime in which cooling remains effective but depletion is no longer negligible. Xenon finally shows that once annihilation becomes sufficiently strong at low momentum, the late-time state is reshaped and the thermal Maxwellian ceases to be the relevant asymptotic reference state.

Chapter 5

Asymptotic eigenmode analysis

5.1 Motivation and scope

The question of the time evolution of the positron momentum distribution at late times is not simply whether xenon cools, but what shape the *surviving* positron distribution approaches when cooling and annihilation compete. The time-dependent results of the previous chapter suggest that, after an initial transient, the positron distribution in xenon evolves towards a nearly time-independent shape while its overall amplitude continues to decay exponentially. This motivates an asymptotic variable-separable description of the form

$$F(p, \tau) \sim e^{-\beta\tau} \phi(p), \quad (5.1)$$

where $\beta > 0$ is the asymptotic late-time decay rate and $\phi(p)$ is the corresponding asymptotic shape.

In the absence of annihilation, the non-decaying mode is the thermal Maxwellian. With annihilation present, the dominant mode is expected to be a distorted version of the thermal shape.¹

The Appendix E develops the structure of the *continuous* late-time eigenproblem and shows how it can be rewritten in a weighted Sturm–Liouville form. The purpose of the present chapter is different: it computes the *discrete* analogue of that asymptotic distribution for the actual

¹If $\lambda(p)$ were independent of momentum, annihilation would contribute only a uniform exponential decay and would not distort the shape. The distortion therefore comes entirely from the momentum dependence of $\lambda(p)$.

finite-volume scheme used in this dissertation, and compares it directly with the late-time distributions produced by time stepping.

To keep the notation manageable, the main objects used in this chapter are summarised here. The matrix L denotes the cooling-only finite-volume operator, A is the corresponding unsplit semidiscrete generator including annihilation, and S is the one-step split-update matrix. Their dominant eigenvalues are denoted by μ_0 for A and ρ_0 for S . The associated effective decay rates are written $\Lambda_{\text{eff}}^{(A)}$ and $\Lambda_{\text{eff}}^{(S)}$, while Λ_{fit} denotes the independent late-time decay rate obtained from fitting the time-evolved survival fraction.

5.2 Discrete eigenproblems

Let

$$\mathbf{F}(\tau) = (F_1(\tau), F_2(\tau), \dots, F_{N_p-1}(\tau))^T \quad (5.2)$$

denote the interior nodal values of the radial distribution, excluding the fixed boundary nodes at $p = 0$ and $p = p_{\text{max}}$. The conservative cooling operator from the finite-volume scheme defines a matrix L through

$$\frac{d\mathbf{F}}{d\tau} = L\mathbf{F} \quad (\text{cooling only}), \quad (5.3)$$

while the full unsplit semidiscrete generator is

$$A = L - \text{diag}(\lambda_i), \quad (5.4)$$

where $\lambda_i = \lambda(p_i)$ on the interior grid.

The matrix L was constructed numerically by applying the discrete cooling right-hand side to each interior basis vector in turn, so that it exactly matches the implemented finite-volume operator.² The dominant eigenmode of the unsplit semidiscrete system is then obtained from

$$A\mathbf{v}_m = \mu_m \mathbf{v}_m. \quad (5.5)$$

If μ_0 is the eigenvalue with the largest real part, then the corresponding late-time decay rate is

$$\Lambda_{\text{eff}}^{(A)} = -\Re(\mu_0). \quad (5.6)$$

²This avoids any mismatch between the algebraic operator used for the eigenanalysis and the one actually used in the time-stepping code.

Since the actual numerical solver uses operator splitting, it is also useful to analyse the one-step update matrix

$$S = E(I + \Delta\tau L), \quad E = \text{diag}(e^{-\lambda_i \Delta\tau}), \quad (5.7)$$

where I is the identity matrix, so that

$$\mathbf{F}^{n+1} = S\mathbf{F}^n. \quad (5.8)$$

Its dominant eigenmode satisfies

$$S\mathbf{w}_m = \rho_m \mathbf{w}_m, \quad (5.9)$$

and the corresponding effective decay rate is

$$\Lambda_{\text{eff}}^{(S)} = -\frac{1}{\Delta\tau} \ln |\rho_0|. \quad (5.10)$$

Because S corresponds exactly to the time-stepping method used in the code, its dominant eigenmode is expected to match the final time-evolved shape particularly closely.

For comparison, the late-time probability decay can also be estimated directly from the numerical solution by fitting a straight line to $\ln N(\tau)$ in the tail region, where

$$N(\tau) = \int_0^{p_{\text{max}}} F(p, \tau) dp. \quad (5.11)$$

If the late-time behaviour is truly exponential, as in Eq. (5.1), this fitted decay rate should agree with the dominant eigenvalue predictions from both A and S .

In practice, all eigenpairs were computed using the dense routine `numpy.linalg.eig` applied to the matrices A and S . For xenon, the dominant eigenvalues were real to numerical precision. Before comparison with the late-time time-stepped distribution, the dominant eigenvectors were sign-adjusted so that their bulk weight was positive, any tiny negative components attributable to roundoff were set to zero, and the resulting shapes were normalized so that

$$\sum_i v_i \Delta p = 1, \quad \sum_i w_i \Delta p = 1. \quad (5.12)$$

where v_i and w_i are the components of the dominant eigenvectors of A and S , respectively.

After reinstating the zero boundary values at $p = 0$ and $p = p_{\text{max}}$, these normalized eigenmodes can be compared directly with the normalized distribution from time stepping.

5.3 Xenon eigenanalysis results

The eigenanalysis was carried out for xenon on the same grid and with the same timestep as used in the corresponding production run:

$$p_{\max} = 0.6 \text{ a.u.}, \quad N_p = 400, \quad \Delta\tau = 10^{-3} \text{ ns amagat}. \quad (5.13)$$

For this case, the annihilation rate ranged from

$$\lambda(p) \in [5.2 \times 10^{-3}, 1.52] \text{ (ns amagat)}^{-1}, \quad (5.14)$$

so that

$$\max(\lambda \Delta\tau) = 1.52 \times 10^{-3}. \quad (5.15)$$

The leading eigenvalue of the unsplit generator A was found to be

$$\mu_0 = -7.97991742 \times 10^{-2}, \quad (5.16)$$

giving

$$\Lambda_{\text{eff}}^{(A)} = 7.97991742 \times 10^{-2} \text{ (ns amagat)}^{-1}. \quad (5.17)$$

For the split-step matrix S , the dominant timestep eigenvalue was

$$\rho_0 = 0.9999201997202827, \quad (5.18)$$

which corresponds to

$$\Lambda_{\text{eff}}^{(S)} = 7.98034639 \times 10^{-2} \text{ (ns amagat)}^{-1}, \quad (5.19)$$

while a direct late-time linear fit to $\ln N(\tau)$ gave

$$\Lambda_{\text{fit}} = 7.98034577 \times 10^{-2} \text{ (ns amagat)}^{-1}. \quad (5.20)$$

The three values are in excellent agreement. In terms of the corresponding asymptotic averaged annihilation parameter, they correspond to $\bar{Z}_{\text{eff}} = 397.13, 397.15$ and 397.15 , respectively. These are also in very close agreement with the direct large- τ value from the time-dependent simulation, for which the late-time average was $\bar{Z}_{\text{eff}}(\tau) \approx 397.12$.

The asymptotic eigenmodes from A and S are compared with the thermal Maxwellian in Fig. 5.1. Both asymptotic modes are clearly broader than the thermal shape and have a higher mean energy. In particular,

$$\frac{\bar{E}_A}{3k_B T/2} = 1.21990122, \quad \frac{\bar{E}_S}{3k_B T/2} = 1.21991760, \quad (5.21)$$

Table 5.1: Summary of the xenon eigenanalysis.

Quantity	Value
Leading generator eigenvalue μ_0	$-7.97991742 \times 10^{-2}$
$\Lambda_{\text{eff}}^{(A)}$	$7.97991742 \times 10^{-2}$
Leading split-step eigenvalue ρ_0	0.9999201997202827
$\Lambda_{\text{eff}}^{(S)}$	$7.98034639 \times 10^{-2}$
Late-time fit Λ_{fit}	$7.98034577 \times 10^{-2}$
Thermal-average estimate Λ_{th}	$9.90007851 \times 10^{-2}$
$\bar{E}_{\text{th}}/(3k_{\text{B}}T/2)$	1.00000000
$\bar{E}_A/(3k_{\text{B}}T/2)$	1.21990122
$\bar{E}_S/(3k_{\text{B}}T/2)$	1.21991760

showing that the late-time positron momentum distribution in xenon is about 22% hotter than the Maxwellian.

A stronger validation is obtained by comparing the final normalized distribution from direct time stepping with the dominant eigenmodes. As shown in Fig. 5.2, the final time-evolved shape lies almost exactly on top of the dominant split-step eigenmode, with the generator eigenmode also essentially indistinguishable on the scale of the figure. This confirms that the long-time numerical solution is indeed controlled by the dominant spectral eigenmode.

This spectral picture is also consistent with the results of the time-dependent calculation. By $\tau = 500$ ns amagat, the surviving fraction had fallen to

$$N(500) = 1.637924 \times 10^{-13}, \quad (5.22)$$

while the mean energy remained above the thermal threshold:

$$\bar{E}(500) = 1.697892 \times 10^{-3} \text{ a.u.} > 1.405727 \times 10^{-3} \text{ a.u.} = E_{\text{thr}}, \quad (5.23)$$

with $\bar{E}(500)/E_{\text{thr}} \approx 1.21$. Thus the positron momentum distribution in xenon does not approach a true thermal Maxwellian shape before all the particles are lost. Instead, the swarm approaches a non-Maxwellian asymptotic shape that decays exponentially in amplitude.

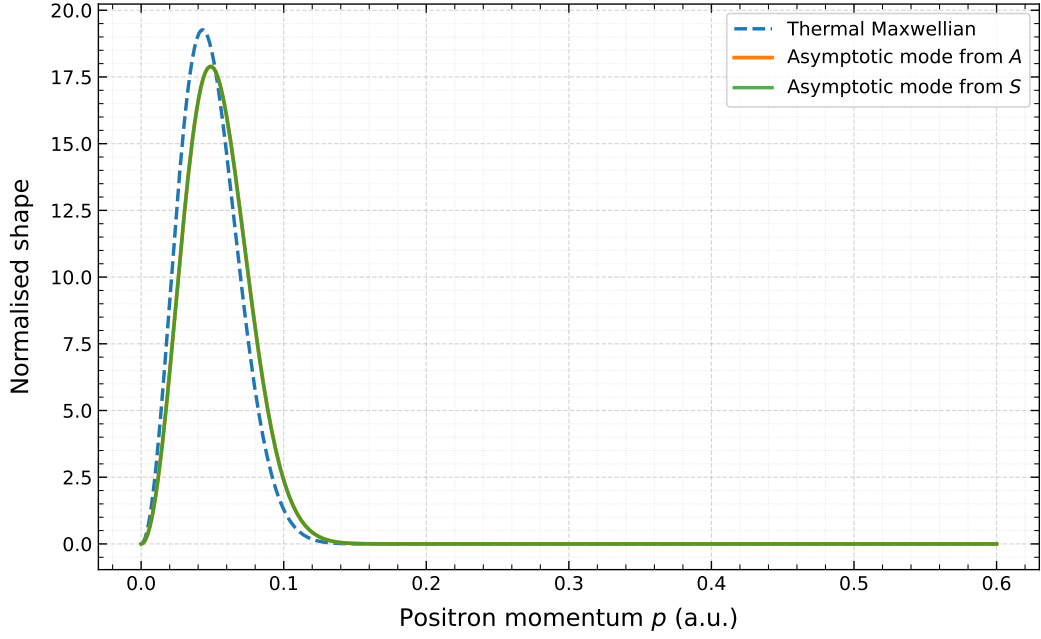


Figure 5.1: Xenon asymptotic eigenmodes compared with the thermal Maxwellian. Both the generator mode and the split-step mode are broader than the thermal distribution and therefore correspond to a hotter late-time state. Note that the asymptotic distributions obtained from A and S are indistinguishable on the graph.

5.4 Physical interpretation

A useful comparison is provided by the weak-annihilation estimate

$$\Lambda_{\text{th}} = \frac{\int_0^{p_{\text{max}}} \lambda(p) \phi_{\text{th}}(p) dp}{\int_0^{p_{\text{max}}} \phi_{\text{th}}(p) dp}, \quad (5.24)$$

where $\phi_{\text{th}}(p) \propto p^2 e^{-p^2/2k_{\text{B}}T}$ is the radial thermal shape. For xenon this gives

$$\Lambda_{\text{th}} = 9.90007851 \times 10^{-2} \text{ (ns amagat)}^{-1}, \quad (5.25)$$

with the corresponding $\bar{Z}_{\text{eff}} \approx 492.688$, which is substantially larger than the actual late-time decay rate,

$$\Lambda_{\text{eff}} \simeq 7.98 \times 10^{-2} \text{ (ns amagat)}^{-1}, \quad (5.26)$$

the latter corresponding to $\bar{Z}_{\text{eff}} \approx 397.133$.

This difference has a clear physical interpretation. The weak-annihilation estimate assumes that the late-time distribution remains close to thermal equilibrium, so that annihilation samples $\lambda(p)$ from a Maxwellian-like distribution. Xenon does not satisfy this assumption. Because

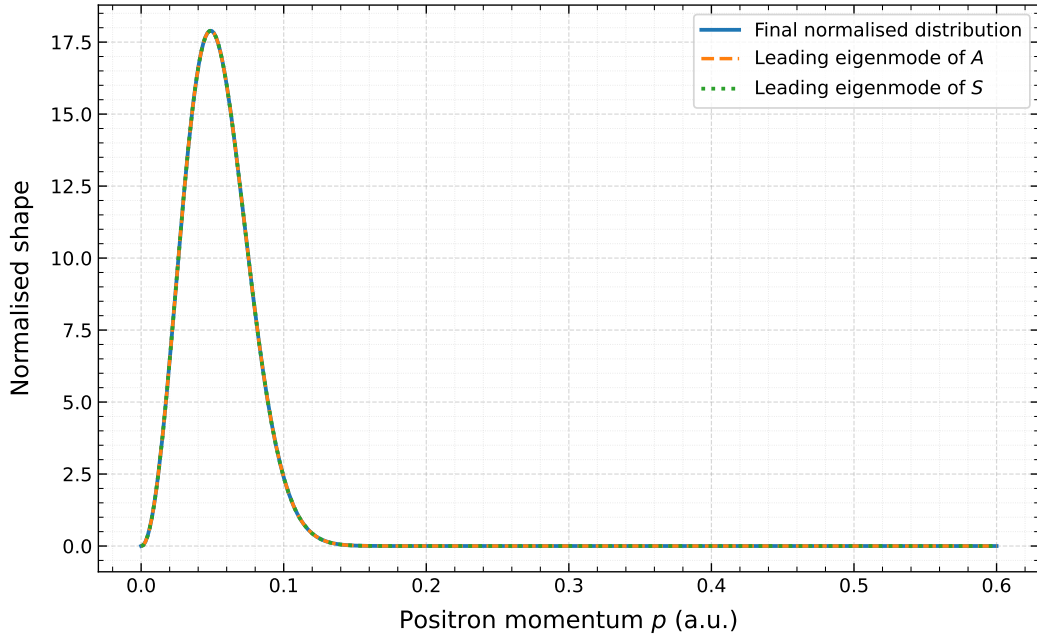


Figure 5.2: Final normalized xenon distribution from direct time stepping compared with the dominant eigenmodes of the generator A and the split-step matrix S . The near-perfect overlap confirms that the late-time solution is governed by the leading asymptotic mode.

$\lambda(p)$ is strongly enhanced at low momentum, the asymptotic mode suppresses weight in precisely the region where annihilation is strongest. The surviving particles are therefore biased away from the most strongly annihilating momenta, which lowers the effective decay rate below the simple thermal average and leaves a hotter-than-thermal shape.

This can be seen directly in Fig. 5.1: relative to the Maxwellian, the asymptotic distribution is depleted at low p and enhanced at somewhat larger p . In other words, the cooling operator continually drives the swarm toward the thermal shape, while the annihilation term selectively removes particles from the low-momentum region. The late-time xenon mode is the compromise between these two effects.³ This interpretation also helps explain the long-standing discrepancy between the lower $\bar{Z}_{\text{eff}}$ values measured for positrons cooling in pure xenon gas and the higher values obtained in Penning–Malmberg trap experiments, where the positrons were cooled to room temperature before xenon was introduced [20]. In pure xenon, incomplete thermalization together with preferential annihilation of the low-momentum part of the distribution lowers the measured $\bar{Z}_{\text{eff}}$, whereas the trap-based measurements probe a distribution much closer to the

³A formal derivation of this balance, including a Rayleigh-quotient representation and a weak-annihilation expansion, is given in Appendix E.

true thermal one.

5.5 Summary

The eigenanalysis shows that the xenon late-time solution is governed by a dominant asymptotic mode. The decay rate extracted from the unsplit generator, the split-step update matrix and a direct fit to the numerical survival fraction all agree to high precision:

$$\Lambda_{\text{eff}}^{(A)} \approx \Lambda_{\text{eff}}^{(S)} \approx \Lambda_{\text{fit}} \approx 7.98 \times 10^{-2} \quad (5.27)$$

corresponding to $\bar{Z}_{\text{eff}} \approx 397$. Moreover, the normalized late-time distribution obtained by time stepping lies almost exactly on the dominant eigenmode of the split-step operator.

The dominant xenon mode is non-Maxwellian and hotter than thermal, with

$$\frac{\bar{E}_{\text{asyp}}}{3k_{\text{B}}T/2} \approx 1.22. \quad (5.28)$$

This explains why xenon does not fully thermalize in the usual sense before the surviving positron population becomes negligible. This also explains the difference between $\bar{Z}_{\text{eff}}$ values measured for xenon in the positron trap [20] and the earlier measurements for positrons cooling in the gas itself. The eigenanalysis therefore provides a compact interpretation of the late-time dynamics: xenon does not approach the thermal Maxwellian, but rather a decaying asymptotic mode shaped by the competition between cooling and momentum-dependent annihilation.

Chapter 6

Conclusion and Future Work

6.1 Future Work

The present work has focused on positron cooling and annihilation in the noble gases, where the dominant low-energy process is elastic momentum transfer and the resulting dynamics can be described accurately by a Fokker–Planck equation with momentum-dependent transport and annihilation input. A natural extension of the present model would be to apply the same framework to molecular gases, where additional internal degrees of freedom provide new cooling channels through rotational and vibrational excitation, and where the annihilation physics can differ qualitatively from the atomic case [24, 25].

The simplest way to do this would be to retain the present elastic Fokker–Planck operator and augment it with extra terms representing inelastic molecular processes. Schematically, one would replace the current evolution equation by

$$\frac{\partial F}{\partial \tau} = \mathcal{L}_{\text{el}}[F] + \sum_j \mathcal{J}_j[F] - \lambda_{\text{ann}}(p) F - \lambda_{\text{Ps}}(p) F, \quad (6.1)$$

where $\mathcal{L}_{\text{el}}$ is the existing elastic transport operator, $\mathcal{J}_j$ denotes the contribution of molecular channel j , $\lambda_{\text{ann}}(p)$ is the annihilation rate, and $\lambda_{\text{Ps}}(p)$ could be included once positronium formation becomes energetically accessible.

For each additional molecular channel, the new physical input would be the threshold energy ΔE_j , the corresponding cross section $\sigma_j(p)$, and, where necessary, the thermal population of the initial molecular state. In a rotational or vibrational excitation event, a positron at momentum p is transferred to a lower momentum determined by the energy loss ΔE_j . Numerically, this could

be implemented as a conservative gain–loss update after the existing finite-volume transport step, so that the present solver would not need to be replaced but only supplemented by an additional operator-split stage. This would preserve the structure of the current code while allowing individual molecular processes to be introduced in a controlled and transparent way.

For rotational excitation, where the energy transfer is often small, it may also be possible to absorb part of the effect into an effective transport coefficient $B_{\text{eff}}(p)$, thereby modifying the drift and diffusion already present in the Fokker–Planck operator. Vibrational excitation is less naturally diffusive, since it produces a finite jump in energy, and would therefore be better treated explicitly through the source–sink terms described above. In this sense, the most natural molecular extension would likely be a hybrid scheme: elastic and possibly rotational cooling handled by the Fokker–Planck part, and vibrational excitation treated through discrete momentum-space transfer terms.

A further extension would be to replace the atomic annihilation input by molecular $Z_{\text{eff}}(p)$ data. This is especially important because positron annihilation on molecules can involve additional mechanisms beyond the simple atomic picture, particularly once vibrational degrees of freedom become relevant [25]. The same numerical framework could then be used to compare candidate molecular buffer gases by tracking both the cooling time and the surviving positron fraction. Such a study would provide a useful bridge between the present noble-gas calculations and applications to positron traps, accumulators, and molecular cooling experiments [2].

6.2 Conclusion

In this dissertation, positron cooling and annihilation in the noble gases helium, neon, argon, krypton and xenon have been studied using a radial momentum-space Fokker–Planck equation with many-body-theory input for the momentum-transfer cross section and annihilation parameter $Z_{\text{eff}}(p)$. The equation was solved numerically using a conservative finite-volume discretisation for the transport operator together with an operator-split exponential update for the annihilation sink. Verification tests showed that the cooling operator conserves total probability to machine precision in the absence of annihilation and that the sink term is implemented consistently.

The calculations show that the balance between cooling and annihilation changes strongly

across the noble-gas sequence. Helium behaves as the clearest near-baseline case: annihilation is weak, the momentum distribution cools smoothly, and the late-time swarm remains close to thermal. Neon is qualitatively different, not because annihilation is especially strong, but because the deep Ramsauer–Townsend minimum in the momentum-transfer cross section produces a pronounced cooling bottleneck. As a result, the positron population lingers at intermediate momenta and is substantially depleted before practical thermalization is achieved. Argon and krypton lie between these extremes. In both gases cooling remains effective, but annihilation increasingly modifies the surviving swarm, with krypton showing the stronger depletion of the two.

Xenon is the most striking case. There, the strong low-momentum enhancement of $Z_{\text{eff}}(p)$ causes annihilation to compete directly with cooling, so that the surviving positron distribution does not approach a true thermal Maxwellian before the population becomes extremely small. Instead, the low-momentum part of the swarm is preferentially depleted, leaving a late-time state that is hotter than thermal. This explains why xenon differs qualitatively from the lighter noble gases and why its measured annihilation behaviour is especially sensitive to whether the positrons are allowed to cool in the gas itself or are pre-thermalized in a trap.

To interpret this behaviour, an eigenanalysis of the discrete xenon operators was carried out. The dominant eigenmodes of both the unsplit semidiscrete generator and the split-step update matrix were found to reproduce the late-time decay rate and the final normalized distribution obtained from direct time stepping. This provides a clear spectral interpretation of the xenon asymptotics: the long-time solution is governed not by the thermal Maxwellian, but by a non-Maxwellian decaying eigenmode determined by the competition between transport and momentum-dependent annihilation.

Overall, the present work shows that the Fokker–Planck framework, when combined with accurate many-body-theory scattering and annihilation data, provides a robust description of positron cooling in noble gases. It captures not only the broad progression from weak to strong annihilation across the sequence, but also the qualitative breakdown of thermal behaviour in xenon. In this sense, the dissertation demonstrates both the usefulness of the Fokker–Planck approach and the importance of realistic microscopic input in understanding low-energy positron cooling and annihilation.

Appendix A

Additional momentum-transfer cross sections

For completeness, the momentum-transfer cross sections for neon, argon and krypton are shown in Figures A.1, A.2 and A.3, respectively. These were generated in the same way as the helium and xenon cross sections discussed in the main text, by interpolating the MBT phase-shift data and evaluating the partial-wave expression for $\sigma_m(p)$ on a uniform momentum grid. Together with Figure 2.1, they provide the full set of cross-section inputs used in the Fokker–Planck simulations.

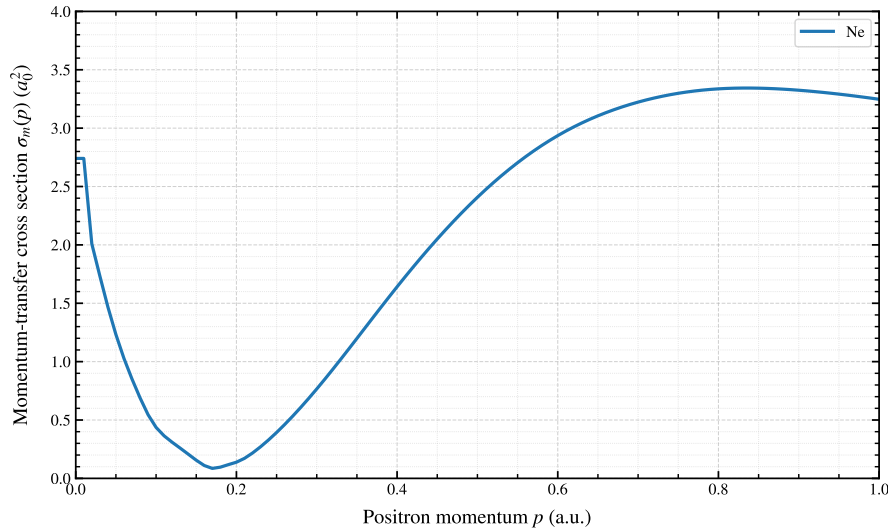


Figure A.1: Momentum-transfer cross section $\sigma_m(p)$ for neon, obtained from the interpolated MBT phase shifts and evaluated on the uniform solver momentum grid. The pronounced minimum at low momentum is the Ramsauer–Townsend minimum discussed in the main text.

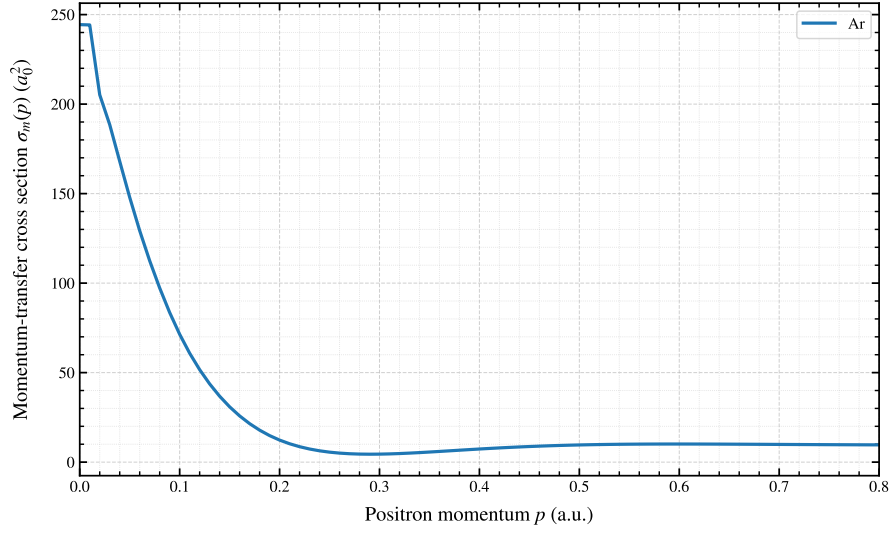


Figure A.2: Momentum-transfer cross section $\sigma_m(p)$ for argon, evaluated from the interpolated MBT phase shifts on the uniform momentum grid used by the solver. Compared with neon, the cross section is substantially larger over most of the plotted range.

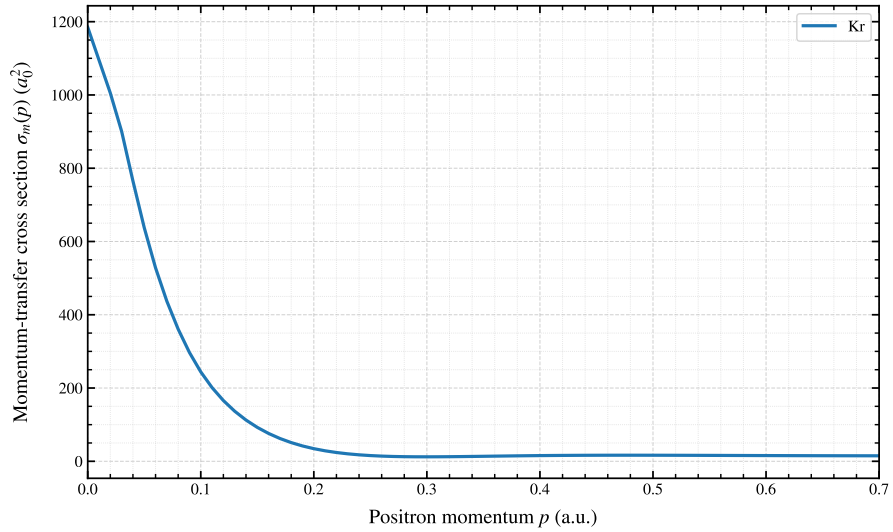


Figure A.3: Momentum-transfer cross section $\sigma_m(p)$ for krypton, evaluated from the interpolated MBT phase shifts on the uniform momentum grid. The low-momentum cross section is larger again than for argon, consistent with the stronger positron–atom interaction in the heavier noble gases.

Appendix B

Closed-form three-point determination of $Z_{\text{eff}}(0)$

As discussed in the main text, the MBT tables provide values of $Z_{\text{eff}}(p)$ only for nonzero momenta, whereas the numerical solver requires a value at $p = 0$. To construct this low-momentum anchor, the first three tabulated points were fitted exactly to the virtual-level form

$$Z(p) = \frac{K}{x^2 + p^2} + A, \quad (\text{B.1})$$

where A , K , and x^2 are constants to be determined. This appendix records the exact algebraic three-point solve used in the calculations.

Suppose that the first three tabulated values are

$$(p_1, Z_1), \quad (p_2, Z_2), \quad (p_3, Z_3), \quad (\text{B.2})$$

with $p_1, p_2, p_3 > 0$. Define

$$s_i = p_i^2, \quad i = 1, 2, 3. \quad (\text{B.3})$$

Then (B.1) may be written as

$$Z(p) - A = \frac{K}{x^2 + p^2}. \quad (\text{B.4})$$

Taking the reciprocal gives

$$\frac{1}{Z(p) - A} = \frac{x^2 + p^2}{K} = \frac{1}{K}p^2 + \frac{x^2}{K}. \quad (\text{B.5})$$

Hence, if the data are described exactly by (B.1), then the transformed quantity

$$y(p) := \frac{1}{Z(p) - A} \quad (\text{B.6})$$

is a straight line in p^2 . Equivalently,

$$y = ms + b, \quad m = \frac{1}{K}, \quad b = \frac{x^2}{K}, \quad (\text{B.7})$$

where $s = p^2$.

Determination of A

At the three given points,

$$y_i = \frac{1}{Z_i - A}, \quad s_i = p_i^2. \quad (\text{B.8})$$

Since y must be linear in s , the slope between points 1 and 2 must equal the slope between points 2 and 3:

$$\frac{y_1 - y_2}{s_1 - s_2} = \frac{y_2 - y_3}{s_2 - s_3}. \quad (\text{B.9})$$

Introduce the shorthand

$$D_{12} = s_1 - s_2 = p_1^2 - p_2^2, \quad D_{23} = s_2 - s_3 = p_2^2 - p_3^2. \quad (\text{B.10})$$

Substituting $y_i = 1/(Z_i - A)$ into (B.9) gives

$$\frac{\frac{1}{Z_1 - A} - \frac{1}{Z_2 - A}}{D_{12}} = \frac{\frac{1}{Z_2 - A} - \frac{1}{Z_3 - A}}{D_{23}}. \quad (\text{B.11})$$

Now use the identity

$$\frac{1}{Z_i - A} - \frac{1}{Z_j - A} = \frac{Z_j - Z_i}{(Z_i - A)(Z_j - A)}. \quad (\text{B.12})$$

Then (B.11) becomes

$$\frac{Z_2 - Z_1}{D_{12}(Z_1 - A)(Z_2 - A)} = \frac{Z_3 - Z_2}{D_{23}(Z_2 - A)(Z_3 - A)}. \quad (\text{B.13})$$

After cancelling the common factor $(Z_2 - A)$ and cross-multiplying, one obtains

$$(Z_2 - Z_1)D_{23}(Z_3 - A) = (Z_3 - Z_2)D_{12}(Z_1 - A). \quad (\text{B.14})$$

Expanding both sides and collecting the terms proportional to A ,

$$-(Z_2 - Z_1)D_{23}A + (Z_3 - Z_2)D_{12}A = (Z_3 - Z_2)D_{12}Z_1 - (Z_2 - Z_1)D_{23}Z_3. \quad (\text{B.15})$$

Therefore,

$$A = \frac{(Z_3 - Z_2)D_{12}Z_1 - (Z_2 - Z_1)D_{23}Z_3}{(Z_3 - Z_2)D_{12} - (Z_2 - Z_1)D_{23}}. \quad (\text{B.16})$$

This is the exact closed-form expression used to determine A from the first three tabulated values.

Determination of K and x^2

Once A is known, the transformed points

$$y_i = \frac{1}{Z_i - A} \quad (\text{B.17})$$

lie exactly on the straight line

$$y = ms + b. \quad (\text{B.18})$$

Hence the slope can be found from any pair of points, for example points 1 and 2:

$$m = \frac{y_1 - y_2}{s_1 - s_2}. \quad (\text{B.19})$$

The intercept is then

$$b = y_1 - ms_1. \quad (\text{B.20})$$

Comparing with (B.7),

$$m = \frac{1}{K}, \quad b = \frac{x^2}{K}, \quad (\text{B.21})$$

so that

$$K = \frac{1}{m}, \quad x^2 = bK. \quad (\text{B.22})$$

Zero-momentum value

Finally, substituting $p = 0$ into (B.1) gives

$$Z_{\text{eff}}(0) = \frac{K}{x^2} + A. \quad (\text{B.23})$$

This value was inserted at $p = 0$ and then used as the low-momentum anchor for the interpolation of the full $Z_{\text{eff}}(p)$ data onto the solver grid.

Remarks

This construction is exact for three data points under the assumption that the local low-momentum behaviour is well represented by (B.1). In the present work, only the resulting anchor value $Z_{\text{eff}}(0)$ was used operationally. For gases in which the first three MBT points are only weakly varying, the intermediate fitted quantities K and x^2 should not necessarily be over-interpreted physically; their role here is simply to provide a smooth and analytically consistent extrapolation to zero momentum.

Appendix C

Additional $Z_{\text{eff}}(p)$ input curves

For completeness, the interpolated total $Z_{\text{eff}}(p)$ curves for neon, argon and krypton are shown in Figures C.1, C.2 and C.3, respectively. These were obtained in the same way as the helium and xenon curves shown in the main text: the partial-wave MBT values were first combined to form the total $Z_{\text{eff}}(p)$, a zero-momentum anchor was inserted using the exact three-point solve described in Appendix B, and the resulting data were then interpolated onto the uniform solver grid. Together with Figure 2.2, these figures provide the full set of annihilation-rate inputs used in the simulations.

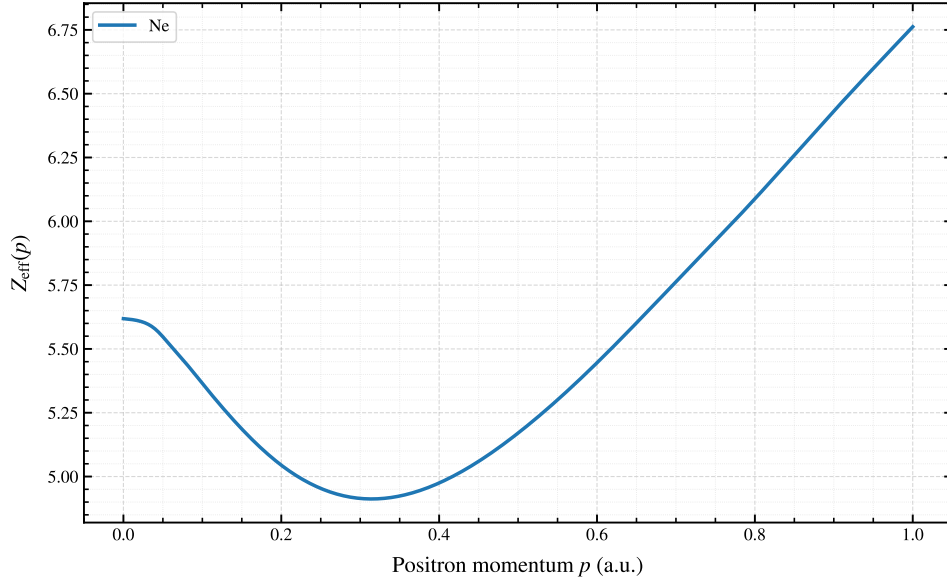


Figure C.1: Interpolated total $Z_{\text{eff}}(p)$ for neon. The curve remains comparatively modest over the full momentum range, with only a weak low-momentum enhancement.

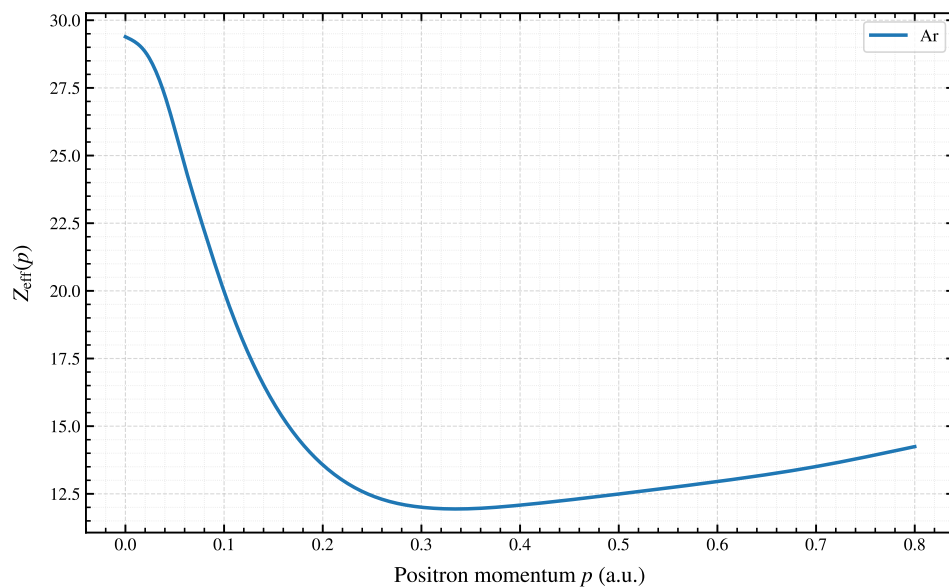


Figure C.2: Interpolated total $Z_{\text{eff}}(p)$ for argon. Compared with neon, the low-momentum values are substantially larger, reflecting stronger positron–atom correlation effects.

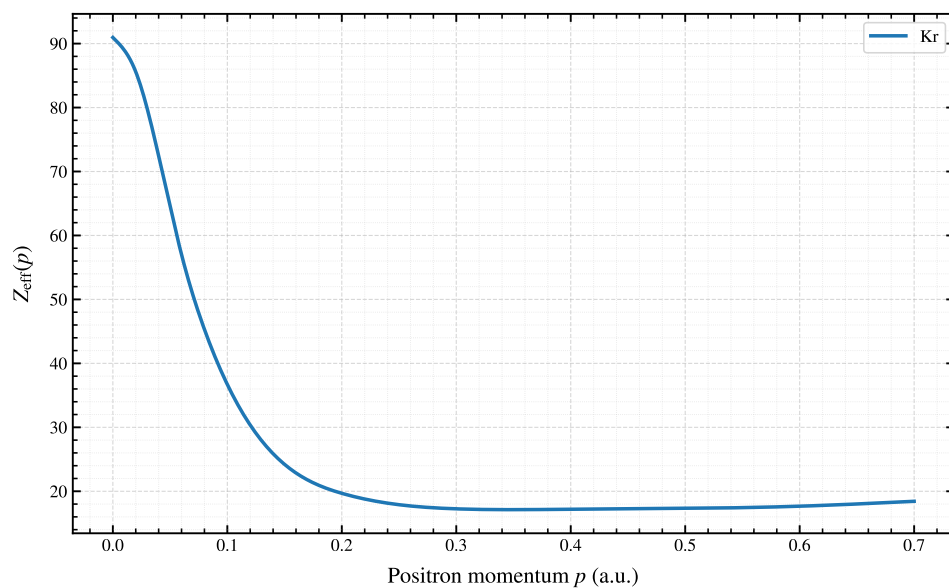


Figure C.3: Interpolated total $Z_{\text{eff}}(p)$ for krypton. The pronounced enhancement at small momentum is stronger again than in argon, consistent with the general increase in annihilation strength along the noble-gas sequence.

Appendix D

Additional diagnostic plots for noble-gas cooling

This appendix collects additional diagnostic plots for helium, neon, argon, krypton and xenon that are not shown in the main text, including the surviving fraction $N(\tau)$, averaged annihilation rate $\bar{\lambda}(\tau)$, mean energy $\bar{E}(\tau)$ and averaged annihilation parameter $\bar{Z}_{\text{eff}}(\tau)$. These are included for completeness and to provide a fuller record of the numerical results discussed in Chapter 4.

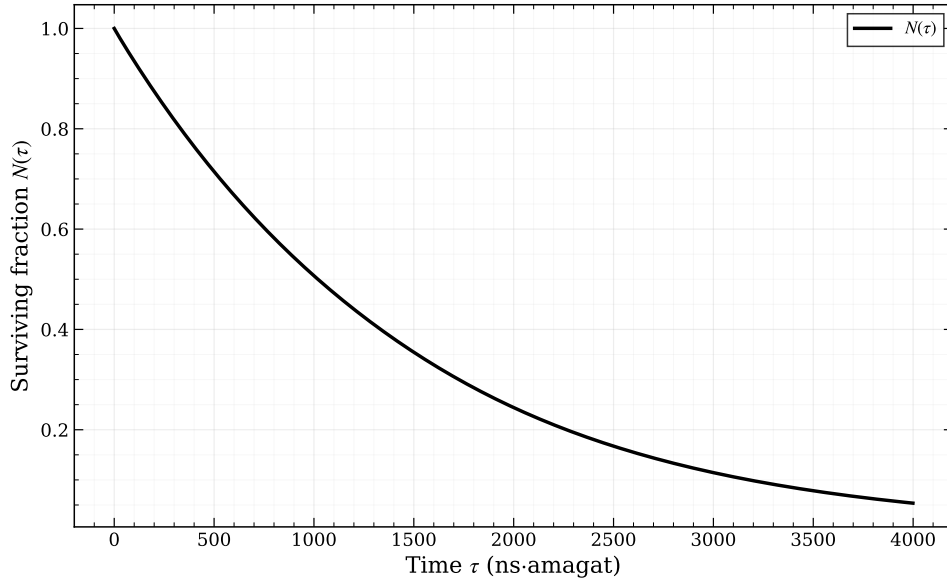


Figure D.1: Surviving fraction $N(\tau)$ for helium. The decay is gradual over the full cooling interval, confirming that annihilation acts only as a weak depletion mechanism in helium compared with the stronger-loss cases discussed later.

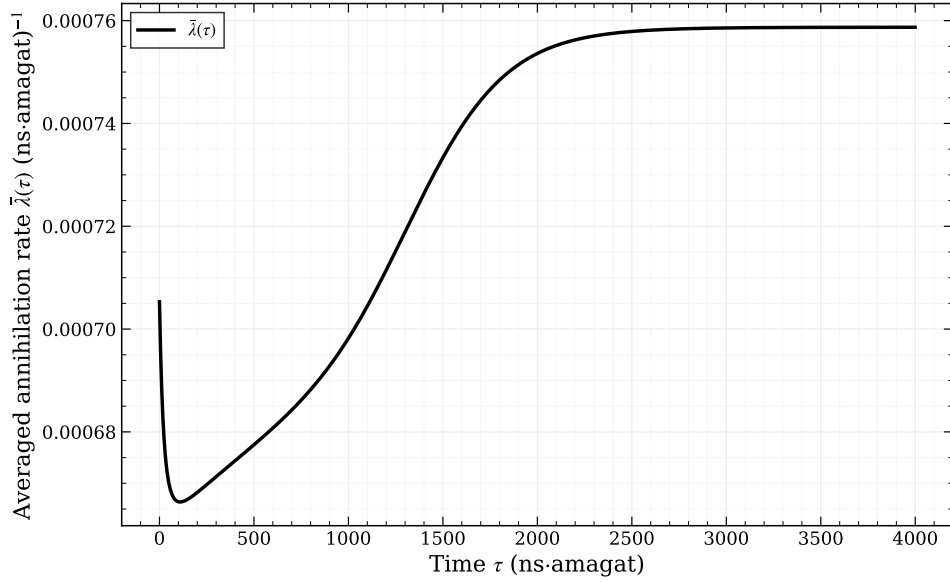


Figure D.2: Averaged annihilation rate $\bar{\lambda}(\tau)$ for helium. After a small initial decrease, the rate rises smoothly towards its late-time value as the positron distribution cools into the lower-momentum region where annihilation is slightly more favourable.

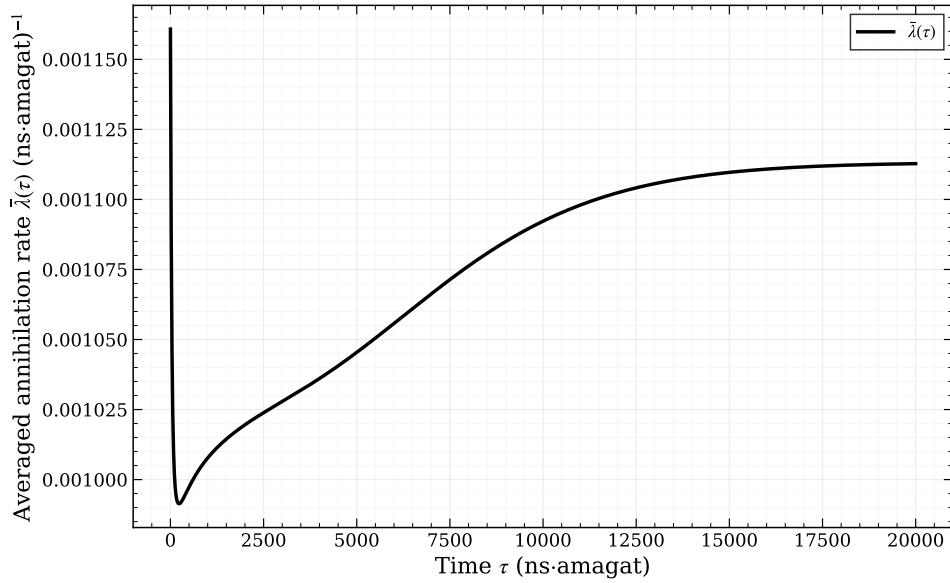


Figure D.3: Averaged annihilation rate $\bar{\lambda}(\tau)$ for neon. After a sharp initial drop, the rate increases only slowly as the positron distribution drifts through the bottleneck region associated with the Ramsauer–Townsend minimum in the momentum-transfer cross section. This behaviour mirrors the very slow evolution of $\bar{Z}_{\text{eff}}(\tau)$ and reflects the transport-limited character of cooling in neon.

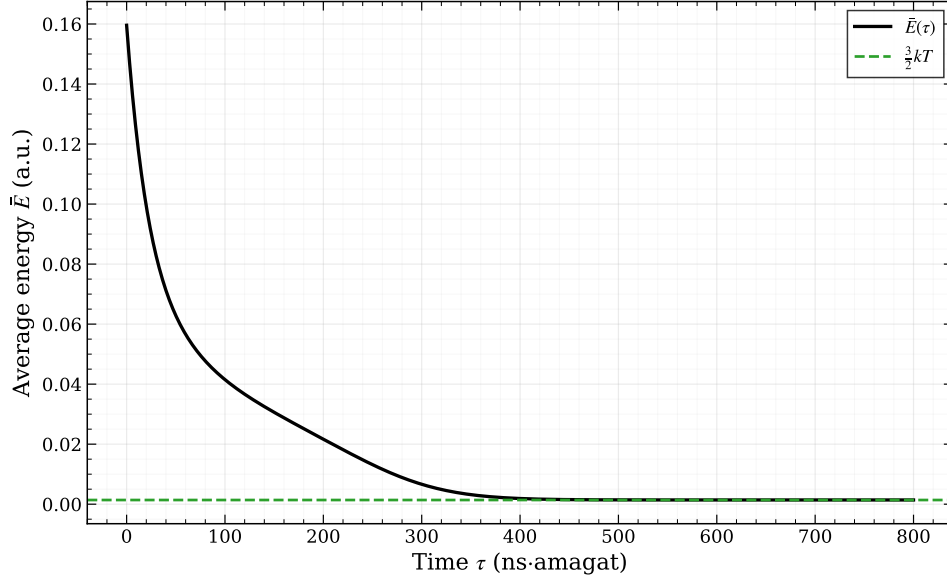


Figure D.4: Mean positron energy $\bar{E}(\tau)$ for argon. The distribution cools rapidly towards the thermal scale, and by the end of the run the mean energy lies very close to $3k_{\text{B}}T/2$, indicating near-thermalization within the simulated time window.

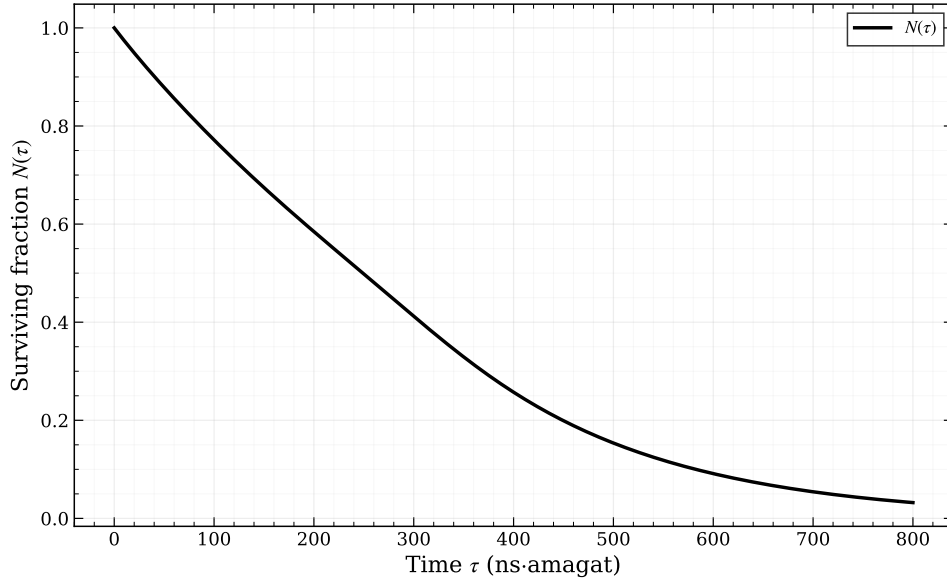


Figure D.5: Surviving fraction $N(\tau)$ for argon. The decay is stronger than in helium but remains moderate enough that a substantial fraction of the swarm survives while cooling proceeds.

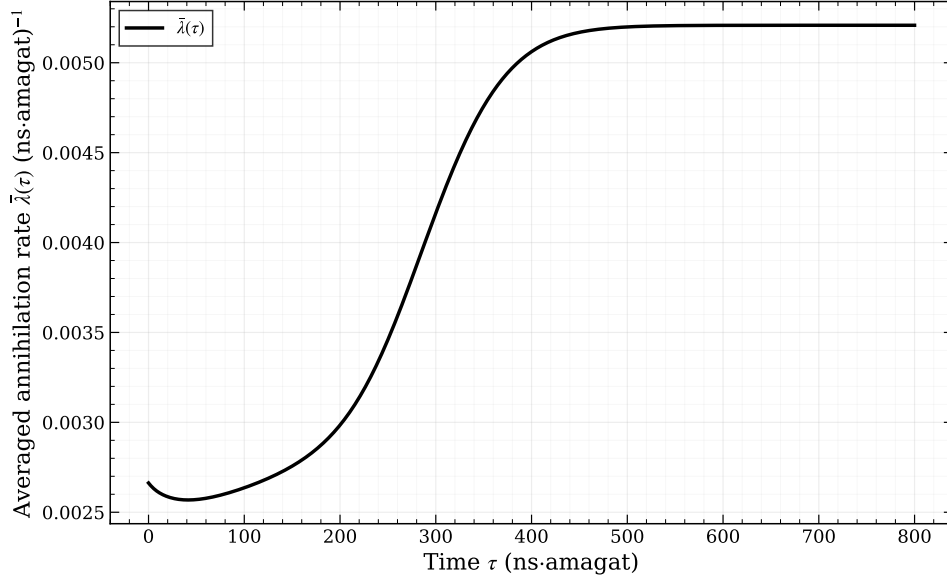


Figure D.6: Averaged annihilation rate $\bar{\lambda}(\tau)$ for argon. After a shallow early decrease, the rate rises smoothly as the positron distribution shifts towards lower momenta where annihilation becomes more probable.

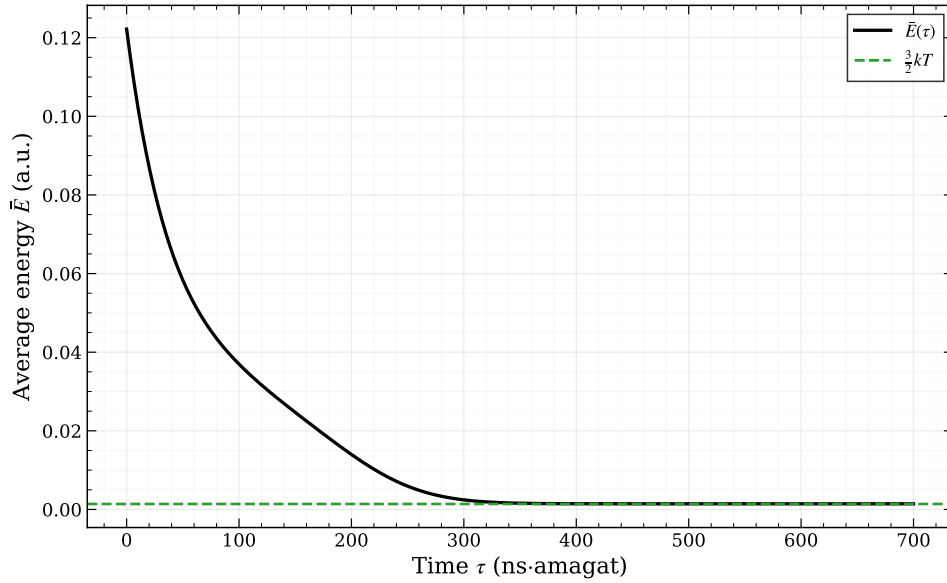


Figure D.7: Mean positron energy $\bar{E}(\tau)$ for krypton. Cooling remains efficient, but the late-time mean energy stays slightly further above the thermal value than in argon, reflecting the stronger influence of annihilation during the final stages of relaxation.

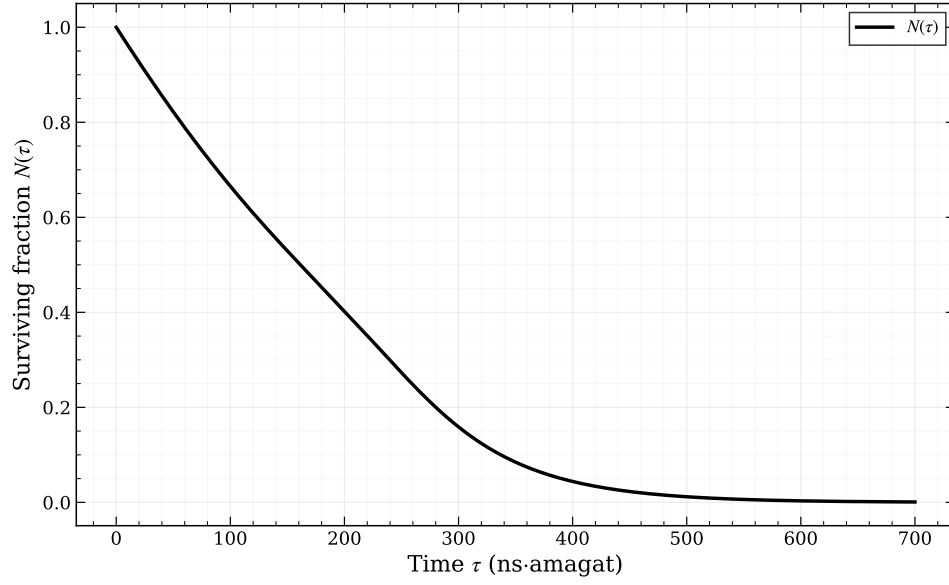


Figure D.8: Surviving fraction $N(\tau)$ for krypton. The surviving population decreases more rapidly than in argon, showing the increasing importance of annihilation across the noble-gas sequence.

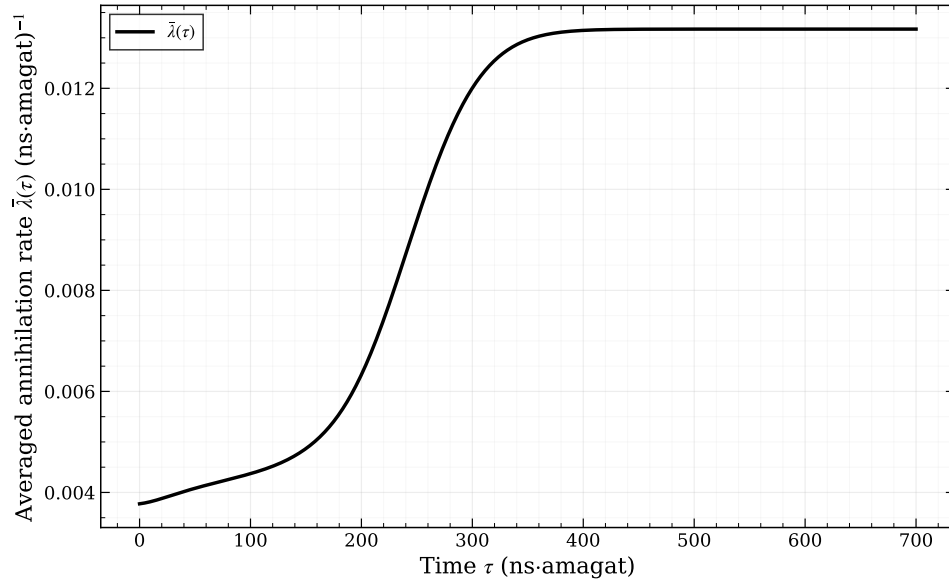


Figure D.9: Averaged annihilation rate $\bar{\lambda}(\tau)$ for krypton. The stronger late-time rise compared with argon reflects the larger annihilation response as the swarm accumulates at low momenta.

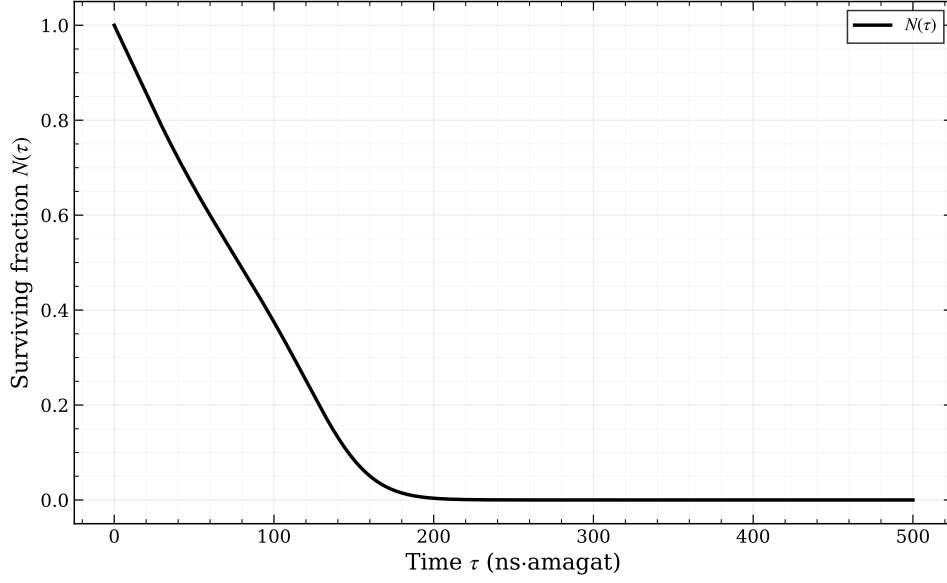


Figure D.10: Surviving fraction $N(\tau)$ for xenon. The rapid decay shows that annihilation removes most of the positron swarm before a thermalized Maxwellian distribution can be established.

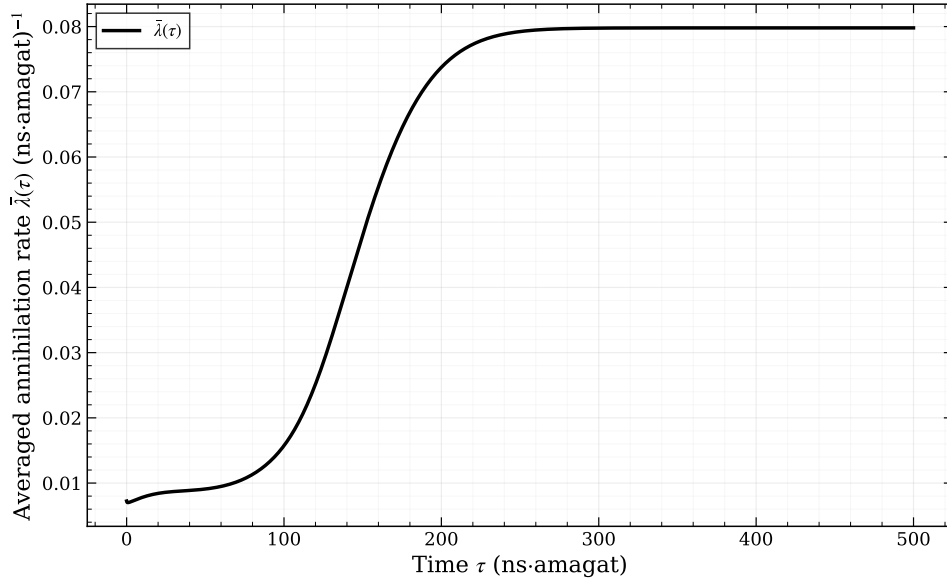


Figure D.11: Averaged annihilation rate $\bar{\lambda}(\tau)$ for xenon. The sharp rise and high late-time plateau reflect the strong enhancement of annihilation as the swarm is driven into the low-momentum region where $Z_{\text{eff}}(p)$ is largest.

Appendix E

Derivation of the asymptotic eigenmode problem

This appendix collects the longer derivations underlying the asymptotic eigenmode analysis of Chapter 5. The starting Fokker–Planck description is the same as that used throughout this dissertation and follows the standard kinetic treatment of positron cooling in gases [7], with the xenon transport and annihilation input data taken from the many-body-theory calculations of Ref. [3]. The transformation of the late-time eigenproblem to weighted Sturm–Liouville form and the associated variational interpretation follow standard ideas from Sturm–Liouville and eigenvalue theory [22, 23].

Continuous asymptotic eigenproblem

Start from the scaled radial Fokker–Planck equation

$$\frac{\partial F}{\partial \tau} = \alpha \frac{\partial}{\partial p} \left\{ \sigma_m(p) p \left[\frac{\partial F}{\partial p} + \left(\frac{p}{k_B T} - \frac{2}{p} \right) F \right] \right\} - \lambda(p) F, \quad (\text{E.1})$$

where the transport coefficient is determined by the momentum-transfer cross section $\sigma_m(p)$, and the annihilation sink is governed by $\lambda(p)$. In the xenon calculations discussed in this dissertation, these quantities were constructed from the many-body-theory data of Ref. [3].

Introduce the shorthand

$$a(p) = \sigma_m(p) p. \quad (\text{E.2})$$

Then Eq. (E.1) may be written as

$$\frac{\partial F}{\partial \tau} = \alpha \frac{d}{dp} \left\{ a(p) \left[\left(\frac{p}{k_B T} - \frac{2}{p} \right) F + \frac{dF}{dp} \right] \right\} - \lambda(p) F. \quad (\text{E.3})$$

Now assume a separable late-time form

$$F(p, \tau) = e^{-\beta \tau} \phi(p). \quad (\text{E.4})$$

Substituting Eq. (E.4) into Eq. (E.3) gives

$$-\beta \phi = \alpha \frac{d}{dp} \left\{ a(p) \left[\left(\frac{p}{k_B T} - \frac{2}{p} \right) \phi + \phi' \right] \right\} - \lambda(p) \phi, \quad (\text{E.5})$$

where the prime denotes differentiation with respect to p .

Equivalently,

$$\alpha \frac{d}{dp} \left\{ a(p) \left[\left(\frac{p}{k_B T} - \frac{2}{p} \right) \phi + \phi' \right] \right\} = [\lambda(p) - \beta] \phi. \quad (\text{E.6})$$

This is the continuous asymptotic eigenvalue problem.

Cooling-only case

When $\lambda(p) = 0$, Eq. (E.5) becomes

$$\alpha \frac{d}{dp} \left\{ a(p) \left[\left(\frac{p}{k_B T} - \frac{2}{p} \right) \phi + \phi' \right] \right\} = -\beta \phi. \quad (\text{E.7})$$

The stationary mode corresponds to $\beta = 0$, so

$$\frac{d}{dp} \left\{ a(p) \left[\left(\frac{p}{k_B T} - \frac{2}{p} \right) \phi + \phi' \right] \right\} = 0. \quad (\text{E.8})$$

Hence

$$a(p) \left[\left(\frac{p}{k_B T} - \frac{2}{p} \right) \phi + \phi' \right] = C. \quad (\text{E.9})$$

For a physical equilibrium state with no flux through the boundaries, one takes $C = 0$. Assuming $a(p) \neq 0$ on the interior of the domain then gives

$$\phi' + \left(\frac{p}{k_B T} - \frac{2}{p} \right) \phi = 0. \quad (\text{E.10})$$

Dividing by ϕ and integrating,

$$\frac{\phi'}{\phi} = -\frac{p}{k_B T} + \frac{2}{p} \implies \ln \phi = -\frac{p^2}{2k_B T} + 2 \ln p + \text{const.} \quad (\text{E.11})$$

so that

$$\phi(p) \propto p^2 e^{-p^2/2k_B T}. \quad (\text{E.12})$$

This is exactly the radial Maxwell–Boltzmann form.

Transformation to Sturm–Liouville form

Write the eigenfunction as

$$\phi(p) = \phi_{\text{th}}(p) u(p), \quad \phi_{\text{th}}(p) = p^2 e^{-p^2/2k_{\text{B}}T}. \quad (\text{E.13})$$

Since ϕ_{th} satisfies

$$\phi'_{\text{th}} + \left(\frac{p}{k_{\text{B}}T} - \frac{2}{p} \right) \phi_{\text{th}} = 0, \quad (\text{E.14})$$

one finds

$$\left(\frac{p}{k_{\text{B}}T} - \frac{2}{p} \right) \phi + \phi' = \phi_{\text{th}} u'. \quad (\text{E.15})$$

Therefore Eq. (E.5) becomes

$$\alpha \frac{d}{dp} [a(p) \phi_{\text{th}}(p) u'(p)] - \lambda(p) \phi_{\text{th}}(p) u(p) = -\beta \phi_{\text{th}}(p) u(p). \quad (\text{E.16})$$

Rearranging,

$$-\frac{d}{dp} [a(p) \phi_{\text{th}}(p) u'(p)] + \frac{\lambda(p)}{\alpha} \phi_{\text{th}}(p) u(p) = \frac{\beta}{\alpha} \phi_{\text{th}}(p) u(p). \quad (\text{E.17})$$

This has the form of a weighted Sturm–Liouville problem [22],

$$-\frac{d}{dp} (P(p) u'(p)) + Q(p) u(p) = \mu W(p) u(p), \quad (\text{E.18})$$

with

$$P(p) = a(p) \phi_{\text{th}}(p), \quad Q(p) = \frac{\lambda(p)}{\alpha} \phi_{\text{th}}(p), \quad W(p) = \phi_{\text{th}}(p), \quad \mu = \frac{\beta}{\alpha}. \quad (\text{E.19})$$

Rayleigh quotient

Multiply Eq. (E.17) by a test function $v(p)$ and integrate over $p \in [0, p_{\text{max}}]$:

$$\int_0^{p_{\text{max}}} v \left[-\frac{d}{dp} (a \phi_{\text{th}} u') + \frac{\lambda}{\alpha} \phi_{\text{th}} u \right] dp = \frac{\beta}{\alpha} \int_0^{p_{\text{max}}} \phi_{\text{th}} u v dp. \quad (\text{E.20})$$

Integrating the derivative term by parts gives

$$\int_0^{p_{\text{max}}} a \phi_{\text{th}} u' v' dp - [v a \phi_{\text{th}} u']_0^{p_{\text{max}}} + \frac{1}{\alpha} \int_0^{p_{\text{max}}} \lambda \phi_{\text{th}} u v dp = \frac{\beta}{\alpha} \int_0^{p_{\text{max}}} \phi_{\text{th}} u v dp. \quad (\text{E.21})$$

Under the natural no-flux/regularity boundary condition,

$$[v a \phi_{\text{th}} u']_0^{p_{\text{max}}} = 0, \quad (\text{E.22})$$

then Eq. (E.21) becomes

$$\int_0^{p_{\max}} a(p) \phi_{\text{th}}(p) u'(p) v(p) dp + \frac{1}{\alpha} \int_0^{p_{\max}} \lambda(p) \phi_{\text{th}}(p) u(p) v(p) dp = \frac{\beta}{\alpha} \int_0^{p_{\max}} \phi_{\text{th}}(p) u(p) v(p) dp. \quad (\text{E.23})$$

Setting $v = u$ yields the corresponding Rayleigh quotient [23]

$$\beta[u] = \alpha \frac{\int_0^{p_{\max}} a(p) \phi_{\text{th}}(p) (u'(p))^2 dp + \int_0^{p_{\max}} \lambda(p) \phi_{\text{th}}(p) (u(p))^2 dp}{\int_0^{p_{\max}} \phi_{\text{th}}(p) (u(p))^2 dp}. \quad (\text{E.24})$$

Special cases and weak-annihilation expansion

No annihilation

If $\lambda(p) = 0$, Eq. (E.24) reduces to

$$\beta[u] = \alpha \frac{\int_0^{p_{\max}} a(p) \phi_{\text{th}}(p) (u'(p))^2 dp}{\int_0^{p_{\max}} \phi_{\text{th}}(p) (u(p))^2 dp}. \quad (\text{E.25})$$

Since both numerator and denominator are non-negative, one has

$$\beta \geq 0. \quad (\text{E.26})$$

The minimum value $\beta_0 = 0$ occurs only when $u'(p) = 0$, i.e. when u is constant. Hence the lowest mode is

$$\phi_0(p) \propto \phi_{\text{th}}(p) \propto p^2 e^{-p^2/2k_{\text{B}}T}, \quad (\text{E.27})$$

which is the thermal Maxwellian, while all higher modes decay exponentially.

Constant annihilation

If $\lambda(p) = \lambda_0$ is constant, then $u = \text{const}$, still solves the eigenproblem and gives

$$\beta_0 = \lambda_0. \quad (\text{E.28})$$

Thus a momentum-independent annihilation rate changes only the exponential decay rate, not the asymptotic shape.

Weak annihilation

For weak annihilation, take $u(p) \approx 1$. Then $u'(p) \approx 0$, so the distortion term in Eq. (E.24) vanishes to leading order, giving

$$\beta_0 \approx \frac{\int_0^{p_{\max}} \lambda(p) \phi_{\text{th}}(p) dp}{\int_0^{p_{\max}} \phi_{\text{th}}(p) dp}. \quad (\text{E.29})$$

Hence the leading-order decay rate is just the thermal average of the annihilation rate.

To obtain the first correction to the shape, write

$$u(p) = 1 + \delta u(p), \quad (\text{E.30})$$

with $|\delta u| \ll 1$. Retaining only first-order terms gives

$$-\frac{d}{dp}[a(p)\phi_{\text{th}}(p)(\delta u)'(p)] \approx \frac{\phi_{\text{th}}(p)}{\alpha}(\beta_0 - \lambda(p)). \quad (\text{E.31})$$

This shows that the correction is driven by the difference $\beta_0 - \lambda(p)$. Where $\lambda(p) > \beta_0$, the asymptotic mode tends to suppress weight; where $\lambda(p) < \beta_0$, it tends to enhance weight.

For xenon, $\lambda(p)$ is especially large at low momentum, so the asymptotic mode is depleted at small p relative to the Maxwellian. This is why the dominant xenon eigenmode is hotter than thermal and why the actual late-time decay rate lies below the simple thermal-average estimate.

Appendix F

Python code: Xenon solver

This appendix presents the xenon implementation of the finite-volume Fokker–Planck solver used in this work. It is included as a representative example, since the corresponding helium, neon, argon and krypton codes are identical in structure and differ only in the tabulated input data and atomic constants. The script evolves the radial momentum distribution $F(p, \tau)$, includes the annihilation sink term, and records the main diagnostics used in the analysis, namely the surviving fraction, the averaged annihilation rate, the averaged $\bar{Z}_{\text{eff}}(\tau)$, and the mean energy.

```
1 import numpy as np
2
3 # =====
4 # Xenon cooling and annihilation solver
5 # -----
6 # This script evolves the radial momentum distribution F(p, tau)
7 # for positrons cooling in xenon, including an annihilation sink.
8 #
9 # Input files:
10 #   - xe_sigma_m_01res.txt
11 #   - xe_zeff_total_on_grid.txt
12 #
13 # Time variable:
14 #   tau = t(ns) * n_g(amagat)
15 # =====
```

```

16
17
18 # -----
19 # 1. Momentum grid
20 # -----
21 pmax = 0.6
22 Np = 400
23
24 p = np.linspace(0.0, pmax, Np + 1)
25 dp = p[1] - p[0]
26
27
28 # -----
29 # 2. Physical parameters
30 # -----
31 Tgas = 293.0
32 kT_au = 3.1668e-6 * Tgas
33 D0 = 5.21e-4
34
35 # Xenon mass in electron masses
36 M_Xe = 131.293 * 1822.888486
37
38 # Diffusion prefactor appearing in the FP equation
39 alpha = D0 * (Tgas / M_Xe)
40
41
42 # -----
43 # 3. Load transport and annihilation data
44 # -----
45 # Momentum-transfer cross section sigma_m(p)
46 sigma_data = np.loadtxt("xe_sigma_m_01res.txt")
47 p_sigma = sigma_data[:, 0]
48 sigma_tab = sigma_data[:, 1]

```

```

49 sigma_node = np.interp(p, p_sigma, sigma_tab)
50
51 # Zeff(p)
52 zeff_data = np.loadtxt("xe_zeff_total_on_grid.txt")
53 p_zeff = zeff_data[:, 0]
54 zeff_tab = zeff_data[:, 1]
55 zeff_node = np.interp(p, p_zeff, zeff_tab)
56
57 # Annihilation rate lambda(p) in (ns amagat)^(-1)
58 r0 = 2.8179403262e-15
59 c = 2.99792458e8
60 n_amg = 2.6867805e25
61
62 lambda_prefactor = np.pi * r0**2 * c * n_amg * 1.0e-9
63 lam = lambda_prefactor * zeff_node
64
65
66 # -----
67 # 4. Initial condition
68 # -----
69 # Uniform in energy corresponds to F(p,0) proportional to p
70 F = 2.0 * p / pmax**2
71 F[0] = 0.0
72 F[-1] = 0.0
73
74 # Normalise to unit total population
75 F /= np.sum(F) * dp
76 F0 = F.copy()
77
78
79 # -----
80 # 5. Time-stepping parameters
81 # -----

```

```

82 tau_end = 500.0
83 dt = 1.0e-3
84 nsteps = int(tau_end / dt) + 1
85
86
87 # -----
88 # 6. Diagnostics
89 # -----
90 tau_hist = []
91 N_hist = []
92 lambda_avg_hist = []
93 zeff_avg_hist = []
94 Ebar_hist = []
95
96 # Thermal average energy and threshold used for "cooling time"
97 E_thermal = 1.5 * kT_au
98 E_thresh = 1.01 * E_thermal
99
100 print(f"lambda prefactor = {lambda_prefactor:.6e} (ns amagat)^(-1)")
101 print(f"lambda range      = [{lam.min():.6e}, {lam.max():.6e}]")
102 print(f"max(lambda dt)      = {np.max(lam * dt):.6e}")
103
104
105 # -----
106 # 7. Main time-stepping loop
107 # -----
108 for step in range(nsteps):
109
110     # Cell-face values
111     p_face = 0.5 * (p[:-1] + p[1:])
112     F_face = 0.5 * (F[:-1] + F[1:])
113     dFdp_face = (F[1:] - F[:-1]) / dp
114

```

```

115     # Interpolate sigma_m onto faces
116     sigma_face = np.interp(p_face, p, sigma_node)
117
118     # Coefficients in the flux
119     G_face = sigma_face * p_face
120     A_face = (p_face / kT_au) - (2.0 / p_face)
121
122     # Conservative flux for the cooling operator
123     J_face = -alpha * G_face * (dFdp_face + A_face * F_face)
124
125     # No-flux boundary conditions
126     J_face[0] = 0.0
127     J_face[-1] = 0.0
128
129     # Conservative update for the transport part
130     rhs = np.zeros_like(F)
131     rhs[1:-1] = -(J_face[1:] - J_face[:-1]) / dp
132
133     F_star = F + dt * rhs
134     F_star[0] = 0.0
135     F_star[-1] = 0.0
136
137     # Annihilation step
138     F = F_star * np.exp(-lam * dt)
139     F[0] = 0.0
140     F[-1] = 0.0
141
142     # Save diagnostics every 100 steps
143     if step % 100 == 0:
144         tau = step * dt
145         N = np.sum(F) * dp
146
147         if N > 0.0:

```



```

148         lambda_avg = np.sum(lam * F) * dp / N
149         zeff_avg = np.sum(zeff_node * F) * dp / N
150         Ebar = np.sum(0.5 * p**2 * F) * dp / N
151     else:
152         lambda_avg = np.nan
153         zeff_avg = np.nan
154         Ebar = np.nan
155
156     tau_hist.append(tau)
157     N_hist.append(N)
158     lambda_avg_hist.append(lambda_avg)
159     zeff_avg_hist.append(zeff_avg)
160     Ebar_hist.append(Ebar)
161
162
163 # Convert to arrays
164 tau_hist = np.array(tau_hist)
165 N_hist = np.array(N_hist)
166 lambda_avg_hist = np.array(lambda_avg_hist)
167 zeff_avg_hist = np.array(zeff_avg_hist)
168 Ebar_hist = np.array(Ebar_hist)
169
170
171 # -----
172 # 8. Cooling time from mean energy
173 # -----
174 cooling_time = None
175
176 valid = np.isfinite(Ebar_hist)
177 tau_valid = tau_hist[valid]
178 Ebar_valid = Ebar_hist[valid]
179
180 crossings = np.where(Ebar_valid <= E_thresh)[0]

```

```

181
182 if len(crossings) > 0:
183     i = crossings[0]
184
185     if i == 0:
186         cooling_time = tau_valid[0]
187     else:
188         t1, t2 = tau_valid[i - 1], tau_valid[i]
189         E1, E2 = Ebar_valid[i - 1], Ebar_valid[i]
190
191         if E2 != E1:
192             cooling_time = t1 + (E_thresh - E1) * (t2 - t1) / (E2 -
193                 E1)
194         else:
195             cooling_time = t2
196
197         print(f"\nCooling threshold reached at tau = {cooling_time:.6f}
198             ns amagat")
199     else:
200         print("\nCooling threshold was not reached in the simulated time
201             window.")
202
203 # -----
204 # 9. Late-time annihilation diagnostics
205 # -----
206 n_tail = max(5, len(zeff_avg_hist) // 10)
207
208 zbar_final = zeff_avg_hist[-1]
209
210 print("\nLate-time diagnostics:")
211 print(f"Final surviving fraction N(tau_end) = {N_hist[-1]:.8e}")
212 print(f"Final Zbar_eff                      = {zbar_final:.8f}")

```

```
211
212
213 # -----
214 # 10. Save output for plotting
215 # -----
216 np.savetxt(
217     "xe_diagnostics.txt",
218     np.column_stack((tau_hist, N_hist, lambda_avg_hist,
219                     zeff_avg_hist, Ebar_hist)),
219     header="tau  N(tau)  lambda_bar(tau)  Zbar_eff(tau)  Ebar(tau)"
220 )
```

Listing F.1: Representative Python implementation of the finite-volume Fokker–Planck solver for positron cooling and annihilation in xenon.

Appendix G

Construction of Xenon MBT Input Data

This appendix records the preprocessing used to generate the xenon input data for the solver. A single representative example is shown, since the corresponding routines for helium, neon, argon and krypton are identical in structure and differ only in the tabulated many-body-theory data and the relevant atomic constants.

Two ingredients were required on the solver momentum grid: the momentum-transfer cross section $\sigma_m(p)$, and the total annihilation parameter $Z_{\text{eff}}(p)$. The first was constructed from the tabulated xenon phase shifts δ_0 , δ_1 and δ_2 reported in Ref. [3], together with the standard polarization-tail approximation for higher partial waves. The second was constructed by summing the tabulated s -, p - and d -wave contributions to $Z_{\text{eff}}(p)$ from Ref. [3], estimating the zero-momentum anchor $Z_{\text{eff}}(0)$ from the first three points using the virtual-level form, and interpolating the result onto the solver grid.

G.1 Construction of the xenon $\sigma_m(p)$

The following script generates the xenon momentum-transfer cross section table used by the solver. It interpolates the tabulated s -, p - and d -wave phase shifts, appends the large- l polarization-tail contribution for $l = 3, \dots, 10$, evaluates $\sigma_m(k)$, and writes the result to file.

```
1 import numpy as np
2 from scipy.interpolate import CubicSpline
```

```

3
4 # -----
5 # Build the xenon momentum-transfer cross section sigma_m(k)
6 # from the MBT phase shifts listed in Table VII.
7 # -----
8
9 # Xenon constants
10 alpha_d = 27.04          # dipole polarisability (a.u.)
11 a_xe = -84.5            # scattering length (a.u.)
12
13 # Solver grid in momentum space
14 k_grid = np.concatenate(([0.0], np.arange(0.02, 0.61, 0.01)))
15
16 # MBT phase shifts for l = 0, 1, 2
17 k_mbt = np.array([0.02, 0.04, 0.06, 0.08, 0.10,
18                  0.20, 0.30, 0.40, 0.50, 0.60])
19
20 delta0_tab = np.array([0.9849, 1.1510, 1.1300, 1.0850, 1.0270,
21                       0.7199, 0.4256, 0.1551, -0.0922, -0.3192])
22
23 delta1_tab = np.array([2.303e-3, 9.418e-3, 2.178e-2, 4.001e-2,
24                       6.450e-2,
25                       0.2666, 0.4784, 0.5624, 0.5354, 0.4503])
26
27 delta2_tab = np.array([3.223e-4, 1.285e-3, 2.856e-3, 5.015e-3,
28                       7.809e-3,
29                       3.572e-2, 9.425e-2, 0.1904, 0.3118, 0.4298])
29
30 # Smooth interpolants for the tabulated phase shifts
31 delta0 = CubicSpline(k_mbt, delta0_tab)
32 delta1 = CubicSpline(k_mbt, delta1_tab)
33 delta2 = CubicSpline(k_mbt, delta2_tab)

```

```

34 sigma_m_vals = []
35
36 for k in k_grid:
37     if k == 0.0:
38         # Threshold limit from the scattering length
39         sigma_m_vals.append(4.0 * np.pi * a_xe**2)
40         continue
41
42     # Use the interpolated MBT phase shifts for l = 0, 1, 2
43     d0 = float(delta0(k))
44     d1 = float(delta1(k))
45     d2 = float(delta2(k))
46
47     deltas = [d0, d1, d2]
48
49     # Add the large-l polarisation tail for l = 3,...,10
50     for l in range(3, 11):
51         dl = (np.pi * alpha_d * k**2) / ((2*l - 1) * (2*l + 1) * (2*l
52             + 3))
53         deltas.append(dl)
54
55     # Momentum-transfer cross section
56     sigma_m = (4.0 * np.pi / k**2) * sum(
57         (l + 1) * np.sin(deltas[l] - deltas[l + 1])**2
58         for l in range(len(deltas) - 1)
59     )
60     sigma_m_vals.append(sigma_m)
61
62 sigma_m_vals = np.array(sigma_m_vals)
63
64 np.savetxt(
65     "xe_sigma_m_01res.txt",

```

```

66     np.column_stack((k_grid, sigma_m_vals)),
67     header="k    sigma_m"
68 )
69
70 print("Saved xe_sigma_m_01res.txt")

```

Listing G.1: Python script used to generate the xenon momentum-transfer cross section table `xe_sigma_m_01res.txt`.

G.2 Construction of the xenon $Z_{\text{eff}}(p)$

The next script generates the total xenon $Z_{\text{eff}}(p)$ table used by the solver. The tabulated s -, p - and d -wave contributions are first summed, after which the zero-momentum value is estimated by fitting the first three data points to the virtual-level form

$$Z_{\text{eff}}(p) = \frac{K}{x^2 + p^2} + A. \quad (\text{G.1})$$

The resulting anchored $Z_{\text{eff}}(p)$ curve is then interpolated onto the solver grid. For xenon, a PCHIP interpolant was used to preserve the shape of the low-momentum data, whereas a standard CubicSpline interpolant was used for the other noble gases.

```

1  import numpy as np
2  from scipy.interpolate import PchipInterpolator
3
4  # -----
5  # Build the xenon total Zeff(p) table used by the solver.
6  # The p = 0 anchor is obtained from the first three points
7  # using the virtual-level form Z(p) = K / (x^2 + p^2) + A.
8  # -----
9
10 def three_point_virtual_level_fit(p_vals, z_vals):
11     """Exact 3-point solve for Z(p) = K/(x^2 + p^2) + A."""
12     p_vals = np.asarray(p_vals, dtype=float)
13     z_vals = np.asarray(z_vals, dtype=float)
14

```

```

15     p2 = p_vals**2
16     p1, p2_, p3 = p2
17     z1, z2, z3 = z_vals
18
19     d12 = p1 - p2_
20     d23 = p2_ - p3
21
22     num = (z3 - z2) * d12 * z1 - (z2 - z1) * d23 * z3
23     den = (z3 - z2) * d12 - (z2 - z1) * d23
24
25     if np.isclose(den, 0.0):
26         raise RuntimeError("3-point fit became degenerate.")
27
28     A = num / den
29
30     # Fit 1 / (Z - A) = b + m p^2
31     f = 1.0 / (z_vals - A)
32     m = (f[0] - f[1]) / (p2[0] - p2[1])
33     b = f[0] - m * p2[0]
34
35     K = 1.0 / m
36     x2 = b * K
37     Z0 = K / x2 + A
38
39     return K, x2, A, Z0
40
41
42 # Tabulated positron momenta
43 p_tab = np.array([0.02, 0.04, 0.06, 0.08, 0.10,
44                  0.20, 0.30, 0.40, 0.50, 0.60], dtype=float)
45
46 # s-wave contributions: valence + core
47 Zs_val = np.array([1227.0, 351.7, 161.0, 93.91, 62.39,

```



```

48         18.88, 10.01, 6.618, 4.906, 3.896])
49
50 Zs_core = np.array([29.71, 8.289, 3.767, 2.195, 1.463,
51                     4.608e-1, 2.623e-1, 1.912e-1, 1.597e-1, 1.453e-1])
52
53 # p-wave contributions: valence + core
54 Zp_val = np.array([1.040e-1, 4.235e-1, 9.763e-1, 1.782, 2.854,
55                    11.22, 16.75, 15.84, 13.28, 11.08])
56
57 Zp_core = np.array([1.085e-3, 4.428e-3, 1.025e-2, 1.880e-2, 3.032e-2,
58                     1.256e-1, 2.023e-1, 2.110e-1, 1.989e-1, 1.896e-1])
59
60 # d-wave contribution
61 Zd = np.array([2.716e-5, 4.350e-4, 2.205e-3, 6.975e-3, 1.704e-2,
62                2.713e-1, 1.325, 3.769, 7.384, 10.99])
63
64 # Total Zeff(p)
65 Z_total = (Zs_val + Zs_core) + (Zp_val + Zp_core) + Zd
66
67 # Estimate Zeff(0) from the first three points
68 K, x2, A, Z0 = three_point_virtual_level_fit(p_tab[:3], Z_total[:3])
69
70 # Add the p = 0 anchor before interpolation
71 p_use = np.insert(p_tab, 0, 0.0)
72 Z_use = np.insert(Z_total, 0, Z0)
73
74 # Shape-preserving interpolation on the solver grid
75 interp = PchipInterpolator(p_use, Z_use)
76
77 p_grid = np.arange(0.0, 0.6000001, 0.01)
78 Z_grid = np.maximum(interp(p_grid), 0.0)
79
80 np.savetxt(

```

```
81     "xe_zeff_total_on_grid.txt",  
82     np.column_stack((p_grid, Z_grid)),  
83     header="p   Zeff_total "  
84 )  
85  
86 print("Saved xe_zeff_total_on_grid.txt")
```

Listing G.2: Python script used to generate the xenon annihilation table `xe_zeff_total_on_grid.txt`.

The files produced by Listings [G.1](#) and [G.2](#) were then used as direct input for the xenon cooling and annihilation solver presented in [Appendix F](#).

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